

**GREEN
CLIMATE
FUND**

Meeting of the Board
10 - 13 July 2023
Incheon, Republic of Korea
Provisional Agenda Item 12

GCF/B.36/02/Add.08

19 June 2023

Consideration of funding proposals – Addendum VIII

Funding proposal package for FP213

Summary

This addendum contains the following six parts:

- a) A funding proposal summary titled “The Blue Green Bank (BGB)” submitted by Pegasus Capital Advisors (PCA);
- b) No-objection letter issued by the national designated authorities or focal point(s);
- c) Environmental and Social report(s) disclosure;
- d) Independent Technical Advisory Panel’s assessment;
- e) Response from the accredited entity to the independent Technical Advisory Panel’s assessment; and
- f) Gender documentation of the funding proposal.

These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled titled “The Blue Green Bank (BGB)” submitted by Pegasus Capital Advisors (PCA) is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged or commercially sensitive information of the entity.

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Funding Proposal

Project/Programme title: The Blue Green Bank (BGB).

Country(ies): Barbados

Accredited Entity: Pegasus Capital Advisors

Date of first submission: [2023/02/02]

Date of current submission 2023/06/14

Version number [V.011]



GREEN
CLIMATE
FUND

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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

A. PROJECT/PROGRAMME SUMMARY				
A.1. Project or programme	Programme	A.2. Public or private sector	Private	
A.3. Request for Proposals (RFP)	If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+. <u>Not applicable</u>			
A.4. Result area(s)	Check the applicable GCF result area(s) that the <u>overall</u> proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.			
		GCF contribution	Co-financers' contribution¹	
	Mitigation total	<u>Enter number</u> %	<u>Enter number</u> %	
	☒ Energy generation and access	50 %	50 %	
	☐ Low-emission transport	<u>Enter number</u> %	<u>Enter number</u> %	
	☐ Buildings, cities, industries and appliances	<u>Enter number</u> %	<u>Enter number</u> %	
	☐ Forestry and land use	<u>Enter number</u> %	<u>Enter number</u> %	
	Adaptation total	<u>Enter number</u> %	<u>Enter number</u> %	
	☐ Most vulnerable people and communities	<u>Enter number</u> %	<u>Enter number</u> %	
	☒ Health and well-being, and food and water security	15 %	15 %	
	☒ Infrastructure and built environment	35 %	35 %	
☐ Ecosystems and ecosystem services	<u>Enter number</u> %	<u>Enter number</u> %		
A.5. Expected mitigation outcome <i>(Core indicator 1: GHG emissions reduced, avoided or removed/sequestered)</i>	Lifetime: 16,736,509 tCO ₂ e reduced emissions after 25 years Implementation Period: 6,946,692 tCO ₂ e reduced emissions after 15 years	A.6. Expected adaptation outcome <i>(Core indicator 2: direct and indirect beneficiaries reached)</i>	Indicate the total number of direct and indirect beneficiaries	
			Direct: 226,000 50% Female	Indirect: ~65,200 51% female
			80% of the total population	Remaining 20% of the total population
A.7. Total financing (GCF + co-finance²)	30,500,000 USD	A.9. Project size	Small (Upto USD 50 million)	
A.8. Total GCF funding requested	<u>15,500,000</u> USD			

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² Refer to the Policy of Co-financing of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	☐ Grant <u>Enter number</u> ☐ Loan ☐ Guarantee the ☒ Equity <u>USD15,500,000</u> ☐ Results-based payment <u>Enter number</u>		
A.11. Implementation period	<u>15 years</u>	A.12. Total lifespan	<u>25 years</u>
A.13. Expected date of AE internal approval	<u>Click or tap to enter a date.</u>	A.14. ESS category	<u>I-2</u>
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g., GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☒ No ☐		
A.20. Executing Entity information	The project will be implemented through four executing entities, Pegasus Capital Advisors (PCA), the Blue Green Bank, GCF BGB Holdings, L.P. and Pegasus BGB Holdings GP, LLC. Specifically, Pegasus Capital Advisors will hold responsibility for Output 1.1, which will include the establishment of the BGB, alongside the finalisation of the shareholder agreements, governance structures and operational processes, as well as gaining regulatory approval. Once fully established and onboarded, the senior management team of the BGB will take over the EE responsibilities for Output 1.2 and 2.1. These responsibilities will include crowding in public and private investment, as well as identifying, structuring and financing eligible climate investments. The BGB's management team will also be responsible for managing, monitoring and reporting on all investments made through the bank. BGB will be registered as a company with the Corporate Affairs and Intellectual Property Office (CAIPO) in Barbados and will receive an operational license from the appropriate financial regulator in Barbados. Two additional entities – to be established – will also be engaged in project execution: an SPV, tentatively named GCF BGB Holdings, L.P., will be formed as a Delaware limited partnership to serve as the holding company for GCF's equity interest in the BGB. Its general partner will be a Delaware LLC, tentatively named Pegasus BGB Holdings GP, LLC, which will be wholly owned by PCA.		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			
The people of Barbados are being severely impacted by climate change, constraining the sustainable development ambitions of the country and threatening the lives and livelihoods of vulnerable communities. These climate impacts are driven largely by the increased frequency and intensity of tropical storms and hurricanes — along with the associated impacts of sea level rise and storm surge — as well as decreasing rainfall with longer dry spells, lower soil moisture and increased aridity. Urgent adaptation action is, therefore, needed to reduce the wide-reaching impacts on key economic sectors and secure a sustainable future for all Bajans. In addition to these core adaptation needs, the country recognises the vital importance of global mitigation efforts to reduce the threat of future climate change exacerbating these impacts. To this end, despite contributing significantly less than 0.01% of global GHG emissions, Barbados' resilience strategy includes highly ambitious mitigation targets, including a transition to 100% renewable energy by 2030. These ambitious targets are intended to demonstrate the country's commitment to climate action, as well as to serve as an example to the rest of the world of the feasibility of the transition when innovative approaches are adopted. Unfortunately, despite these commitments, progress has been slow — especially in adopting renewable energies and water conservation initiatives by middle- to low-income households. This relates primarily to the structure of ownership of capital in the Barbados economy and the financial ecosystem (see <i>Barriers</i> below for more detail).			

Given the existing fiscal constraints, Barbados' resilience strategy — encompassed in the Roofs to Reefs Programme (R2RP) — recognises the need to partner with the private sector to mobilise finance for innovative climate investments that are, or can be made, bankable. Such an approach will enable the government to direct the limited concessional finance available towards public-good resilience measures that do not have direct financial returns.

There is a discernible gap in the market in Barbados and the Caribbean for a financial institution that can provide opportunities for existing financial institutions and capital markets to mobilise funds, whether through conventional bank lending or alternative securitisation of revenue streams deriving from climate change mitigation and adaptation activities, thereby creating stronger and more diversified capital markets.

To address the urgent need for innovative finance in Barbados, the proposed project will establish a sustainable climate finance vehicle — the Blue Green Bank (BGB) — to stimulate innovation in the financial sector with the goal of mobilizing the securities and loan markets in order to unlock additional funding for climate-responsive initiatives. The BGB will be a joint public-private sector effort to create an improved financing infrastructure that works closely with other financing players — such as commercial banks, credit unions, pension funds, and insurance companies — to crowd in private capital alongside public capital, rather than crowding out the significant investment opportunities offered by these key financial institutions. For such institutions to become involved in climate change mitigation and adaptation activities, the leadership should be provided by a third party putting the deal together, for which it must establish the appropriate capacities in technical and HR terms, as well as appropriate management and institutional arrangements.

The BGB is designed to become a significant institution for practical and strategic financing of climate change adaptation and mitigation policies and measures, supporting the Government of Barbados to achieve national resilience and emissions reduction targets. As a partnership which brings together public (Government of Barbados), private and multilateral institutions at the level of equity, the BGB can present a holistic view of the market and the potential for green financing by ensuring that new projects align with Barbados' Nationally Determined Contributions (NDC) targets. Because the BGB's goal is national climate adaptation and mitigation, it will seek not to replace other financial institutions but to act as a partner and support system for them in this underdeveloped market segment. To this end, the bank will be wholesale in operation, providing support in the form of funds, guarantees or other mechanisms for eligible projects directly as well as through intermediaries such as existing lenders, whether banks, credit unions, insurance companies etc to facilitate the funding of blue/green climate-responsive programs contemplated by government climate policy. The BGB will also serve as a conduit between local financial institutions and the Government of Barbados, allowing the BGB to identify market opportunities, and provide feedback and data to inform policy decisions more effectively.

Once established, the BGB will engage with investors, including private investment firms, development banks, philanthropic organisations and other sources of finance to continue to capitalise on the bank over the long term. The bank will maintain a debt-finance to equity ratio of 7:1, aiming to mobilise at least USD 5.8 billion of investment in climate projects over the next 15 years³.

Overall, the BGB will promote a paradigm shift towards investment in resilient livelihoods and low-carbon technologies, with emphasis on low-income, vulnerable households and the energy, housing, agriculture and transport sectors. The project will also build the wider community's awareness of climate investment opportunities in support of a more resilient and sustainable Caribbean. Finally, by focusing on climate resilience and adaptation for middle to low-income households — including a specific focus on women and other highly vulnerable groups — the BGB would aim to attract more concessional or philanthropic providers of capital, expanding the goals of climate action in Barbados without the need for further injections of money by the Government and GCF. This will create long-term sustainability and scalability in the banking sector, with the potential to scale beyond Barbados into the broader Caribbean region over time.

³ With potential to be double that if the additional targeted equity is successfully raised

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Climate Change Problem

Barbados — a small island developing state (SIDS) located in the Atlantic Ocean outside the southern reaches of the Lesser Antilles in the Caribbean (Figure 1) — is highly vulnerable to the impacts of climate change, which is constraining the sustainable development ambitions of the country and threatening the lives and livelihoods of vulnerable communities. Full details of the observed and projected future climate change hazards and impacts for Barbados can be found in Annex 2.1. These climate impacts are driven largely by the increased frequency and intensity of tropical storms and hurricanes — along with the associated impacts of sea level rise and storm surge — as well as decreasing rainfall with longer dry spells, lower soil moisture and increased aridity (Figure 2). It is globally recognised that such rapid changes observed in the world's climate system are overwhelmingly due to human influence through greenhouse gas (GHG) emissions⁴. In response to the impending risks posed by climate change — not to mention the impacts already being felt — the Government of Barbados (GoB) recognises the vital importance of global mitigation efforts to reduce the threat of future climate change exacerbating these impacts. To this end, despite contributing significantly less than 0.01% of global GHG emissions, Barbados' Nationally Determined Contribution (NDC) includes highly ambitious mitigation targets, including a transition to 100% renewable energy by 2030. These ambitious targets are intended to demonstrate the country's commitment to climate action, as well as to serve as an example to the rest of the world of the feasibility of the transition when innovative approaches are adopted. At the same time, the country recognises the need to urgently adapt to the inevitable impacts of those changes which have already occurred, along with those anticipated in the future. Figure 2 below provides an overview of the climate change impacts being experienced in Barbados, with further details on the science basis for these climate change impacts presented in Annex 2.1.



Figure 1. Map of Barbados and its relative position in the Caribbean.

⁴ IPCC 2022. The Physical Science Basis

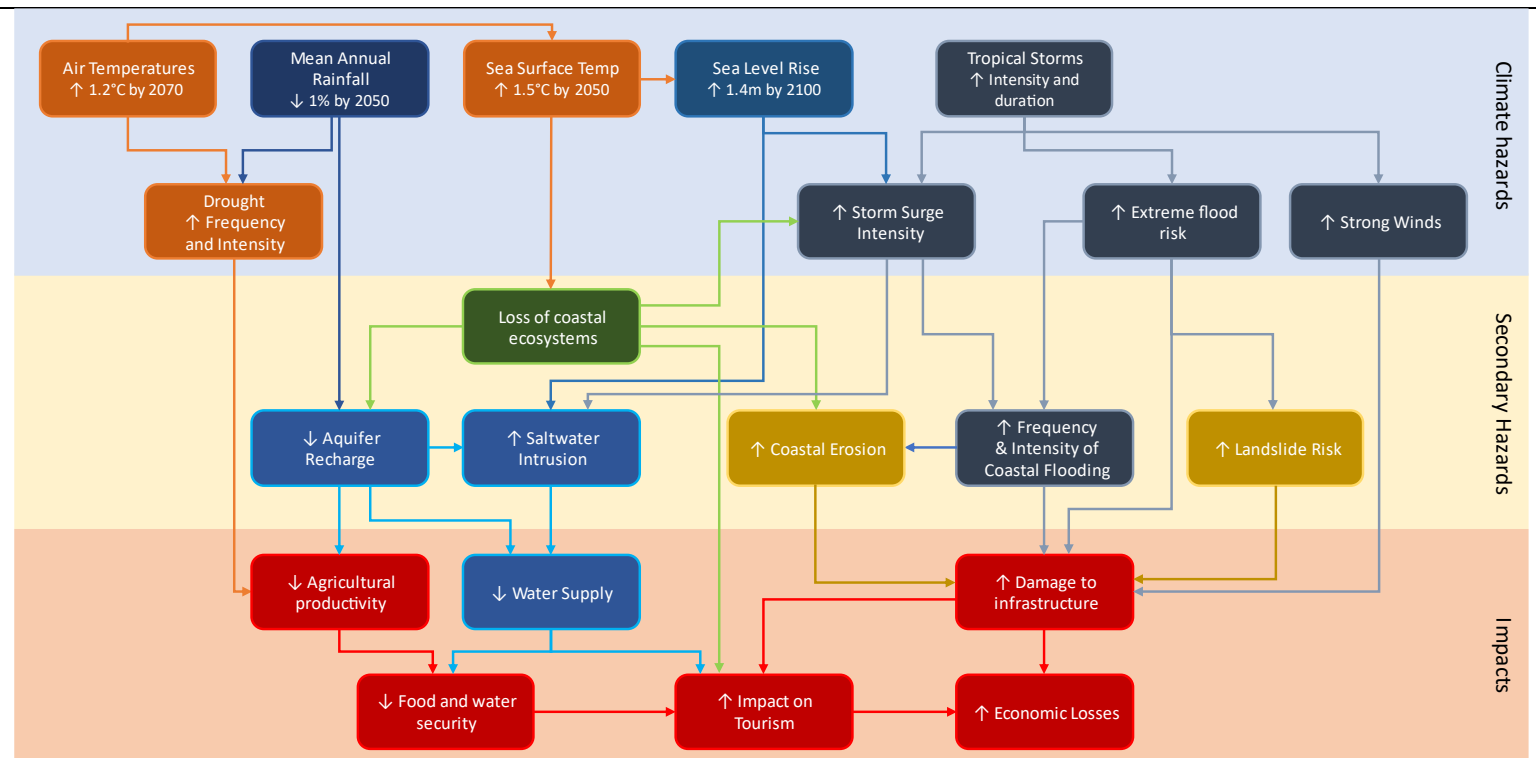


Figure 2. Climate change impact pathways in Barbados under the SSP2-4.5 scenario, as identified in the Barbados Climate Country Report (Annex 2.1).

Country context

Physical environment context

Barbados is a SIDS with a land area of only 432 km² and a population of 281 200 (2021) at a density of 661 persons per km² — making it one of the world's most densely populated islands. Most of the population and infrastructure are located along the coastal zone, which given the low-lying topography of the island makes them highly vulnerable to coastal hazards such as sea level rise and storm surges.

Natural Environment Context

Barbados has rich marine biodiversity; although terrestrial diversity is limited as an artefact of the islands' remote location outside of prevailing currents and winds from the nearest landmass, exacerbated by clear-cutting associated with a colonial past and plantation agriculture. Several plant and animal species are endangered due to existing threats and climate change will exacerbate conditions, especially for coral reefs. The National Biodiversity Strategy and Action Plan (NBSAP) 2020 for Barbados⁵ provides a full account of the flora and fauna of the country, including drivers of biodiversity loss which include habitat fragmentation, invasive species, pollution and over-harvesting of species.

Barbados is home to several types of marine/coastal ecosystems which contribute greatly to environmental health and tourism on the Island. Marine biodiversity is particularly under threat with all three species of marine turtles, two of which are nesting in Barbados, listed as endangered or critically endangered by the International Union for Conservation of Nature (IUCN).

The country has four species of seagrasses which have been impacted by coastal and land-based sources of pollution. There are 10 species of soft coral and 31 species of hard coral, all of which are greatly under threat by climate change. Coral reefs support more than 25% of all marine species, making them the most biologically diverse of marine ecosystems and serve as natural breakwaters along the coasts. Furthermore, the coral reefs generate significant revenues for local fishermen and the tourism sector through snorkelling and scuba diving.

In addition to biodiversity, the NBSAP identifies several critical ecosystem services which may be impacted or exacerbated by climate change. Of provisioning services, fisheries which are critical to livelihoods, tourism and food security are threatened by overfishing, pollution and climate change. The tourism sector in Barbados employs about 26,000 persons (9% of the total population) and indirectly contributes to about 40% of Barbados' earnings, making

⁵ [National Biodiversity Strategy and Action Plan \(NBSAP\) 2020 for Barbados](#)

climate change a great threat to the island's main economic earner which derives its value from the natural economy. Supporting services such as erosion control are also critical for adapting to climate change however habitat loss and conversion can reduce the effectiveness of these natural solutions for climate change adaptation.

Water is another critical provisioning service due to the reliance on groundwater as the only source of potable water on the island. Fresh, potable water resources are scarce in Barbados. Barbados is considered a highly water-stressed country as 88% of the renewable water resources are being withdrawn (SDG indicator 6.4.2)⁶. Barbados is among the most water-scarce countries in the world with only 305-310 m³ per person per year of water available.⁷ Surface water sources, including rivers, streams, dams and ponds, are few and the majority of water is abstracted from subsurface groundwater via wells and boreholes. These water resources exist deep below ground level. As a small coral limestone island, the water supply in Barbados is limited. With ~85% of the island covered by Pleistocene reef limestone, a succession of terraces, slopes and scarps create a landscape of steep gullies and ravines from the central highland to the coastline that strongly influence the island's drainage system and result in limited surface water. Consequently, 98.6% of the island's potable water comes from groundwater abstractions. Natural reservoirs within the unconfined aquifer – which serve as collection points for groundwater – are supplied with fresh water by underground streams which are in turn recharged from the surface through sinkholes and fissures. Water availability in Barbados is affected by the periods and intensities of the wet and dry seasons and by extreme weather systems such that even short dry spells of only 7-10 days affect water availability. As the effects of climate change worsen and the sea level rises, these underground water sources are increasingly exposed to salt infiltration threatening Barbados' water supply.^[OBJ]

The Barbados Water Authority (BWA) is responsible for supplying fresh water to the island and providing services for wastewater treatment and disposal for sewered regions⁸. Water resource volumes, demand and withdrawal data were retrieved from the UN FAO AQUASTAT Country Profile Report for Barbados⁹ as shown in Section 3.1.2 of Annex 2: Feasibility Study.

Socio-Economic Context

Barbados has transformed itself from a low-income economy reliant almost solely on sugar production to having one of the highest per capita incomes in the Caribbean at USD 17 225, driving the economy to upper-middle-income status. Although Barbados' economy has shown fairly consistent growth over the last half-century — driven largely by a shift from a reliance on primary production focussed on sugar cane to tertiary industries, such as tourism, real estate and the finance sector — the country's GDP growth rate has been negative since 2019 with its most recent annual growth rate being -0.2 in 2021 (World Bank, 2023). The most significant recent contributors to GDP include the financial services sector, government, distribution and tourism as shown in Figure 3¹⁰, whilst it is expected there will be additional contributions to economic growth from the construction and renewable energy sectors in coming years.

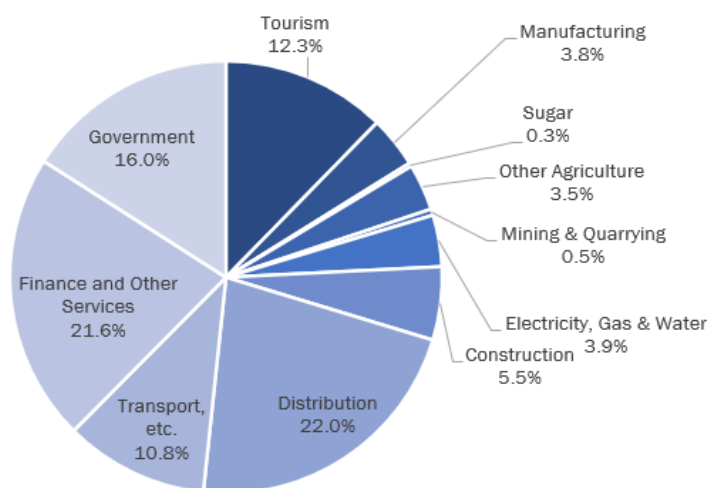


Figure 3. GDP contribution by sector in Barbados

Despite the relatively high GDP per capita, the Barbados economy exhibits characteristics commensurate with many SIDS. These characteristics include a lack of economic diversification, vulnerability to natural disasters and climatic events, limited natural resources and a high reliance on imports and exports. Given its dependence on external markets and economies, it is highly susceptible to market fluctuations, limiting its potential for economic growth and international competition.

⁶ [UN Water SDG 6 Snapshot in Barbados](#)

⁷ Barbados Second National Communication (2018)

⁸ Barbados Water Authority, 2023

⁹ UN FAO, 2015

¹⁰ Central Bank of Barbados, 2017

In addition to these external influences, the legacy of colonialism in Barbados has had an impact on the country's social and economic structures — resulting in persistent disparities in wealth and income. The country's post-colonial society has seen the development of an economic system that is heavily skewed towards a small group of individuals (primarily the descendants of the plantocracy) and companies, with a significant percentage of the population falling behind in terms of economic development. This legacy has led to a situation in which a small portion of the population controls the majority of the wealth. This is exemplified when considering Barbados' Human Development Index (HDI)¹¹ — which is relatively high at 0.790¹², but drops 16.8% when adjusted for inequality.

Based on the 2010 Population and Housing Census, almost 80% of houses have corrugated metal sheets for roofing and over 90% of homes have piped water directly into the dwelling. That Census also found that almost 75% of the dwellings are owner-occupied, with the remaining houses privately rented or subsidized government-rented housing units.

Mitigation Needs

Barbados' greenhouse gas (GHG) emissions are negligible on the global scale, totalling only 1,558 GgCO_{2e} in 2018 — significantly less than 0.01% of global emissions. The majority of Barbados' GHG emissions come from the energy sector, which was responsible for ~54% of GHG emissions in 2020, followed by the waste sector with a 20% contribution. The domestic transport sector accounts for 15% of the total emissions, while the industrial processes sector contributed 10% and the Land-use, Land-use Change, and Forestry (LULUCF) sector contributed less than 1% (Figure 4)¹³. Barbados' GHG emissions decreased by 26% between 2008 to 2020, this is primarily due to the decrease in the domestic transport sector emissions. Further details on the country's GHG emissions profile are presented in Annex 2.1: Barbados Country Climate Change Analysis Report.

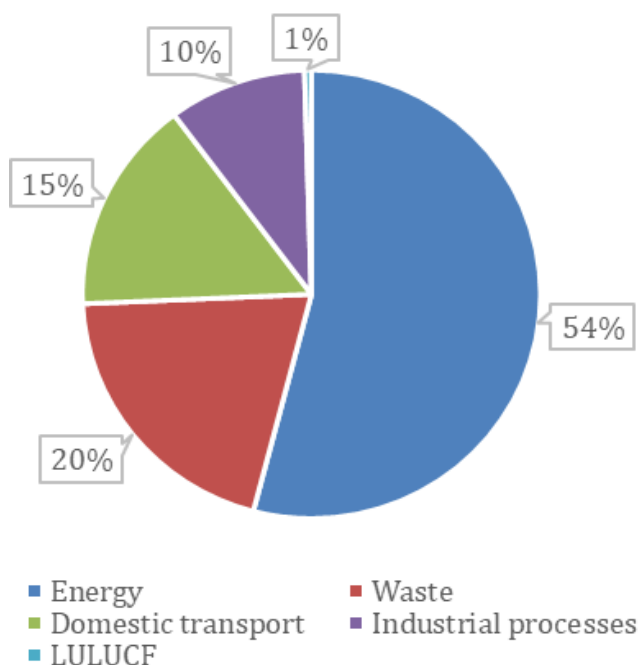


Figure 4: Barbados' Greenhouse Gas Emissions/Removals by Sector in 2020

Despite the country's relatively low contribution to global GHG emissions, Barbados continues to pursue an ambitious mitigation pathway. Barbados' first intended Nationally Determined Contribution (NDC) was submitted in September 2015 and updated in 2021 to better reflect Barbados' commitment to the 1.5 degrees objective of the Paris Agreement. With this NDC update, Barbados is accelerating its decoupling of economic development from emissions growth, providing targets that are significantly more ambitious than the initial targets. This includes the unconditional commitment to a fossil fuel-free electricity sector by 2030, combined with the actions to significantly enhance the resilience of its people and economy — as reflected in the Barbados National Energy Policy (BNEP).

To meet the ambitions of Barbados' NDC and the BNEP, significant additional investments are urgently needed for the energy sector. To this end, an Integrated Resource and Resilience Plan (IRRP) was finalised in 2022 to guide the implementation of the BNEP in the electricity sector. The IRRP provides a roadmap and action plan for the energy transition, which includes the need for utility-scale solar (180MW by 2030), along with rooftop solar (100MW by 2030) and solar thermal energy (60MW by 2030). The plan also identifies a number of pilot projects as a key area for budget allocation, including lithium-ion batteries, solar PV, vehicle-to-grid (V2G) technology-embedded electric vehicle chargers, and high-power DC vehicle chargers.

¹¹ UNDP Human Development reports – <https://hdr.undp.org/data-center/specific-country-data/#/countries/BRB>

¹² Ranking 70th globally, 9th in the Latin America and Caribbean (LAC) region

¹³ Government of Barbados. (2020). Barbados GHG profile-NDC.

Adaptation Needs

Given the multitude of potential climate change hazards combined with the vulnerabilities and exposure that the country faces due to the socio-economic and natural environment context as a SIDS, the magnitude of climate change impacts in Barbados is clear and urgent support is required to help vulnerable communities adapt. The government has estimated that Barbados needs ~\$2 billion (40% of GDP) to achieve climate resilience.

Although there are multiple climate change hazards at play in Barbados, as outlined above, this project focuses on two specific hazard and risk areas – hurricanes/storms and increased water stress – as these are both critical areas, with the potential for high impact from interventions and identified as priorities by the government as part of the Roofs to Reefs Programme (R2RP, outlined below). The adaptation needs identified recognize the importance of interventions that build synergies between adaptation, mitigation and development. This approach reflects the recent IPCC AR6 reports focus on Climate Resilient Development and the need to build not only adaptive and absorptive capacity, but also transformational capacity as outlined in the NDC of Barbados.

Targeted adaptation needs (Table 1) have been identified for Barbados based on the risks determined as a function of climate change hazards and vulnerability and exposure as a function of socio-economic and natural environment context (Figure 5). This considers the historical conditions as well as projected changes to the climate and society, including population and economic growth and ensures that interventions respond to the two key risks – water stress and hurricanes/storms — in ways that are compatible with the development vision and priority of the country and benefit vulnerable low- and middle-income households. The interventions identified, focused on hurricane-resilient housing and water conservation, target not only adaptation needs but also several Sustainable Development Goals (SDGs; see Section D.4 below) and simultaneously address critical economic stresses that can occur as a result of hurricanes which incur major economic losses¹⁴. Moreover, investments in hurricane-resilient roofs provide a foundation for rooftop PV — a critical component for a resilient energy transition.

¹⁴ Recent hurricanes that affected neighbouring countries such as Dominica, Barbuda, St. Maarten, British Virgin Islands and The Bahamas amounted to loss and damage of over 200% of GDP for those countries.

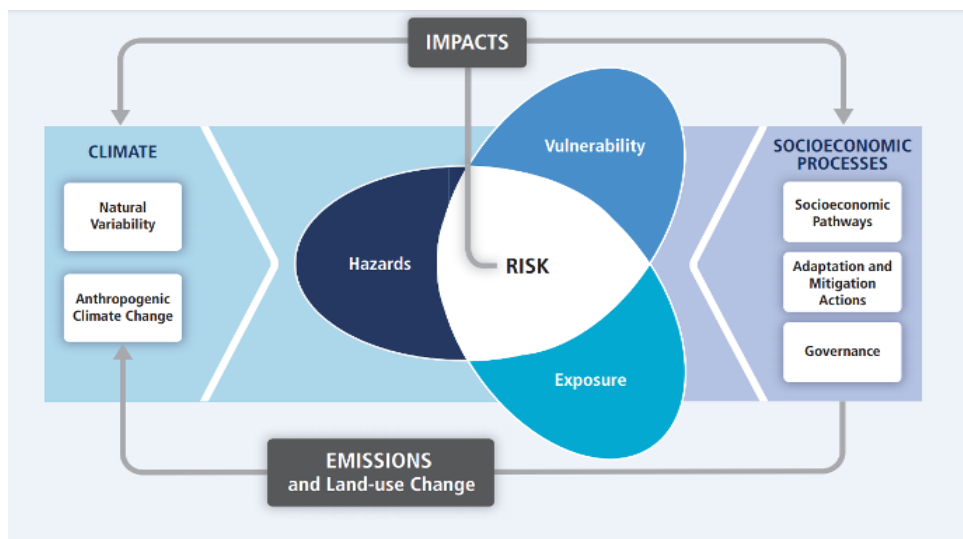


Figure 5. Climate risk nexus (IPCC)

Table 1: Priority risks and adaptation needs in Barbados

Risk	Climate change hazards ¹⁵	Vulnerability and Exposure (Socio-economic and Environmental)	Impacts	Adaptation Needs
Increased Water Stress	Changes in precipitation (shorter wet season and prolonged dry season) and more frequent drought	- Population reliant on scarce groundwater reserves. - Available water resources are ~305-310m³ per person per year. - Groundwater contamination from pollution. - Lack of water saving/conservation at household and small business levels. - Upfront financial burden of water saving/conservation for households and small businesses.	- Aquifer recharge is directly affected by cycles and intensities of the wet and dry seasons and by increases in the frequency and intensity of extreme weather systems such as tropical storms and hurricanes in the region. - A projected increase in intense rainfall events will result in increased surface runoff and reduced aquifer recharge, with much water being lost to the sea.	- Reducing water demand through improved household-level conservation measures and in-situ rainwater storage to minimize water stress - Reducing run-off and groundwater contamination from pollution
	Increasing air temperatures at a rate of 0.24°C/decade	Water demand for major economic sectors – agriculture and tourism.	- Increased water demand for domestic and industrial use. - Increased evapotranspiration, leading to increased water demand for natural environments and agriculture.	- Alternative irrigation water supplies using recycled water and harvested stormwater. - Increased water augmentation and recharge.

¹⁵ Details on the extent and the science basis of these hazards are presented in Annex 2.1: Barbados Climate Country Report

	Sea Level Rise averaging 1.8 mm per year, rising by 0.5 to 1.4 meters by 2100		Rising sea levels and decreasing aquifer recharge are resulting in saltwater intrusion into groundwater aquifers, threatening critical freshwater reserves.	Increasing water collection and recycling from non-groundwater sources to improve water supply.
Increased damage from tropical storms and hurricanes	Increased intensity and frequency of hurricanes	- Many buildings, infrastructure and utilities are not built to withstand hurricane-strength winds. Those that are hurricane resilient are generally only rated to Category 3. - Majority of roofs in residential dwellings are made from corrugated metal sheets and lack hurricane strapping. - Most buildings and infrastructure are in highly exposed low-lying coastal zones.	- Floods and strong winds resulting in damage to buildings, and loss of roofs. - Projected future increase in exposure and risk. - Strong winds and floods associated with tropical storms and hurricanes damage utility infrastructure, impacting centralised energy supplies, as well as water, gas and telecommunications infrastructure.	Increasing the climate resiliency of the housing stock, infrastructure and utilities.
	Increased sea surface temperatures and sea level rise	- Lower- to middle-income groups lack access to affordable finance to purchase hurricane-resilient homes or to retrofit existing structures. - Higher levels of vulnerability of low-income households. - Unsustainable land use practices result in runoff during extreme rainfall events, polluting and degrading marine ecosystems	- Negative impact on health from being displaced by storms. - Negative impact on household finances if rebuilding post-hurricane is needed. - Damage to marine ecosystems from strong winds and storm surge, as well as runoff pollution, reducing ecosystem service provision, thereby increasing coastal community exposure to storm surge and reducing fish breeding grounds.	- Increased financial assistance for vulnerable households to minimize the financial impact of hurricanes - Increasing protections of marine ecosystems to build their climate resilience

Barbados' Resilience Strategy: The "Roofs to Reefs Programme" (R2RP)

To address the country's urgent adaptation needs, the Government of Barbados launched the Roofs to Reefs Programme (R2RP) — an ambitious and comprehensive resilience agenda that aims to protect private and community roofs of Barbados from extreme weather events, while encouraging the use of renewable energy sources, rainwater harvesting and nature-based solutions to build climate resilience. The R2RP responds at the individual, community and country levels to improve living conditions, social and environmental conditions and improving post-disaster recovery and response. Responding directly to the Barbados Physical Development Plan (PDP), the R2RP aims to create a roadmap for the sustainable growth and development of Barbados, while facilitating and guiding public and private investment. It is also included in the latest NDCs as well as the Marine Spatial Plan for Barbados, demonstrating the integrated nature of the project across sectors and considering its systemic impact on both people and nature. While the R2RP provides the overarching framework that allows the integrated approach to addressing climate change impacts, the structure of the programme allows for projects to be prepared at the sector level. The Programme works to identify critical projects and programs that address the country's climate change mitigation and adaptation needs, along with their associated costs, and to find and take advantage of funding opportunities to bring them to fruition. The R2RP also provides the vehicle through which public investment will be directed.

The main objectives of the R2RP are:

- to make low- and middle-income homes more resilient to extreme weather events and their impacts, such as possible loss of access to electricity and potable water distribution systems;
- to increase freshwater storage capacity and water use efficiency and reduce emissions through the deployment of distributed renewable energy generation;
- to decrease land-based sources of marine pollution through more sustainable land use practices;
- to make critical utility, water and sanitation and road infrastructure climate resilient; and,

- to restore vulnerable coral reef ecosystems, particularly on the west and south coasts of the island.

These objectives have been categorised into six target sectors, namely shelter (for example, housing upgrades); water (including rainwater harvesting, stormwater capture, groundwater, potable water, and wastewater); energy; waste; land use (including use of the EEZ and terrestrial agriculture); and ecosystems (including gully and wetland ecosystems, terrestrial biodiversity, coral reefs and other marine ecosystems and biodiversity). These sectors were chosen because they are interlinked and fundamental to human well-being. The shelter sector also reinforces bringing the programme to the level of the individual household. Table 1 on the priority risks and adaptation needs demonstrates how for two hazards, there are vulnerability, exposure, impact and adaptation needs related to shelter, water, waste, land use and ecosystems for increased water stress and all six sectors are affected by increased damage from tropical storms and hurricanes. Further details of the R2RP and the pipeline of projects identified to date are presented under Section 3.5.1 of Annex 2: Feasibility Study.

Complementarity with ongoing and planned initiatives

As described above, Barbados has developed detailed and ambitious climate change adaption and mitigation strategies. To achieve these strategies, the GoB has paid close attention to the available options, including public, private and donor finance, to create a roadmap towards climate resilience. This roadmap is encapsulated in the R2RP. The proposed programme forms a key component of the GoB's overarching strategy, specifically designed to fill a finance niche for potentially bankable initiatives that will free up public fiscal space to invest in climate action where the potential for financial returns is constrained by the public good nature of the benefits. By designing the programme to fill this niche, it will be inherently complementary to other ongoing initiatives in the country, including those funded by the GCF. A summary of the country's GCF pipeline and the complementarity of the proposed programme is provided in Table 2 below.

Table 2. Alignment with ongoing and planned GCF initiatives in Barbados.

Project	Objectives	Key Considerations for Complementarity
Three Rs (Reduce, Reuse and Recycle) for Climate-Resilient Wastewater Systems in Barbados (3R-CReWS) AE: CCCCC Approved 2022	• Promote the use of carbon-neutral and climate-resilient water and energy management technologies and strategies that ensure water is protected, managed, recycled, reused, and conserved. This will include an upgrade of the Bridgetown Sewage Treatment Plant (BSTP) to a tertiary treatment system and the installation of wastewater treatment systems in two Zone A locations in Barbados to harness reclaimed water, which will be utilized for aquifer recharge and agricultural irrigation. • Incentive programme to encourage water conservation, recycling and re-use in the private sector, supported by the expansion of the Revolving Adaptation Fund Facility (RAFF).	Wastewater recycling under the HOPE initiative will complement the utility-scale processes promoted under Three Rs – contributing to the overarching objective of creating a circular economy for wastewater. Moreover, the innovative water conservation initiatives identified as an initial pipeline activity for the BGB will be directly complemented by the efforts under Three-Rs to strengthen the enabling environment for private sector adoption of wastewater technologies and reuse options. To this end, the BGB will leverage the nationwide public education and awareness campaign on the nexus between climate change and water availability to mobilise private finance in the sector.
E-Mobility Programme for Sustainable Cities in Latin America and the Caribbean (Regional) AE: IDB Approved 2022	The E-Mobility Programme links EV deployment with sustainable low-carbon urban development and increased climate-resilience of urban transport infrastructure and the grid. Programme components include climate-resilient transport infrastructure; increased climate-resilience of the grid with green hydrogen and vehicle-to-grid (V2G) pilot projects; and electrified integrated urban mobility.	The BGB has the potential to introduce innovative financial products to support the roll-out of e-mobility programmes in Barbados. In particular, V2G and associated EV charging infrastructure has been identified as a potential area where BGB involvement could mobilise private sector investment.

Water Sector Resilience
Nexus for Sustainability in
Barbados (WSRN S-
Barbados)

AE: CCCCC

Approved 2018

The project aims to: i) build greater resilience to extreme storm events and drought conditions by utilizing cleaner energy sources, decentralising water storage, promoting rainwater harvesting at the household and community level, and improving the efficiency with which rainwater runoff replenish aquifers in Barbados; ii) further advance adaptation and mitigation initiatives in the water sector of Barbados by redirecting and mobilising local funds through a revolving adaptation fund; iii) reduce the greenhouse gas emissions intensity of water provision; iv) capacity building via knowledge sharing and lessons learnt platforms; v) support the review and development of a legislative framework to supports climate-smart development and water sector resilience; vi) collate and disseminate lessons learnt for use in developing further adaptation and mitigation initiatives and raising public awareness about climate change and water conservation.

As with the e-mobility programme above, the BGB has the potential to introduce innovative financial products to support the roll-out of water conservation initiatives introduced through the WSRN S-Barbados project – including by promoting private investment in the nexus of water and energy. The BGB will also draw on the lessons learned to identify scalable options that support water conservation efforts for lower- to middle-income groups.

Operation Coastal
Resilience (OCR):
Innovative financing for
nature-based solutions that
support climate-resilient
coastal areas in the
Caribbean.

Regional

AE: TNC

CN Under Development

OCR is a regional programme aiming to establish innovative financial mechanisms (IFMs) for nature-based solutions in the coastal zones of four Caribbean countries. The programme will use a combination of regional and national funds to channel finance to projects that use nature-based solutions to generate public goods by leveraging innovative finance options.

The BGB is targeting a separate niche in the financial space for climate action in Barbados from that targeted by OCR. Specifically, it is envisioned that OCR will act more in the public space, identifying options for funding nature-based solutions to coastal resilience that do not yield direct financial returns, instead using financial mechanisms such as payments for ecosystem services and philanthropic/donor funding to finance public-good initiatives. By contrast, the BGB will strengthen the capital market and package initiatives to improve bankability, with the aim of mobilising private finance for climate action targeting lower- to middle-income groups.

R2RP Readiness

Ministry of the Environment
and National Beautification

Approved Readiness

As a response to these climate change challenges, the GOB launched the Roof to Reef Programme (R2RP). The R2RP provides the overarching framework that allows the integrated approach to addressing the negative impacts of climate change. The R2RP is the Government's sustainable development model for the next decade and represents the country programme for Barbados. The primary focus is on improving the social and environmental circumstances of the people in Barbados. The R2RP will enhance the country's ability to recover from climatic events. The R2RP is hinged on six thematic areas: Shelter, Water, Energy, Waste, Land Use, and Ecosystems Management. The thematic areas under the R2RP are aligned with Barbados' NDC priority areas. Barbados has developed and is seeking to develop a pipeline of projects under the R2RP that will improve the country's resilience to

The BGB has been designed to align closely with the R2RP, intended to fill a niche gap in the financing of initiatives under the R2RP. For example, the initial pipeline investments through the HOPE Programme are directly listed as a priority intervention under the R2RP. To this end, the BGB will engage with the R2RP implementation teams to draw on the readiness work being conducted to identify investment opportunities and to package projects to create bankable opportunities.

climate change as well as increase its push to become carbon neutral by 2030.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Problem Statement

The people of Barbados are being severely impacted by climate change, constraining the sustainable development ambitions of the country and threatening the lives and livelihoods of vulnerable communities. These climate impacts are driven largely by the increased frequency and intensity of tropical storms and hurricanes — along with the associated impacts of sea level rise and storm surge — as well as decreasing rainfall with longer dry spells, lower soil moisture and increased aridity. Urgent adaptation action is, therefore, needed to reduce the wide-reaching impacts on key economic sectors and secure a sustainable future for all Bajans. In addition to these core adaptation needs, the country recognises the vital importance of global mitigation efforts to reduce the threat of future climate change exacerbating these impacts. To this end, despite contributing significantly less than 0.01% of global GHG emissions, Barbados' resilience strategy includes highly ambitious mitigation targets, including a transition to 100% renewable energy by 2030. These ambitious targets are intended to demonstrate the country's commitment to climate action, as well as to serve as an example to the rest of the world of the feasibility of the transition when innovative approaches are adopted.

Unfortunately, despite these commitments, progress has been slow — especially in adopting renewable energies and water conservation initiatives by middle- to low-income households. This relates primarily to the structure of ownership of capital in the Barbados economy and the financial ecosystem (see *Barriers* below for more detail). Given the existing fiscal constraints, Barbados' resilience strategy — encompassed in the Roofs to Reefs Programme (R2RP) — recognises the need to partner with the private sector to mobilise finance for innovative climate investments that are, or can be made, bankable. Such an approach will enable the government to direct the limited concessional finance available towards public-good resilience measures that do not have direct financial returns.

Goal Statement

If innovative and affordable climate finance is mobilised in Barbados through the establishment of the Blue Green Bank, **then** the transition to a fossil fuel-free energy supply, as well as hurricane-resilient infrastructure and a secure water supply will be accelerated **because** climate-responsive investments from the private sector will be securitised, de-risked and packaged, increasing private capital and opening the climate finance market to lower-income groups and small businesses that are otherwise unable to access loan finance from commercial banks.

Barriers

Several barriers currently constrain the financing of potentially bankable climate investments in Barbados – particularly when considering the potential involvement of low- to middle-income households. These barriers are detailed below.

Barrier 1: The government of Barbados is constrained in its capacity to take on additional debt to finance climate action.

It is estimated that Barbados needs to invest approximately USD 2 billion in climate-responsive initiatives to achieve resilience. Given the size of investment required relative to the country's GDP (~40% of GDP), Barbados currently does not have the financial resources to meet its adaptation needs and mitigation ambitions. This financial shortfall is exacerbated by Barbados being classified as a highly indebted country, constraining the government's ability to take on additional debt to meet climate investment needs. The total government debt reached ~144% of GDP at the end of 2020, with the most significant contributor to increased debt levels recently having been the loss and damage arising from climate-related and other natural disasters alongside the necessary expense to protect the land and marine environment. Given the growing cost of loss and damage from natural disasters, it is recognised that a failure to invest in resilience will worsen the country's debt position.

To help with managing this debt, Barbados has entered into an agreement under the Extended Fund Facility of the International Monetary Fund, the commitments of which limit the ability of the country to take on additional debt. Moreover, because its GDP per capita is above the eligibility threshold, Barbados does not qualify for concessionary loans from the World Bank and other Multilateral Development Banks (MDBs).

As a result of these constraints, the GoB is unable to access adequate, affordable debt to finance climate action. Those limited funds that are available are needed for investments in public-good nature interventions, highlighting the need for private investment to be mobilized to bridge the finance gap and meet the adaptation needs and mitigation ambitions of Barbados.

Barrier 2: Limited capital markets for climate investments in Barbados.

While there has been an increase in the issuance of green bonds in Barbados in recent years¹⁶, the capital market for climate investments remains limited. These limitations on the capital market are largely driven by the small number of participants in the market, relying largely on a relatively small banking sector (see Barrier 3 below), that exhibits considerable market overlap and limited portfolio diversification. As a result, market innovation is largely stifled, limiting private investment in climate action. As noted above, such private investment is essential to bridge the finance gap to meet the country's adaptation needs and mitigation ambitions.

Barrier 3: Commercial banks in Barbados are typically risk-averse and focus on collateralised loans.

The banking sector in Barbados is limited, with only six banks¹⁷ operating in the country, all of which are headquartered abroad — including several in Canada and one in Trinidad & Tobago — and take a "branch banking" approach. While the banks operate legally as local subsidiaries subject to local regulation, the banks are controlled and regulated from head offices based abroad, which influences their operations and appetite for risk in a significant way. This risk aversion is reinforced by an international bank de-risking process initially spurred by global anti-money laundering efforts. While this is a worldwide phenomenon, de-risking has hit the Caribbean region more heavily than any other, evidenced by the 25% loss of correspondent banking relationships. Perceptions of risk and the resulting enhanced due diligence have raised the cost of banking and reduced the attractiveness of small jurisdictions for international banks. Moreover, the increasing dependence on a narrowing set of correspondent banks for cross-border payments has led to further general risk aversion by the banks.

As a result of the factors above, commercial banks in Barbados typically focus on providing mainly collateralised loans such as mortgages and car loans to a narrow set of borrowers. These loans primarily target high-income households, which are seen as the primary source of near-term profits. The near-term focus of commercial banks in Barbados has been linked to the possibility of the sale of subsidiaries, with each of the Canadian-headquartered banks having sold sizeable parts of their Caribbean subsidiaries over the past decade. In addition to these factors, the high lending interest rates and relatively short tenors available significantly constrain the banking sector's ability to finance resilience and sustainable projects, especially those that target low-income households or small businesses¹⁸. This overall situation means that the local financial market cannot match the scale of funding required to deliver the green transformational impacts planned and required to build resilience to future climate scenarios and transition the energy sector. In particular, by targeting high-income households, the banking sector has largely excluded potential investment from the sizeable lower- to middle-income groups, restricting the adaptive capacity of much of the population.

Barrier 4: Lower and middle-income groups in Barbados have limited collateral and limited credit histories to access loan finance from commercial banks.

Following decades of free public education, health and other social investments, many Barbadians have the material comforts of a "middle-income" economy and relatively low poverty levels. However, like in other post-colonial societies, the ownership of capital remains highly skewed, with some in the lower- to middle-income groups not owning land or other major capital assets. This lack of capital ownership impedes access to finance, as they lack the collateral required by the risk-averse banking sector in Barbados (see Barrier 3). While this bank risk aversion can be offset by savings

¹⁶ For example, a green bond valued at BB\$ 14 M was issued by RF Merchant Bank & Trust Limited in 2021 for Williams Renewable Energy Limited (WREL), which formed part of a larger BB\$ 34 M programme.

¹⁷ Royal Bank of Canada; Scotiabank; CIBC FirstCaribbean; First Citizens; Republic Bank; and Sagicor Digital

¹⁸ Costs of finance and access to financial services are noted among the most significant constraints facing Barbadian firms in terms of their productivity and performance. — [IDB 2020](#). Barriers to Financial Access in Barbados — Key Challenges and Focal Areas for Reform.

and investment institutions — such as mutual funds and private equity firms — in the Caribbean and other small, developing states with skewed ownership of capital, the market is not there for this to happen on its own. A combination of weak credit profiles and a low savings ratio which characterizes lower- to middle-income groups in Barbados means that these groups have difficulty in securing finance for investments at a household level which would otherwise make economic sense. As a result, this important sector of the economy is largely excluded from the climate finance landscape, preventing investment at the scale needed to address climate change impacts in Barbados. This exclusion is particularly problematic given that these lower- to middle-income groups are also the most vulnerable to climate impacts and most in need of urgent action.

Barrier 5: Limited familiarity among financial institutions with opportunities provided through climate finance.

The packaging of climate investments with other complementary initiatives has been identified as an innovative solution for the financing of what may otherwise appear as un-bankable initiatives when viewed in isolation. For example, packaging solar PV with hurricane resilience and land ownership support programmes, as is being done with the Government's HOPE programme, can help securitise and de-risk loans for investment in green homes. However, commercial banks in Barbados have a limited understanding and technical capacity to identify opportunities for packaging initiatives. Furthermore, such packaging often requires specialised bridging finance to establish securities and initiate/unlock the bankability of such investments. The role of packing initiatives to increase bankability extends to the potential of leveraging economies of scale — for example through bulk procurement. This is particularly important for SIDSs, where emerging technologies (such as solar PV and battery storage) generally need to be imported, often at high costs. Bulk procurement of such technologies can significantly decrease costs and increase potential returns; however commercial banks lack the capacity to initiate and operate the schemes necessary to manage bulk procurement.

Market Opportunity

There is a discernible gap in the market in Barbados and the Caribbean for a financial institution that can provide opportunities for existing financial institutions and capital markets to mobilise funds, whether through conventional bank lending or alternative securitisation of revenue streams deriving from climate change mitigation and adaptation activities, thereby creating stronger and more diversified capital markets. Experience to date has demonstrated that the absence of such a market player has left banking liquidity underutilised. Such an institution could be a partnership between public and private stakeholders, ensuring that all parties value and recognise the financing potential of such investments driven by the local government policy via a regulatory regime for the promotion of climate resilience and transition to low-carbon futures. Such an institution should equally serve to focus efforts on low- to middle-income households in the interests of wider national economic and social development. By showing that climate investments are bankable and profitable, disseminating market knowledge, and encouraging best practices, a PPP financial institution can close the existing climate finance gap — not merely as a tool for deploying capital, but playing a crucial role in changing market behaviour, educating key stakeholders, and informing future policy goals.

Although international initiatives directed at the provision of finance for climate change mitigation and adaptation have been present for over twenty years, the national commercial banking market in small economies has, despite possessing adequate liquidity, often remained on the sidelines, with much of the running being made by multilateral institutions such as the IADB or the World Bank. Aside from a certain institutional prudence described earlier, the low participation of the Barbadian banking sector in activities of this kind can also be ascribed to the historical absence of a broad, coherent package of measures driven by the government to introduce the necessary reforms. Such programs as HOPE and the broader ambitions of R2RP mean that financing opportunities will grow both numerically in terms of individual operations as well as in size, given the more significant funding requirements relating to housing, flood control, desalination plants, Waste To Energy (WTE), or ecosystem protection. Not all of these projects will be inherently financially viable and appropriate project finance structuring will need to be undertaken. While some of the projects in the R2RP portfolio can generate income and therefore are viable in financing terms, others (for example, coastal management activities) will require more innovative solutions and forms of partnership with the private sector, as providers of services as well as inward investment.

Project Approach

The proposed project will establish a sustainable climate finance vehicle — the Blue Green Bank (BGB) — in Barbados to stimulate innovation in the financial sector with the goal of mobilizing the securities and loan markets in order to unlock additional funding for climate-responsive initiatives. The BGB will be a joint public-private sector effort to create an improved financing infrastructure that works closely with other financing players — such as commercial banks, credit unions, pension funds, and insurance companies — to crowd in private capital alongside public capital, rather than crowding out the significant investment opportunities offered by these key financial institutions. For such institutions to become involved in climate change mitigation and adaptation activities, the leadership should be provided by a third party putting the deal together, for which it must establish the appropriate capacities in technical and HR terms, as well as appropriate management and institutional arrangements.

The BGB is designed to become a significant institution for practical and strategic financing of climate change adaptation and mitigation policies and measures, supporting the Government of Barbados to achieve national resilience and emissions reduction targets. As a public-private partnership (PPP) that brings together public (Government of Barbados), private and multilateral institutions at the level of equity, the BGB can present a holistic view of the market and the potential for green financing by ensuring that new projects align with Barbados' NDC targets. Because the BGB's goal is national climate adaptation and mitigation, it will seek not to replace other financial institutions but to act as a partner and support system for them in this underdeveloped market segment. To this end, the bank will be wholesale in operation, providing support in the form of funds, guarantees or other mechanisms to existing lenders, whether banks, credit unions, insurance companies etc to facilitate the funding of blue/green programs contemplated by government policy. The BGB will also serve as a conduit between local financial institutions and the Government of Barbados, allowing the BGB to identify market opportunities and provide feedback and data to inform policy decisions more effectively. Details on the BGB's barrier removal strategy are presented in Table 3.

Table 3: Barrier removal strategy for the BGB.

Barrier	Barrier removal strategy
Barrier 1: The government of Barbados is constrained in its capacity to take on additional debt to finance climate action.	The establishment of the BGB (Output 1.1) will mobilise finance for bankable climate initiatives, thereby reducing pressure on the country's constrained fiscal space, and enabling the government to direct limited concessional finance towards non-bankable adaptation initiatives, for which the benefits are largely public goods.
Barrier 2: Limited capital markets for climate investments in Barbados.	The nature of the BGB will enable it to strengthen the capital market for climate investments. By playing a catalytic role in support of the financing of climate change mitigation and adaptation activities (Output 2.1), the BGB should find ways to mobilise and enhance capital market liquidity directed at this area. This will include, for example, the trading of existing blue/green bonds with financial institutions on condition that the mobilised funds are used to buy new blue/green bonds, thereby enabling institutions to diversify portfolios while expanding the capital market. The BGB will also create an enabling environment to foster innovation in the climate finance sector.
Barrier 3: Commercial banks in Barbados are typically risk-averse and focus on collateralised loans.	The BGB will fund initiatives to demonstrate the bankability of such investments, including through the: i) packaging and project structuring of programmes and initiatives to improve bankability (Activity 2.1.3); ii) provision of bridging finance to support packaged initiatives in securing collateral for low- to middle-income households (Activity 2.1.4); and iii) provision of guarantees or other securities for supported initiatives (Activity 2.1.3).
Barrier 4: Lower and middle-income groups in Barbados have limited collateral and limited credit histories to access loan finance from commercial banks.	By partnering directly with commercial banks, who will act as intermediaries for investment in the GoB projects or programme (Activity 2.1.2), the BGB will unlock financial resources for low- to middle-income households as well as small businesses to invest in climate-responsive initiatives — thereby significantly increasing the potential for climate action in Barbados.
Barrier 5: Limited familiarity among financial institutions with opportunities provided through climate finance.	The BGB will work with various partners, including government-run programmes, to package climate-responsive initiatives and create bankable projects, thereby strengthening a “learning by doing” approach. This can include the financial structuring/design of individual projects (Activity 2.1.3) or programmes as well as the provision of bridging finance where necessary to support the securitisation of the packaged investment options (Activity 2.1.4). Such an approach will leverage the

combined benefits of packaged investments to maximise the cost efficiency and effectiveness of climate investments made through the BGB.
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Once established, the BGB will engage with investors, including private investment firms, development banks, philanthropic organisations and other sources of finance to continue to capitalise the bank over the long term. The bank will maintain a debt-finance to equity ratio of 7:1, aiming to mobilise at least an additional USD 5.8 billion of investment in climate projects over the next 15 years.

Initially, the BGB will target three priority sectors that have been identified under the GoB's resilience strategy, namely: i) renewable energy; ii) climate-resilient infrastructure and housing; and iii) water conservation. These sectors have been assessed to have significant potential for establishing bankable initiatives that target low- and middle-income households when partnered with appropriate initiatives, such as the government's HOPE and SHIFT initiatives. Additionally, the BGB will explore options for providing additional security to such lenders (possibly in the form of full or partial guarantees to the banks for their funding of specific projects or programmes) to ensure competitive pricing of their loans.

While it is expected that the initial BGB investment portfolio will be focused on elements such as land acquisition, solar installations and mortgage finance promotion relating to the HOPE programme, along with retrofitting of existing buildings for hurricane resilience and water-saving devices through the SHIFT and water conservation programmes, this technically simple approach would serve as the gateway to a broader spectrum of climate-finance related operations and more complex project finance structuring. This business progression should also allow for the generation of early income, whilst the BGB's HR and technical capacities were being refined in line with the institution's evaluation of the market potential. The HOPE programme would also open the door to a dialogue between BGB and the Barbadian financing community, with a view to developing a proactive dialogue with the latter in order to identify and develop projects and programmes which, on the basis of project finance structuring, would enable this community to deliver its full potential. This expected expansion of the investment portfolio will be overseen by the BGB's Investment Committee (see Section B.4 below), following the explicit due diligence processes and investment eligibility criteria developed specifically for BGB.

The BGB will be able to target other sectors specifically or more generally through the financing of green bonds and other capital market instruments. Through this approach, the BGB could provide finance for private and public initiatives for green, affordable, gender-inclusive housing, food security and low-carbon transport, in addition to the primary sectors identified above. The eligibility of initiatives and activities to be funded through the BGB will be determined based on four criteria, which will be assessed in detail by the BGB management team and Investment Committee (see Section B.4 for details on the assessment roles and responsibilities)¹⁹:

- *Climate basis*: how does the initiative address a climate change problem, considering both adaptation and mitigation benefits?
- *Financial feasibility*: is the proposed initiative financially viable, with suitable risk mitigation and potential return on investment to be considered bankable?
- *Environmental and Social Safeguards*: does the proposed initiative align with the bank's environmental and social safeguard policies and management systems?
- *Gender Responsiveness*: Is the proposed initiative gender-responsive, equitable and aligned with the bank's gender action plan?

Overall, the BGB will promote a paradigm shift towards investment in resilient livelihoods and low-carbon technologies, with emphasis on low-income, vulnerable households and the energy, housing, agriculture and transport sectors. The project will also build the wider community's awareness of climate investment opportunities in support of a more resilient and sustainable Caribbean. Finally, by focusing on climate resilience and adaptation for middle to low-income households — including a specific focus on women and other highly vulnerable groups — the BGB would aim to attract more concessional or philanthropic providers of capital, expanding the goals of climate action in Barbados without the

¹⁹ Further details on the eligibility criteria are presented in Appendix C to Annex 2: Feasibility Study. Please note, while the high-level criteria are set, the specific eligibility factors considered under each of these categories will be finalised by the BGB Board at the early stages of the establishment of BGB.

need for further injections of money by the Government and GCF. This will create long-term sustainability and scalability in the banking sector, with the potential to scale beyond Barbados into the broader Caribbean region over time.

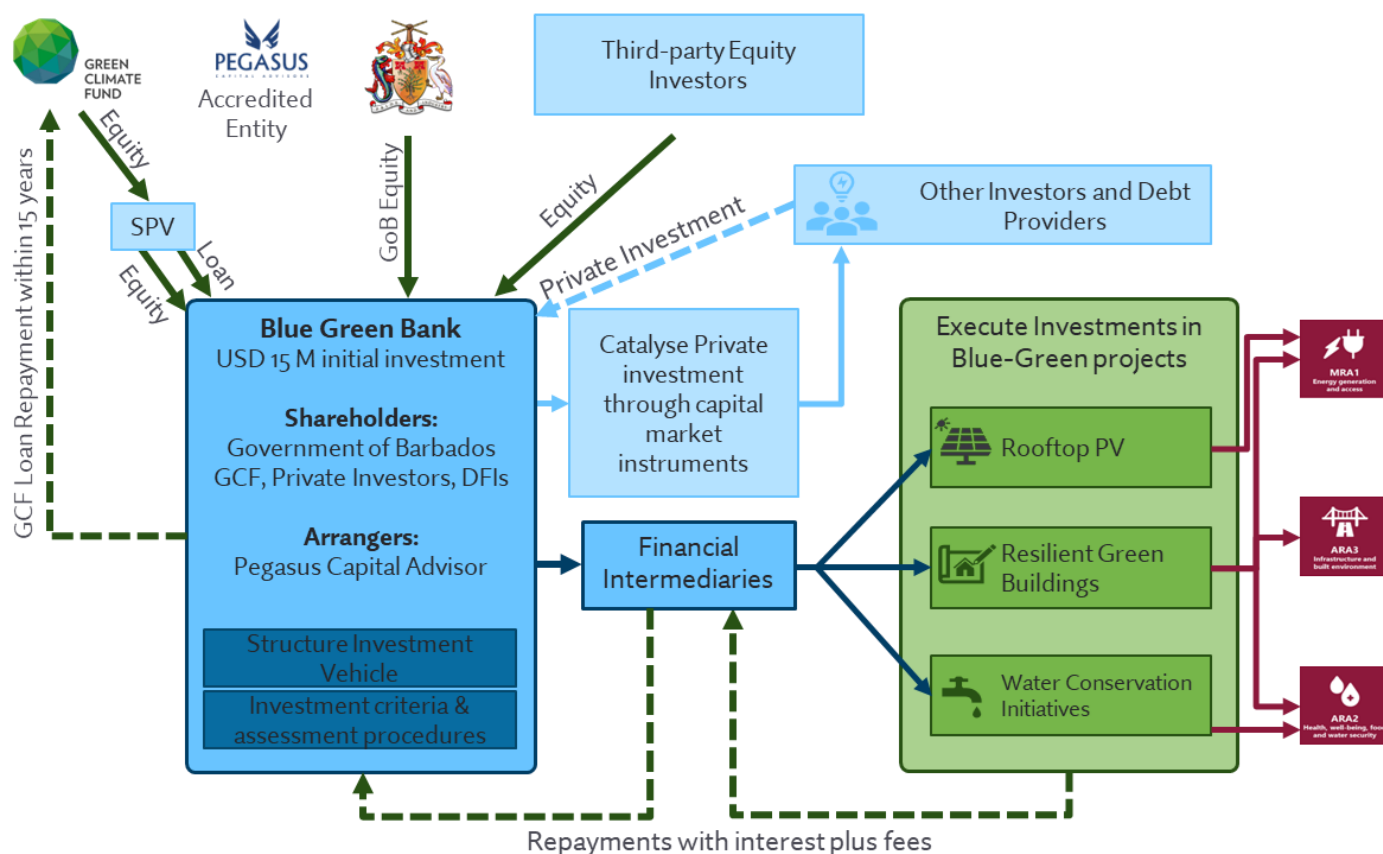


Figure 6. Strategic overview of the BGB investment framework.

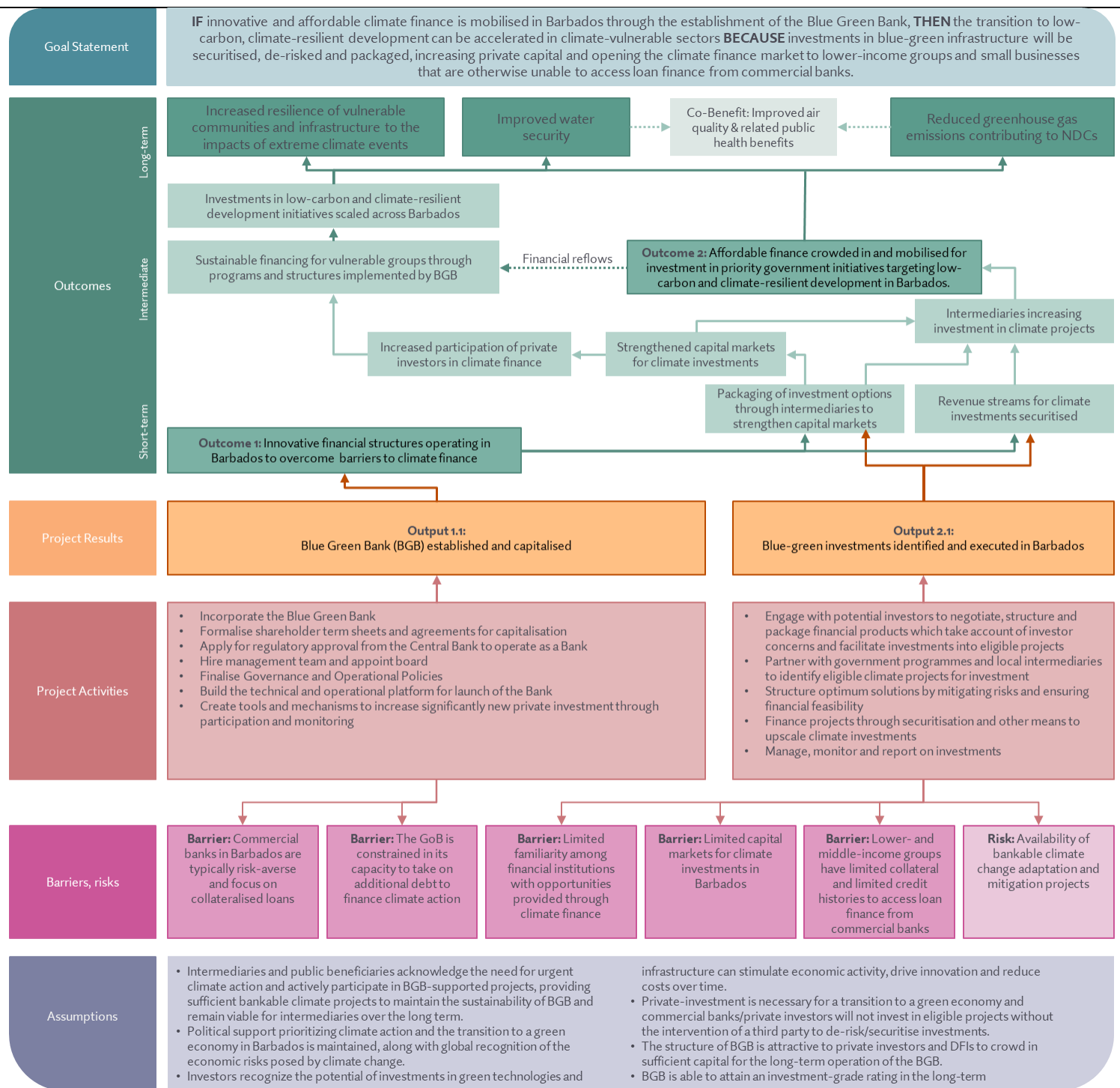


Figure 7. Theory of Change of the GCF-supported project to establish the BGB.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1	☒	☐	☐	☐	☐	☒	☒	☐
Outcome 2	☒	☐	☐	☐	☐	☒	☒	☐

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Reduction in air pollution and its associated impacts on public health	☒	☒	☐	☐	☐	☐
Gender mainstreaming into climate finance	☒	☐	☐	☒	☐	☐
Improved groundwater reserves and quality	☒	☐	☐	☐	☒	☐
Job Creation	☐	☐	☒	☐	☐	☐
Accelerated energy transition	☐	☐	☐	☐	☐	☒

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

The proposed GCF-funded project will establish the Blue Green Bank (BGB) in Barbados to stimulate innovation in the financial sector with the goal of unlocking the financial market and mobilising additional investment in climate-responsive initiatives. The project is structured with two primary Outputs that will achieve two paradigm-shifting Outcomes. The first Output will focus on the establishment of the BGB, including formalising the operational arrangements, setting up the policy environment and building the technical and operational platforms. The second Output will focus on crowding in additional private investment to strengthen the local capital market for blue-green investments, by acting as a catalyst to develop bankable projects. Finally, activities under the third Output will structure and package climate solutions to securitise and ultimately execute blue-green investments.

Outcome 1: Innovative financial structures operating in Barbados to overcome barriers to climate finance.

The establishment of the Blue Green Bank under Outcome 1 will create an enabling space for financial innovation in the climate change adaptation and mitigation spaces. The public-private partnership (PPP) nature of the bank, combining the local interests of the Government of Barbados with the ambitious focus of GCF, will enable the bank to take the lead and incentivize the commercial banks to play a fuller role, via the development of innovative financial structures that will crowd in private investment for climate-responsive initiatives. By catalysing innovation in the financial market, the BGB will demonstrate the potential for significant GHG emission reduction and investment in climate resilience. The foreseeable impact of such investments is far above what is typical of many GCF projects and programs due to the bank's revolving nature and its indefinite lifetime.

While the BGB aims to enable GCF to exit within a 15-year timeframe, the bank will continue to exist in the long term, attracting funding from the public and private sectors to drive transformational projects. Overall, the establishment of the BGB and the resulting crowding in of private investment is expected to overcome two key barriers to climate action in Barbados, specifically those related to the historical reluctance or inability of commercial banks to play their full part, and the constraints on the Government of Barbados' capacity to take on additional debt to finance climate action (see Barriers 1 and 2 in Section B2 above).

Output 1.1: Blue Green Bank (BGB) established and capitalised

The BGB will be incorporated as a corporation under the Companies Act of Barbados²⁰. Once incorporated, the bank will apply for regulatory approval as a wholesale bank from the Central Bank of Barbados²¹. The BGB will be established as a digital²², wholesale²³ bank, with a shareholding structure that will maximise the benefits of the PPP model. This includes the Government of Barbados holding non-majority equity, in its capacity as the prime promoter of opportunities for climate change investment and potentially facilitating future expansion across the Caribbean. The BGB's Board will be made up of 5 — 9 members (see Section B.4: Implementation Arrangements below). Details of the initial capitalisation are as follows:

Partner	Contribution
Government of Barbados	US\$ 10,000,000 Equity
GCF BGB Holdings	US\$ 5,000,000 Equity US\$ 10,000,000 Loan (Output 2)
Other 3 rd party investors	US\$ 5,000,000 Equity

²⁰ Draft Incorporation documents are provided as Annex 9

²¹ Preliminary discussions have been held with the Central Bank during stakeholder engagement process for the development of the project. The CBB

²² A digital bank is a financial institution that operates primarily online, offering services such as online banking, mobile banking, and other digital financial services. Refer to Annex 21 for further information.

²³ As a wholesale bank, the BGB will not deal directly with retail customers.

Following incorporation and regulatory approval, which will involve the formalisation of the shareholder term sheets and agreements for capitalisation, appointment of board and senior management as well as the finalisation of the Bank's Governance and Operational Policies. Finally, the technical and operational platforms will be built ahead of the launch of the bank – including setting up and staffing necessary operational departments within the bank. Full details on the planned legal and operational arrangements are presented in Annexes 9 and 21, respectively.

To enable the Bank to structure and package bankable financial products, suitable legal contracts and document templates will be prepared to address borrower needs as well as meet bank and investor requirements (Activity 1.1.7). This will help operationalise the funding activities of the Bank and unlock access to local and international capital markets for incremental funding for eligible climate-responsive projects. Through this process, the Bank will develop a variety of tools and mechanisms to facilitate and channelise incremental funding including: managing investor engagements and negotiations, preparation of Info memo and documentation related to financing options, enabling the issuance of bonds or other capital market instruments, managing operational activities and fund transfer process, developing suitable monitoring and reporting mechanisms for the investors.

Activity 1.1.1: Incorporate the Blue Green Bank

The BGB will be incorporated as a company under the Barbados Companies Act, CAP308, 2002. The incorporation will be done through an application to the Corporate Affairs and Intellectual Property Office (CAIPO). This Activity will be executed by PCA, together with GCF BGB Holdings and Pegasus BGB Holdings GP.

Activity 1.1.2: Formalise shareholder term sheets and agreements for capitalisation.

Shareholders committing equity to BGB will be formally onboarded. The processes will include the finalisation and formalisation of the shareholder term sheets as well as finalising agreements for the capitalisation of the bank. This Activity will be executed by PCA, together with GCF BGB Holdings and Pegasus BGB Holdings GP.

Activity 1.1.3: Apply for regulatory approval from the Central Bank to operate as a Bank.

Once registered as a company, the BGB will be licenced as a merchant bank with the appropriate regulator, providing the necessary regulatory approval to operate as a bank in Barbados. This Activity will be executed by PCA and BGB.

Activity 1.1.4: Hire a management team and appoint a board.

The BGB recruit experienced professionals with a strong track record of managing operational risks and financial risks to form the management team of the BGB. A board of between 5 and 7 members will be appointed, ensuring compositional alignment with the guidelines provided by the CBB. This Activity will be executed by PCA and BGB.

Activity 1.1.5: Finalise Governance and Operational Policies.

Building on the draft policy recommendations developed during project preparation, the BGB's management team will review and finalise all governance and operational policies. This includes all policies identified under the Legal and Policy review to ensure compliance with regulatory processes. The board will be responsible for the final sign-off and adoption of all policies. This Activity will be executed by BGB.

Activity 1.1.6: Build the technical and operational platform for the launch of the Bank.

As a digital bank, the BGB will require a robust technical platform for its operation. Professional services will be procured to work alongside the BGB management team to develop the technical platform and train all staff on its operation. Once operational, the BGB will be publicly launched. This Activity will be executed by BGB.

Activity 1.1.7: Significantly increase new private investment by creating the right tools and mechanisms for their participation and monitoring.

The bank will use the information generated through initial engagements with investors under Activity 2.1.1 to structure and package financial products that address investor concerns and unlock the capacities of the local capital market. Through this process, the project will develop a variety of products, tools and mechanisms for non-governmental

financing of investments, as well as negotiate with potential investors to co-develop packages that will mobilise private finance for eligible projects. This Activity will be executed by BGB.

Outcome 2: Affordable finance crowded in and mobilised for investment in priority government initiatives targeting low-carbon and climate-resilient development in Barbados.

The core role of the BGB will be to mobilise finance for climate-responsive projects by securitising investments and strengthening local capital markets. While part of this function will be achieved through the structuring and trading of financial products (such as Green Bonds) using tools developed under Output 1.1, the bank will support the direct financing of eligible climate-responsive initiatives under Output 2.1. The project pipeline for Output 2.1 will be approached in two phases: an initial phase of initiatives that have been identified as high priority under Barbados' Roofs to Reefs Programme (R2RP); followed by a second phase of projects that will expand the impact of the bank by opening to new, innovative initiatives. The mandate of the bank will be specifically to mobilise *additional* private finance for climate-responsive projects – including adaptation, mitigation, and cross-cutting initiatives – working alongside existing financial institutions to not crowd out, but rather collaborate and crowd in additional investment.

In addition to the initial equity contributions, GCF BGB Holdings²⁴ will provide USD10 million towards initial capitalization in the form of a convertible loan, with the BGB then raising additional finance that will strengthen and upscale the balance sheet to enable the bank to structure project financing beyond the initial pipeline of projects identified at the initiation of the bank (see the outline of initial pipeline investments presented below). A total of at least USD 5.8 billion is anticipated to be mobilised within the first 15 years of the funding cycle, with the potential to increase if additional equity is successfully raised. Furthermore, the long-term mission of the BGB, along with governance policies that ensure significant reinvestment of profits, will enable the bank to recover and re-invest funds in the medium or long term.

Output 2.1: Blue-green investments identified and executed in Barbados

Under Output 2.1, the BGB will engage with government programmes and other intermediaries to identify, develop and execute eligible climate projects. Once the BGB has been established and is fully operational, the bank's management team will engage in activities to unlock financial resources to fund climate-responsive projects, programmes and activities. This will include actively engaging with potential investors and identifying specific needs, concerns and barriers to investment (Activity 2.1.1). The BGB will then work with intermediaries to structure solutions to optimise the initiative, packaging options to mitigate risks and ensure the bankability of individual investments. All projects will be screened against the bank's specific eligibility criteria which have been developed to ensure that all initiatives financed by the bank are fully aligned with its core objective and operational policies, as well as those of the key equity shareholders – including the GCF (Activity 2.1.2). These criteria include four core categories of assessment, namely the climate change basis; the financial feasibility; the environmental and social safeguard considerations; and the gender responsiveness (described in more detail under Section B.2 above). Further details on the proposed investment criteria are presented as an appendix under Annex 2: Feasibility Study. While intermediaries can develop and propose projects²⁵ to the BGB, the selection of projects for funding will be entirely controlled by the BGB. Candidate projects will be screened by the BGB's investment committee, which will be responsible for all necessary due diligence on behalf of the bank. In addition to the BGB's due diligence processes, all intermediaries will be expected to have suitable internal due diligence in their own planning processes. The bank will also be responsible for monitoring and recording lessons learned from individual investments, creating a best practice database that will demonstrate successful initiatives that can be scaled up or replicated in other areas.

Two initiatives have been identified under the R2RP for the initial pipeline — specifically the HOPE and SHIFT programmes. In addition to these two core programmes, opportunities will be sought to finance water conservation initiatives, which have been identified as a key adaptation need with the potential for creating bankable investments. These three initiatives provide a stepwise progression for BGB investment, from HOPE, which has a clear model ready to go that will kickstart the bank's operations, to SHIFT which has a relatively defined model, but needs support from the BGB to formalise the packaging of the project. Finally, the water conservation initiative is relatively theoretical at

²⁴ Funded by the GCF equity contribution

²⁵ Projects in this sense are considered as the deal between the BGB and the intermediary, rather than between the intermediary and the final beneficiary.

this point, without a specific project operational at this time. In this instance, the BGB will play more of a deal-maker role to get an intermediary on board and develop the final model.

In addition to the core projects prioritised for the initial phase of investment, several investment areas have been preliminarily identified for expanding the impact of the bank. These include initiatives that reduce the reliance on, and reduce emissions from food imports (particularly for the hotel and the tourism industries), climate-resilience agriculture to increase food security in the context of future climate events and vehicle-to-grid electric vehicle charging programmes to promote grid stability. In such sectors, the BGB will work with various partners to identify and package financial solutions that unlock private investment. Depending on the nature of the project, different financial solutions may be applied (Activity 2.1.3). The details of what solutions are best suited to each project, particularly for the second

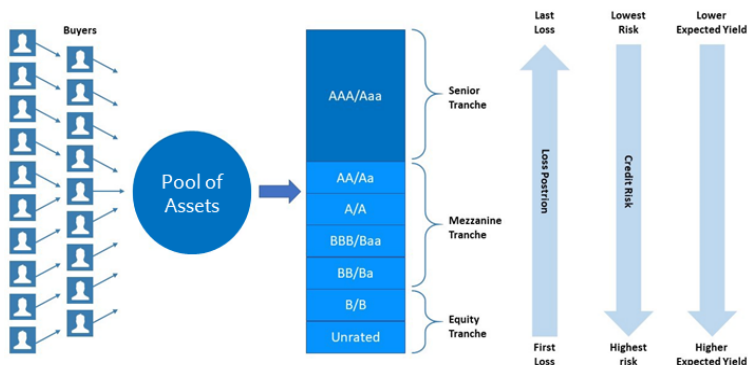


Figure 8: Tranched capital market model for securitised assets.

phase of the BGB's project pipeline, will be determined by the bank's management team, taking into consideration the state of the capital market and what options can be best implemented by the BGB²⁶. One key solution that has been identified is the selling of cash-flow generating assets via securitization to create capital optimization for the BGB, whereby the risk can be tranced for the capital markets (Figure 8). Other financial solutions could include: i) Public Private Partnership (PPP) structures; ii) guarantees, including credit guarantees, performance risk guarantees, or partial risk guarantees; iii) interest make-up arrangements; and iv) insurance.

An overview of the initial pipeline projects is presented in the sections that follow. Further details are provided in Annex 2: Feasibility Study.

Activity 2.1.1: Engage with potential investors to negotiate, structure and package financial products which take account of investor concerns and facilitate investments into eligible projects

The BGB Management team will actively engage with potential investors to identify specific needs, concerns and barriers to investment. This information will inform bank operations to enhance the role of BGB as an arranger in the financial sector. The BGB will then structure and package financial products to crowd in investment for eligible projects. The BGB will negotiate terms with investors and enter agreements for investments using the tools/mechanisms developed under Activity 1.1.7. This Activity will be executed by BGB.

Activity 2.1.2: Partner with government programmes and local intermediaries to identify eligible climate projects for investment.

The BGB investment teams will engage with government programmes and local intermediaries to identify eligible climate projects for investment by the bank. These efforts will initially focus on the three priority pipeline initiatives with HOPE, SHIFT and BWA, but will eventually branch out to identify additional opportunities. A key role of BGB in this process will be as a dealmaker, actively seeking opportunities to package deals that improve the bankability of projects. This Activity will be executed by BGB.

Activity 2.1.3: Structure optimum solutions by mitigating risks and ensuring financial feasibility.

Once opportunities are identified, the BGB investment teams will engage with intermediaries to structure solutions to optimise their bankability. This process will start with HOPE, where the model is already in an advanced stage – looking for opportunities to strengthen the existing model to maximise the benefit offered by BGB over alternative sources. This will be followed by firming up arrangements for SHIFT, and finally taking a broader deal-maker role for water conservation initiatives with BWA as an intermediary partner. In phase 2, the BGB will branch out beyond these initial pipeline initiatives, seeking opportunities to strengthen the capital market and securitise investments for a broader range of climate adaptation and mitigation options. This Activity will be executed by BGB.

²⁶ Note that the project pipeline phases are not mutually exclusive in time, and the second phase will likely begin before the projects under the initial phase have been concluded. Instead, the initial phase is intended to provide a suitable pipeline to initiate the BGB's activities while the additional pipeline for the second phase is developed.

Activity 2.1.4: Finance projects through securitisation and other means to upscale climate investments, initiating the process with the HOPE and SHIFT programmes.

Once deals are identified and packaged, BGB will mobilise funds to finance eligible and approved projects. Initial agreements will be made with HOPE, which provides a ready-to-finance project model to kickstart bank operations, followed by SHIFT. This Activity will be executed by BGB.

Activity 2.1.5: Manage, monitor and report on investments.

The BGB management team will take responsibility for actively managing the accounts of all financed projects, including through regular monitoring and reporting on investments. This Activity will be executed by BGB.

HOPE Programme²⁷: HOPE Inc. is a Barbados government state-owned enterprise established in November 2020 to fill an important social need for hurricane-resilient houses for lower-income markets. Through the HOPE programme, the land is purchased²⁸ from public or private sources and developed as a housing estate along with all associated infrastructure — including water, roads and sewage. The houses, which are built to withstand the impacts of at least a Category 3 hurricane and are fully kitted with rooftop PV, are then sold to low-income, first-time buyers, with the proceeds going to repay the initial investment costs. A transformative aspect of HOPE's business model is the use of feed-in tariffs for energy generated by the rooftop PV to subsidise the developments, allowing HOPE to sell the houses at affordable prices (below market value), while simultaneously contributing to the Government of Barbados' ambitious mitigation targets. Moreover, in exchange for maintaining the rights to the income generated from the rooftop PV²⁹, HOPE is able to provide the land to eligible buyers free of charge, providing them with the collateral needed to access finance from commercial banks for the purchase of the house.



Figure 9. Artist's rendering and site layout of a HOPE housing estate. (Source: HOPE Inc)

The role of BGB in the HOPE programme will be to provide the finance needed to purchase land, as well as to purchase, install and maintain the rooftop PV systems. To this end, HOPE will partner with the soon-to-be-established Legacy PV Inc³⁰ which will be responsible for the installation and management of rooftop PV systems under the HOPE programme. The PV systems will then be funded through 15-year loans issued by the BGB.

The HOPE programme targets 10,000 hurricane-resilient homes powered by renewable energy. In addition to hurricane-proof roofs and rooftop PV, several other climate-responsive considerations have gone into the design of the houses being built through the HOPE programme. For example, the houses are designed to allow natural lighting and cooling, thereby improving energy efficiency, while water-saving toilets and fittings are being used to promote water conservation. The water conservation potential of HOPE is further enhanced through the programmes Water Reclamation Initiative. Under this initiative, grey and black water is channelled separately from stormwater into a central sewage treatment plant with tertiary treatment capability. After treatment, the clean water is pumped into large catchment ponds established by HOPE near each development, along with rainwater runoff from the houses and stormwater drainage systems. The water in these catchment ponds can then be used for irrigation systems for nearby

²⁷ <https://www.hopebarbados.com/>

²⁸ As far as is feasible under the constraints of land availability, the selection of land for HOPE developments will account for potential climate change hazards, particularly for flood and coastal hazards. Where possible, high risk areas will be avoided. However, when alternative locations are constrained and some level of hazard is identified for a particular location, the design of the development will account for the hazard and implement suitable risk-reduction measures, including through adequate stormwater systems.

²⁹ Note: the power generated through the rooftop PV will be on a separate grid connection to the household usage, with the full value of the feed-in tariff from BLPC directed to Legacy PV, and not offset against household usage.

³⁰ Legacy PV Inc will be a new state-owned enterprise.

agricultural projects, drawing on government-regulated consumption-based tariffs to help sustainably fund the operation.

Ultimately, the HOPE business model is designed to fill the need for hurricane-resilient housing for lower-income households without any permanent call on limited government resources — with co-benefits in terms of emissions reduction, water conservation and food security (via the irrigation supply to agriculture from captures and reclaimed water).

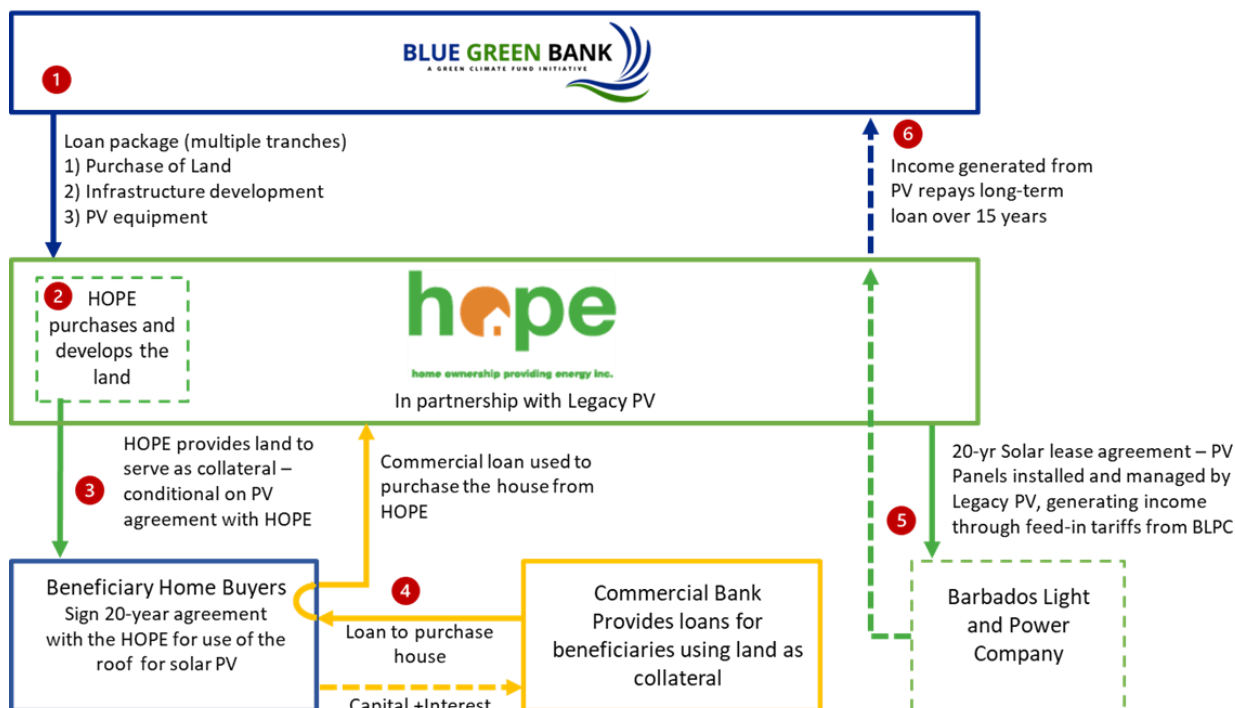


Figure 10. Schematic diagram of the HOPE model. The sequence of steps, labelled in red, is presented below.

1. BGB provides a tranchied loan package to HOPE to purchase land, develop utilities and houses, and install solar PV systems.
2. HOPE uses the initial loan to develop the land.
3. HOPE provides land to eligible buyers to serve as collateral for accessing a commercial home loan, on condition that HOPE retains the right to produce power from rooftop PV of beneficiary houses.
4. Beneficiaries get a commercial loan to purchase a house from HOPE, repaying the loan through private funds.
5. HOPE signs a 20-year power purchase agreement with BLPC to generate income from the power produced through PV installed on beneficiary houses.
6. Income generated from the PV is used to repay the remaining debt over 15 years.

Table 4: Typical cost breakdown of a HOPE house in USD

Finance Mechanism	Cost Category	Two-bedroom house		Three-bedroom house	
		Actual Cost	Assessed Value	Actual Cost	Assessed Value
BGB Loan	Purchase Land	7,800	46,000	7,800	46,000
	Install Infrastructure	14,375		14,375	
	Purchase and install PV	17,060	17,060	22,117	22,117
Mortgage Loan	Construction costs, legal costs & bridging interest	90,000	129,000 (excl. land)	121,000	161,500 (excl. land)

Small Home Income Generating Facility Trust (SHIFT): The SHIFT programme, still in the conceptual stage, is a government initiative to support homeowners to invest in climate resilience and renewable energy. Closely linked to the GoB's Householders' Right to Renewable Energy (RTRE) policy, SHIFT will address a core constraint on rooftop PV – the hurricane resilience of roofs. The nexus of energy production and hurricane-resilient roofs is at the core of R2RP, and SHIFT will form part of the action plan to bridge the gap between R2RP and RTRE. Specifically, SHIFT will support existing homeowners to retrofit their houses for hurricane resilience, leveraging feed-in tariffs for rooftop PV to finance the efforts. This approach will be based on the second model of BGB investment (Figure 11) outlined in Section 7.2 of Annex 2: Feasibility Study, whereby SHIFT signs a deal with homeowners to retrofit houses for hurricane resilience, in exchange for the rights to put PV on the roof. SHIFT takes a loan from BGB to finance the retrofitting and

PV installation, and structures the repayment around the income from feed-in tariffs for the solar energy generated. Under this model, SHIFT targets equipping 50,000 homes with climate-resilient roofs and rooftop PV.

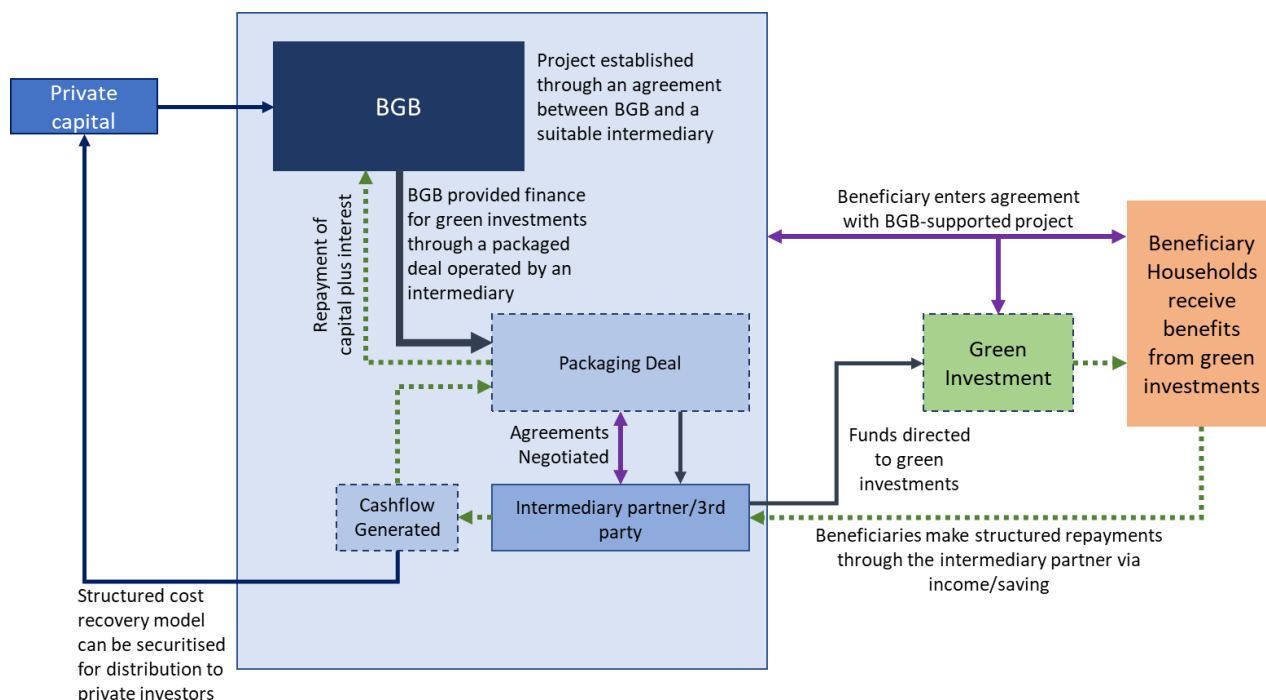


Figure 11. Schematic diagram of the theoretical framework of the BGB model for packaging projects to create bankable repayment models through intermediaries, on which SHIFT will be based.

Water Conservation: The proposed water conservation initiative is the least developed of the initial pipeline projects, although ongoing work is being conducted by the Barbados Water Authority (BWA) to set the foundation for water conservation through the Greening and Conservation Initiative Programme (GCIP). This programme includes several response areas for water conservation, including water-saving devices, personal storage tanks and aquifer recharge, alongside comprehensive awareness initiatives. It is anticipated that the BGB will engage with the GCIP to identify innovative financing options to support the implementation of these initiatives through potentially income-generating activities, with a particular focus on water-saving devices and storage. The conceptual framework will be based on Model 2 of the BGB investment options (as prescribed for SHIFT, Figure 11). Under this model, BGB would partner with the Barbados Water Authority (BWA) as an intermediary, providing a loan for the installation of water conservation devices. These devices could include low-flow showers and toilets, faucet aerators, and rainwater harvesting tanks. BWA would then enter into agreements with homeowners and small businesses to install the equipment, with the cost being added on as a small additional tariff on their utility bill, which would be used to repay the investment from BGB. The cost to the end user would be offset by the water savings, minimising the financial impact on target households that are otherwise unable to finance such investments.

As part of the ongoing GCIP efforts, a feasibility assessment is being conducted of financing modalities for supporting the introduction of water savings devices and the estimation of potential BWA cost savings through a reduction in water demand. This process will also identify evaluation protocols for water-saving initiatives. The BGB will work closely with BWA in developing a bankable initiative, building on the foundations being set, including leveraging the impact of the planning awareness initiatives. To ensure the application of best practices, the development of the water conservation pipeline project will draw from experiences of other countries and programmes, both regionally and further abroad. For example, the project will consider the progress under the ongoing GCF-supported Climate-Resilient Water Sector in Grenada (G-CREWS) project that is establishing a challenge fund targeting heavy waters from the tourism and agriculture sectors. The project will also explore the lessons learned from incentive schemes that promoted the uptake of water-efficient devices, such as the successful Water Smart Homes initiative from Sydney, Australia, which provided a rebate on water tariffs to users who adopted water-efficient technologies. The proposed project will innovate on these

models to leverage the opportunity for the BGB to support the upfront costs for communities that are unable to invest themselves but would be able to repay investments through ongoing tariffs (offset by water savings).

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Accredited Entity

Overview

Pegasus Capital Advisors, L.P. (Pegasus) – Pegasus is a private, alternative asset management firm founded by Craig Cogut that provides strategic growth capital to middle-market companies focused on the sustainability and wellness sectors. Pegasus has invested approximately \$2.6 billion, across five private equity funds, and has completed over 90 private equity transactions. Pegasus is also currently the first and only private equity fund manager accredited by the Green Climate Fund (GCF). In this role, Pegasus is currently partnering with GCF on several impact initiatives, including:

- The Sub-National Fund (SnCF) – a USD 750M target impact fund focussed on sustainable and resilient infrastructure development;
- The Global Fund for Coral Reefs (GFCR) – a USD 500M target impact fund focused on investments that enhance the resilience of coral reefs and the populations that depend on them; and
- Catalytic Capital for First Private Investment Fund for Adaptation Technologies in Developing Countries (CRAFT) – an impact fund to scale up technologies for climate resilience and adaptation and apply them in developing countries using a south-south technology transfer mechanism.

AE Role in Implementation

The Programme will be implemented through a Funded Activity Agreement (FAA) between GCF and Pegasus. Pegasus will enter into an Implementation Agreement with Blue Green Bank, whereby Pegasus as the AE will pass down obligations in the FAA to the Blue Green Bank as the Executing Entity. This includes the establishment of a special purpose vehicle (SPV) and associated general partner, managed by Pegasus, to serve as the holding company for GCF's equity interest in the BGB (see below for more detail). The SPV will, through a subsidiary agreement, be a party to the implementation agreements outlined below.

Pegasus, as per its fiduciary agreement with GCF will be responsible for:

- Overall implementation, oversight and coordination of GCF's commitment to the BGB;
- Proper use of GCF's commitment to the BGB;
- Ensure compliance of BGB activities with the FAA;
- Ensure investments performed by the BGB respect GCF policies, as well as the project's gender, social and environmental safeguard management plans; and
- Reporting to the GCF on the BGB.

To this end, Pegasus, as an Accredited Entity for the GCF, will

- take on the oversight role for the equity funds received;
- further administer the GCF funds (in the form of both equity and subordinated loan) and provide for the appropriate implementing arrangements with the EE (see below for details on the SPV to be established for this purpose);
- review progress as reported by the EE to determine how best to maximize project performance;
- review project progress against work plans, procurement and disbursement plans as submitted;
- implement a team with requisite skills that further analyses technical and financial issues; and
- provide a support and resource function to the EE to discuss ways to strengthen implementation and ensure its direction enhances the likelihood of achieving outcomes and impact, and efficiency.

Executing Entities

The project will be implemented through four executing entities, Pegasus Capital Advisors, the Blue Green Bank, GCF BGB Holdings, L.P., and Pegasus BGB Holdings GP, LLC – the latter three of which have yet to be established. The tentatively named GCF BGB Holdings, L.P. is an SPV (see below) that will be formed as a Delaware limited partnership to serve as the holding company for GCF's equity interest in the BGB. Its general partner will be a Delaware LLC, tentatively named Pegasus BGB Holdings GP, LLC, which will be wholly owned by PCA and will manage the GCF BGB Holdings.

Pegasus will hold responsibility for the first two activities under Output 1.1, which will include the establishment of the BGB, alongside the finalisation of the shareholder agreements, governance structures and operational processes, as well as gaining regulatory approval. Once fully established and onboarded, the senior management team of the BGB will take over the EE responsibilities for the rest of Output 1.1 and for 2.1. These responsibilities will include crowding in public investment, as well as identifying, structuring and financing eligible climate investments. The BGB's management team will also be responsible for managing, monitoring and reporting on all investments made through the bank.

BGB Governance Structure

BGB Board of Directors (The Board):

The Board will comprise 7 – 9 Directors, initially commencing with a board of 5 directors³¹, with the ability to increase the number of directors depending on a number of factors such as the growth of BGB or any potential recommendations from CBB to appoint more Directors. The initial Board will target at least 30% female representation, with the aim to increase it to 50% in time. In line with regulatory guidelines, the Board will comprise a balance of executive and non-executive directors, and will include independent directors that have the ability to exercise objective, independent judgment after fair consideration of all relevant information and views without undue influence from executives or inappropriate external parties or interests. The Board will also be required to have the right balance of expertise, which could include finance, issuing loans, governance, legal, technology, renewable markets and climate policy to operating in the Caribbean.

The Board will meet at least quarterly, while maintaining the option to have additional meetings as needed from time to time.

Senior Management:

A senior management team will be appointed by the Board, comprising a core group of individuals who are responsible, and should be held accountable, for overseeing the day-to-day management of the BGB — serving as a link between the Board and staff. Responsibilities of senior management will include:

- Providing adequate oversight of those they manage, and ensuring that the BGB's activities are consistent with the business strategy, risk tolerance/appetite and policies approved by the Board;
- Keeping the Board adequately informed about the performance of the institution through financial and management reports and such other reports prepared for example by internal auditors, external auditors and the compliance officer;
- Advising the Board on the appropriate organisational structure, and ensuring that the quantity and quality of staff resources are available to carry out all tasks, including internal audit and compliance;
- Ensuring that legal and regulatory requirements related to human resources are met within the required timelines;
- Implementing and maintaining risk management systems appropriate to the scale, nature and complexity of the institution;
- Delineating and documenting the areas of responsibility for each staff member;
- Communicating the strategic direction, reporting lines and risk tolerances throughout the organisation; and
- Overseeing management information systems to enable timely and accurate implementation and reporting.

Committees of BGB

The bank governance structure will include several board and management sub-committees, in line with CBB guidelines. These committees include:

- *Investment Committee:* Comprised of senior members of the Board, as well as potentially members of Senior Management. The composition of the Investment Committee will be set out in its Charter under Output 1.1. The Investment Committee will include two sub-committees, namely the Screening Committee (responsible for assessing the pipeline investments and doing initial due diligence to assess red flags) and the Final Investment Committee (responsible for final decision-making on all investments). The decision-making power of the Investment

³¹ PCA confirms its intention to have a Board seat. This arrangement builds on the fact that PCA is sponsoring the BGB and will have an equity interest in the BGB going forward.

Committee will be agreed upon with the Board, whereby a monetary threshold will be set for investments and other monetary-related matters within its mandate, above which Board approval will be required.

- **Audit Committee:** An Audit Committee will be established, comprised of all non-executive Directors — the majority of whom are independent. This committee will be responsible for overseeing the BGB's financial reporting process, including its financial disclosure obligations; overseeing the internal and external audit functions, including approving the respective audit plans and engagement letters; meeting with and receiving reports from the BGB's internal and external auditors; and monitor management's response to regulatory, internal and external audit reports/letters ensuring timely and appropriate resolution of all matters.
- **Risk Committee:** The Risk Committee is mandated by CBB and will be responsible for advising the Board on the overall current and future risk tolerance/appetite and strategy, and for overseeing Senior Management's implementation of that strategy. This will include strategies for capital and liquidity management, as well as for credit, market, operational, compliance and reputational risk.
- **Additional Committees:** In addition to the above mandatory committees, the BGB Board will, as required, establish several other committees, potentially including: Credit; Human resources; Asset and liability management; Social and Ethics; and Group Remuneration committees.

The Governance structure is summarized in Figure 12 below, with the flow of funds demonstrated in Figure 6 above. GCF funds will be provided in the form of a USD15 million equity contribution to an SPV (see below), which will then channel the funds to the BGB via a USD5M equity contribution and a USD10 million convertible subordinated loan. The subordinated loan will be structured such that it counts towards tier 2 regulatory capital for BGB. This loan will contain provisions for automatic conversion to equity shares such that as far as possible, GCF (acting through the SPV) maintains a 25% equity interest in the BGB whenever additional equity is raised by BGB.

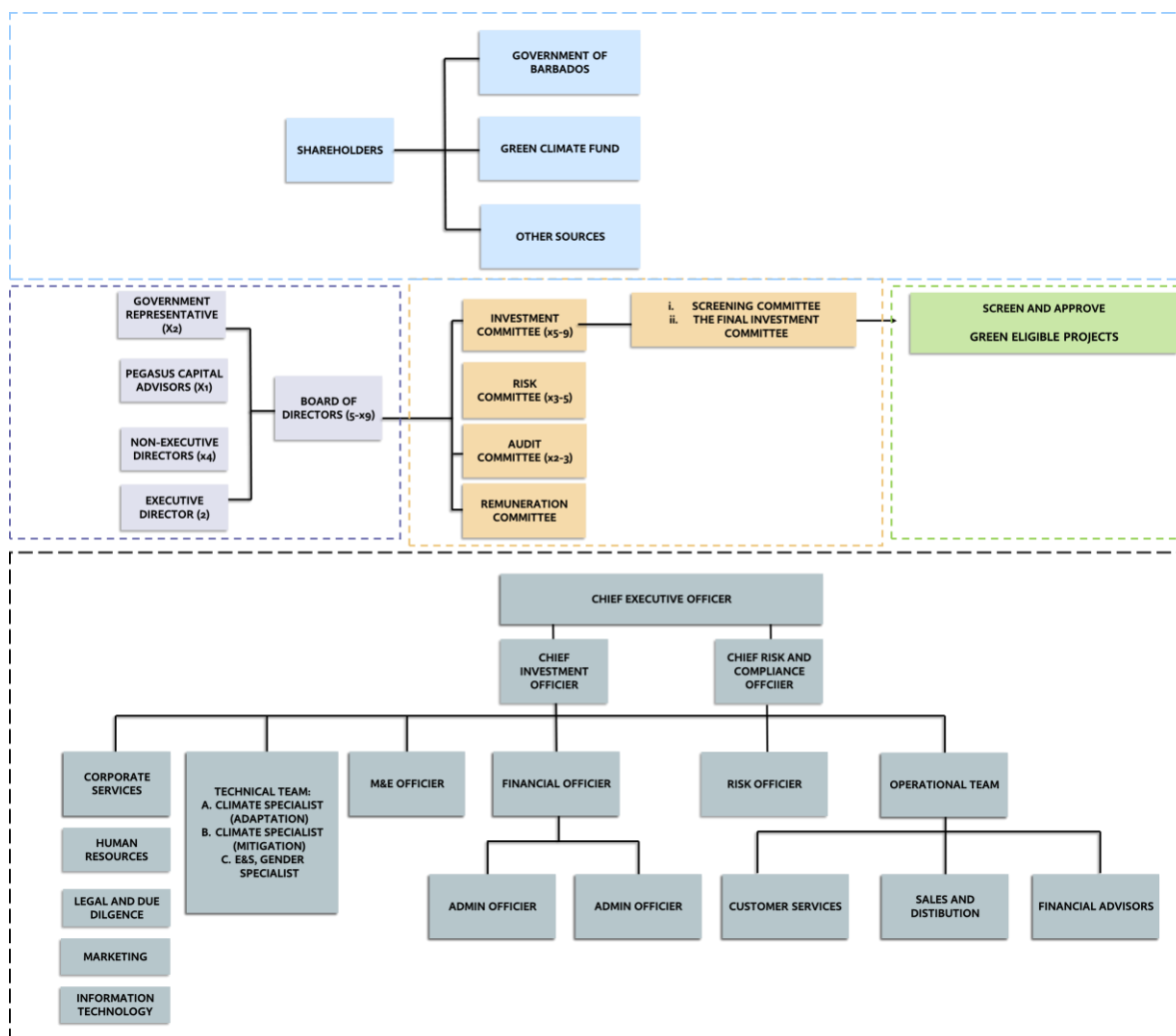


Figure 12. The governance structure for BGB

Portfolio Management and Investment Eligibility

The management of the BGB's project portfolio will be a joint effort between the Senior Management team and the Investment Committee, which itself is comprised of two sub-committees: the Screening Committee and the Final Investment Committee. Detailed investment eligibility criteria have been set (see Annex 2.2) for the BGB and will be formalised during project inception. The Investment Committee will take primary responsibility for screening projects against these eligibility criteria and formalising the bank's project pipeline. Senior Management, together with the teams they oversee, will support the Screening Committee in identifying and packaging eligible projects for submission to the Final Investment Committee for approval. Once operational, the Senior Management team will be responsible for overseeing the implementation, monitoring and reporting on bank-supported projects.

Investment process (preliminary)

The evaluation process will be led by the investment committee consisting of a technical team of subject matter experts to assist with the deliberation including for Climate, gender, E&S or a sector-specific proposal (e.g., a housing or renewable energy expert). The process for determining a generally includes the following steps:

1. **Deal Sourcing:** The investment committee receives investment proposals from the intermediaries to the BGB or referrals (where applicable).
2. **Initial Screening:** The BGB's screening committee conducts an initial review of the investment proposals to determine if they meet the organization's investment criteria (see below). This can involve a high-level analysis of financials, market trends, a climate assessment and other relevant factors.
3. **Due Diligence:** If the investment proposal passes the initial screening, the Steering Committee conducts a more detailed due diligence process. This can include reviewing financial statements, assessing the climate impact, reviewing the E&S considerations, unpacking the gender assessment and action plan, interviewing management, assessing market trends, and evaluating other relevant data.
4. **Valuation:** The committee uses financial models to determine the value of the investment opportunity and assess its potential returns. This involves analysing the company's financials and projections to determine its future growth potential.
5. **Investment Decision:** After completing due diligence and valuation, the proposal is passed on to the Investment Committee which makes a final investment decision. This decision is based on the analysis conducted throughout the previous steps by the Steering Committee, and takes into account the committee's investment criteria, the potential returns, and the risks associated with the investment.
6. **Post-Investment Monitoring:** After making the investment, the committee monitors the investment to ensure that it is performing as expected. This can involve regular reporting, reviewing financials, and making adjustments as necessary.

Investment criteria:

The investment criteria for BGB will be used to evaluate the potential investments. These criteria are based on a combination of financial and non-financial factors that will assist the governing authorities to assess the risk of the investment. At the proposal stage, indicative investment criteria have been developed, which will be reviewed and refined by the BGB Board immediately after the bank's inception. It is anticipated that the broad categories will not change, but the specific details of what qualifies under each will be refined. The initial eligibility criteria are as follows:

- Geography: The project or programme located in Barbados
- Financial: The amount requested has a positive financial return
- Governance: the project does not fall into the prohibited or exclusionary investment list³²
- Climate: The project or programme is contributing to SDG 13–Climate Action by addressing a material climate risk and impact in its area of operation and/or reducing GHG emissions.

³² See ESMS, Annex 6

- Adaptation (if applicable): Proposals must clearly identify the specific climate-related risk/hazards — such as drought, floods or hurricane risk — being targeted and demonstrate the climate change attribution (current or future) to the increasing risk. The intervention strategy must show how the project or programme's activities will provide direct and/or indirect benefits to communities by increasing their ability to adapt to, or absorb, climate change stresses, shocks and variability and/or by helping reduce exposure to them.
- Mitigation (if applicable): Proposals must present a clear emissions baseline from which the project will depart and demonstrate how the project or programme's activities contribute to the reduction of GHG emissions against the baseline and respective NDC targets.
- Environmental and Social Safeguards (ESS): The project or programme's ESS risk is Category B or C
- SDG: the project/programme is aligned or meets the SDG targets

Climate expertise:

The impetus for starting the BGB is to commit to exploring avenues to strengthen climate adaptation and mitigation in Barbados. Climate expertise is becoming increasingly important for financial institutions as climate change and sustainability are now major drivers of financial performance. **BGB will thus ensure that a comprehensive assessment is undertaken by a climate expert with experience in climate-related issues in Barbados.** A core responsibility of the climate specialist will be to technically assess all project proposals against the BGB's climate investment criteria (see above) and make recommendations to the Investment Committee on climate eligibility. To this end, the climate specialist will not only form part of the technical team but will also be part of the Investment Committee, providing approval authority on the climate validity of the proposal.

The investment committee is responsible for making investment decisions and is typically composed of experienced professionals with expertise in various areas of finance, including portfolio management, investment analysis, risk management, and compliance. The investment committee's primary role is to ensure that the bank's investment strategies are aligned with the bank's goals and objectives, as well as with the goals and objectives of its clients. This includes developing and implementing investment policies and strategies, setting investment targets, and monitoring investment performance. The climate specialist will be responsible for providing specialized knowledge and expertise and can help the investment committee to make informed and responsible investment decisions that align with the bank's and clients' goals and objectives. Overall, the climate specialist at the Bank is a crucial component of the Bank's investment management process. Their expertise and experience help to ensure that the bank's investments are managed prudently and responsibly, with the goal of maximizing returns while minimizing risk as they relate to climate-related issues.

In addition to the climate specialist noted above, the BGB will ensure that at least one Board Member is a climate finance expert.

Special Purpose Vehicle

A special purpose vehicle (SPV) is a separate legal entity created for a specific purpose or transaction, typically with limited liability and separate from its sponsors or investors. Pegasus Capital Advisors is the accredited entity (AE) that is authorised to receive funds from GCF and could invest in BGB on behalf of GCF. Subject to the approval of the following by the Board, a separate SPV will be established to channel funds from the GCF to BGB through Pegasus Capital Advisors. The main advantage of establishing the SPV is to ring-fence the GCF funds to finance BGB in a separate structure and thereby minimise the potential risk and liability of GCF and Pegasus Capital Advisors. It is noted that BGB will be established as a limited liability company and therefore all shareholders' liability will be limited to the extent of their capital contribution. However, any of BGB's creditors may still try to look to shareholders for outstanding debts and the SPV offers GCF (and Pegasus Capital Advisors) a level of protection from creditors and from reputational damage should BGB be unable to pay any of its debts. But the SPV is particularly relevant in the circumstances that a loan is made to BGB. An SPV would ring-fence the debt and protect GCF (and Pegasus Capital Advisors) from the increased risk of bankruptcy/insolvency should BGB fail to repay the loan.

Contracting arrangements

GCF and the Accredited Entity will enter into a Funded Activity Agreement (FAA), which will reflect the terms and conditions described in this Term Sheet and the AMA. Subject to the terms of the FAA and the AMA, GCF will make disbursements under the FAA as and when the Accredited Entity submits a Request for Disbursement.

GCF BGB Holdings, acting for the benefit of GCF, will enter into:

- a short-form purchase agreement between BGB and GCF BGB Holdings for the subscription of shares to the value of USD 5 million in the share capital of BGB. The terms of the equity interest will be set forth in the articles of incorporation of the BGB.
- a Loan Agreement between the BGB and GCF BGB Holdings, pursuant to which GCF BGB Holdings will lend BGB an amount of up to USD10 million. Portions of the loan will automatically convert to equity as and when the BGB raises additional equity capital such that GCF's equity share is maintained at 33%.
- a Shareholders Agreement with the BGB, Pegasus and the Government of Barbados that will set forth certain rights with respect to its equity interests.

The Purchase Agreement, the Loan Agreement and the Shareholders' Agreement will be executed by a representative of the Accredited Entity in such individual's capacity as an authorized signatory of GCF BGB Holdings.

GCF will subscribe to a limited partnership interest in GCF BGB Holdings pursuant to the terms of a Subscription Agreement to be entered into between GCF and GCF BGB Holdings. GCF will also be a party, together with the Holdings GP, to the Limited Partnership Agreement of GCF BGB Holdings. The Accredited Entity will also enter into an Implementation Agreement (the "Implementation Agreement") with the BGB, GCF BGB Holdings and Pegasus BGB Holdings GP with respect to the implementation of the Funded Activity to pass down the relevant obligations under the AMA and FAA to such entities. The Holdings Subscription Agreement and Holdings LPA will be executed on GCF's behalf by a representative of the Accredited Entity pursuant to a power of attorney granted by the GCF to the Accredited Entity, which will be included as part of the FAA.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Barbados' triple challenges to climate financing

Context

Barbados has one of the highest per capita incomes in the Caribbean with a total GDP of USD 4.84 billion and a per capita GDP of USD 17,225. Its economy has been able to transform itself from a low-income country, reliant largely on sugar production, to an upper-middle-income country with an economy that relies on a range of primary and tertiary sectors. These sectors include distribution (22%), financial services (21%), government (16%), tourism (12%), transport (10%), construction (5%) and others (14%).

Despite a relatively high GDP per capita, the Barbadian economy exhibits characteristics commensurate with many SIDS. These characteristics include vulnerability to natural disasters and climatic events, limited natural resources and a high reliance on imports and exports. Given its dependence on external markets and economies, Barbados is highly susceptible to market fluctuations, limiting the country's potential for economic growth and international competition. Consequently, Barbados faces several challenges to financing climate action, outlined below, necessitating the need for external support.

Challenge 1: National Debt

In recent years Barbados has found itself fiscally constrained with a debt-to-GDP ratio reaching a high of 158% in 2017. In response, the Government implemented the Barbados Economic Recovery and Transformation (BERT) plan, aimed at restoring fiscal sustainability, increasing reserves, and unlocking growth potential, which has been effective in reducing the country's debt to 118% in 2022 (IMF, 2023). However, in the context of growing development and climate requirements, Barbados' level of indebtedness inhibits the government's ability to respond adequately. Furthermore, the country's status as a middle-income economy means that Barbados does not qualify for concessionary loans from the World Bank and other Multilateral Development Banks (MDBs) owing to a per capita

funding threshold of USD 1,205. This further exacerbates funding gaps for the national response to climate change and development.

Challenge 2: Accessing MDB financing.

Barriers faced by Barbados in accessing multi-lateral funding due to GDP per capita thresholds are typical of small states on the front line of the climate crisis. The averaging effect of measures such as GDP per capita income has become increasingly unrepresentative of countries with small populations, as is the case with Barbados. GDP per capita also fails to consider that the level of income before a crisis does not capture the fragility and volatility of individual or national income, nor does it capture the state's capacity to respond after a disaster. The concept of using a threshold as a measure of need is based on the experience of previous generations which were typically smaller than the size of the economy - unlike the costs of adaptation to climate change. These factors place SIDS at a substantial disadvantage when trying to access funding from MDBs despite there being a clear need in these countries. Those funds which are accessed by the Government of Barbados from MDBs generally need to be directed to urgent development needs in the country, mostly targeting investments for the public good. According to the Ministry of Finance and Economic Affairs, Barbados has taken loans totalling ~USD1.8 billion since 2018, which included funds for water infrastructure development, COVID-19 response, public transport, tourism sector development, social development and economic recovery.

Challenge 3: Absence of local appetite for climate finance.

In addition to barriers in accessing financing from MDBs, global financial architecture also plays a role in disadvantaging SIDS like Barbados' ability to access funds for climate-related projects. Local and regional banks find themselves overexposed to local customer risk and have little appetite to extend finance to small customers, especially in untraditional finance areas such as renewable energy projects. Furthermore, where lending for climate and mitigation projects does exist, this is limited to the most prominent commercial borrowers and there is little investment in developing products and standards that would allow finance to reach lower-income households.

Private financial institutions in Barbados are few with almost all of them being foreign owned resulting in a risk being evaluated from a distance. The more distant and theoretical risks are, the more uncertain and higher they appear. The result is a more risk-averse lending environment which has been reinforced by a substantial restructuring of Government debt in 2018, which resulted in banks head-officed in Barbados making significant losses. Consequently, Barbados has seen little appetite from financial institutions to go beyond the traditional bounds of personal mortgages, car loans and credit cards.

The reluctance of Barbados' institutions to support non-traditional funding practices includes its hesitancy towards public-private partnerships (PPPs), which are often useful in attracting private-sector funding to public initiatives like climate activities. This remains the case in Barbados even where certainty over public sector revenues exists. As a consequence, there are few model agreements and agreed technical specifications that can be applied to fund Barbados' green and climate-related infrastructure.

The BGB

In the face of fiscal, multi-lateral, and traditional lending barriers, new financial institutions such as the BGB are required to facilitate investment towards Barbados' climate action in a financially sustainable manner that does not add to the Government's existing fiscal burden. The BGB will act as a commercial lender responding to government incentives in climate adaptation and mitigation areas, while crowding in additional funding, including from private and multi-lateral sources. The Bank's lending practices and products, will demonstrate that lending to climate-related projects and low-income households can be profitable to draw in private lenders. The BGB's shareholders will mandate this focus and these priorities in a way that the management of the branches of foreign-owned banks does not; thereby mobilising Barbados' capital markets towards investment in the country's climate agenda while overcoming the triple barriers of the national debt, GDP per capita status and financial institutions.

The total initial capital requirements of the BGB are determined based on an assessment of the immediate and long-term needs of the government of Barbados. Initial equity of USD 23.25 million from multiple sources with an additional USD 6.75 million subordinated loan from GCF will cover the immediate investment needs of the BGB with USD 5

million for banking infrastructure and the remaining USD 25 million to be used for initial lending activities. This activity is forecast to increase to at least USD 5.8 billion over 15 years to meet Barbados' climate funding gap.

A 1:1 ratio of GCF to co-finance is expected in the initial investment with the Government of Barbados committing USD 10 million, other third-party investors at least another USD 5 million and the GCF matching these commitments at USD 15 million. This will ensure that the GoB and the GCF remain core shareholders in the BGB. Financing from the GCF will help the project overcome barriers to accessing concessional finance linked to the convention of focusing on levels of debt as a per cent of GDP as the basis of the IMF's debt sustainability limits or ratings from the credit rating agencies.

The revolving nature of the BGB will ensure its long-term operation, crowding in additional loan finance from private sources to expand the potential impact in the long term. It is anticipated that the GCF investment in the BGB will mobilise over USD 5.8 billion in additional finance, providing a ratio of leveraged finance of 7:1 against the initial GCF investment, including additional equity raised from third parties.

Through the GCF investment, the BGB will be the primary investment vehicle for climate projects, partnering with government programmes and the private sector to roll out substantial investments in adaptation and mitigation across Barbados, and potentially across the broader Caribbean region over time.

B.6. Exit strategy (max. 500 words, approximately 1 page)

In addition to supporting the establishment of the fund, the purpose of the GCF's investment is to enable the long-term sustainability of BGB. It is recognised that the institution's sustainability depends on the shareholding and governance structure, business model, and effectiveness and efficiency of its operations. The exit strategy for the proposed project has been considered from two main perspectives — ensuring the long-term sustainability of the BGB and creating an exit strategy for GCF's equity investment.

BGB Long-term Sustainability

It is envisioned that the BGB will persist beyond the lifespan of the initial investment period to become a self-sufficient and sustainable finance vehicle for Barbados in the long term. To this end, sustainability has been a central design consideration for the BGB, including in its governance structure and business model, as well as in the monitoring and accountability systems. Key design considerations for the longevity of BGB are discussed below.

Governance

The BGB will apply international best practices in terms of corporate governance, with all governance structures having been developed in line with the guidelines provided by the Central Bank of Barbados – the regulatory body responsible for issuing licenses to all banks in the country. These structures are designed to ensure long-term sustainability and transparency in the operation of the bank, and to effectively manage and mitigate risk. The Board, appointed by the common shareholders, and the executive management team, appointed by the Board, will be responsible for daily operations. The AE, on behalf of the GCF and the Government of Barbados, will be actively involved in monitoring the BGB's processes and outcomes to ensure shareholders' interests are represented. These structures will ensure the BGB is able to operate independently over the long term.

Business Model

The BGB will be a financier that arranges, structures, and lends to climate-responsive projects in Barbados. The need for such an institution was identified through a thorough market assessment and consultative process, detailed in Annex 2 and 7, respectively. Lending will be wholesale in nature —not lending directly to the public but rather partnering with a range of intermediaries, such as the HOPE housing agency, to deliver impactful projects that maximise impact while limiting operational and credit risks. BGB will earn income through fees and net interest, with the income potential being boosted by a lower cost of capital resulting from its focus on climate- and gender-responsive projects targeting low-income households, which will allow the bank to capture funding from impact investors and MDB funding. Moreover, BGB will operate as a digital bank, ensuring that the operational costs will be modest. This combination of cheap financing, low costs, a range of fees, and net interest income will secure financial sustainability and strength.

The longevity of BGB has also been considered in the bank's reinvestment model. The Board will be responsible for distributing dividends for common shares based on financial performance, with all undistributed profits being re-

invested into the capital of the BGB. All repayments and repurchases of equity investment will be in the same currency as the initial investment.

The BGB programme considers the bank's role as a conduit between local financial institutions and the Government of Barbados, allowing the BGB to identify market opportunities and provide feedback and data to inform policy decisions more effectively. By showing that climate investments are bankable and profitable, disseminating market knowledge, and encouraging best practices, the BGB will close the existing climate finance gap — not merely as a tool for deploying capital, but playing a crucial role in changing market behaviour, educating key stakeholders, and informing future policy goals. This will ensure the continuation of the program's impact beyond the life of the GCF's involvement.

Monitoring and Accountability

Through the active involvement of the Government of Barbados and the AE, the BGB will ensure that policy and investment objectives are appropriately translated into investment, risk and monitoring frameworks. BGB shall monitor the compliance of the agencies and financing partners with the eligibility criteria and programme requirements of BGB and its obligations. A Reporting Framework will be in place and transmitted to all partners. Formalisation of the monitoring and accountability structures into the operational systems of BGB during the initial phases of the programme inception will ensure they continue beyond the formal monitoring period of the proposed GCF programme. Details on the monitoring and evaluation framework are provided in Annex 11.

GCF Exit Strategy

At the end of the programme lifecycle, the GCF will have the option to exit its investment in the BGB by selling its equity to other interested parties. The BGB management team and the AE will provide assistance to GCF in disposing of their interest, particularly in terms of identifying and engaging third parties. The Shareholder's Agreement — which will be finalised under Output 1.1 — will include a provision for GCF to exit its position after 15 years.

C. FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	15.5			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter</u> %		
(ii) Subordinated loans	<u>Enter amount</u>			<u>Enter</u> %		
(iii) Equity	15.5			Sustainable target RoE of 6%+		
(iv) Guarantees	<u>Enter amount</u>	<u>Enter years</u>				
(v) Reimbursable grants	<u>Enter amount</u>					
(vi) Grants	<u>Enter amount</u>					
(vii) Results-based payments	<u>Enter amount</u>					
(b) Co-financing information	Total amount			Currency		
	15			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Government of Barbados	<u>Equity</u>	<u>10</u>	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter</u> %	<u>Options</u>
Third Party Shareholders	<u>Equity</u>	<u>5</u>	<u>million USD (\$)</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter</u> %	<u>Options</u>
Click here to enter text.	<u>Options</u>	<u>Enter amount</u>	<u>Options</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter</u> %	<u>Options</u>
Click here to enter text.	<u>Options</u>	<u>Enter amount</u>	<u>Options</u>	<u>Enter years</u> <u>Enter years</u>	<u>Enter</u> %	<u>Options</u>
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	30.5			<u>million USD (\$)</u>		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	In addition to the GCF and GoB equity contributions listed above, the BGB is seeking to raise between US\$5 million and US\$20 million in additional equity from third-party investors, ideally targeting US\$15 million. Several potential investors have already been identified and indicated interest, with discussions underway to finalise contributions. These contributions will increase the total equity to between US\$30 million and US\$50 million. Following the initial equity contribution, the BGB will raise additional debt finance from various sources, targeting a debt-to-equity ratio not exceeding 7:1. The BGB's own balance sheet lending, combined with the crowding in of private investment through the issuance of green bonds by the BGB and other instruments, will facilitate the mobilization of climate finance of an average of \$342 million per annum over the course of 15 years.					

C.2. Financing by component

Please provide an estimate of the total cost per component and output as outlined in section B.3. above and disaggregate by source of financing. More than one co-financing institution can fund a single component or output. Provide the summarised cost estimates in the table below and the detailed budget plan as annex 4.

Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
Outcome 1: Innovative financial structures operating in Barbados to overcome barriers to climate finance.	1.1: Blue Green Bank (BGB) established	<u>5.5</u>	<u>2.16</u>	<u>Equity</u>	<u>3.34</u>	<u>Equity</u>	<u>GoB</u> <u>3rd Party investors</u>
Outcome 2: Affordable finance mobilised for investment in priority government initiatives targeting low-carbon and climate-resilient development in Barbados.	2.1: Blue-green investments identified and executed in Barbados	<u>25</u>	<u>13.34</u>	<u>Equity</u>	<u>7.66</u> <u>4</u>	<u>Equity</u> <u>Equity</u>	<u>GoB</u> <u>3rd Party investors</u>
Indicative total cost (million USD)		<u>30.5</u>	<u>15.5</u>		<u>15</u>		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☐ No ☒

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☐ No ☒

If the project/programme is expected to support capacity building and technology development/transfer, please provide a brief description of these activities and quantify the total requested GCF funding amount for these activities, to the extent possible.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

The BGB is expected to yield considerable long-term positive impact, both in terms of mitigation and adaptation — with many of the pipeline investments providing cross-cutting benefits. The adaptation impacts are expected to predominantly benefit vulnerable groups who otherwise lack access to the funding required to adapt to climate change — particularly in terms of access to hurricane-resilient housing as well as participation in water conservation initiatives. Given the nature of the bank, the initial pipeline of investments is indicative and will likely expand even within the GCF project lifespan — resulting in what can be considered conservative estimates of the bank's impact presented below. While the potential benefits of the initial equity investments alone will be considerable, the revolving nature of the bank, combined with the potential to crowd in significant private investment to drive transformational projects and the institution's indefinite lifetime, means that the ultimate impact potential of BGB could be significantly higher than these conservative estimates. The role of the BGB as a 'dealmaker' within the existing financial landscape is essential to fill a gap in the financial market for the extensive range of potential climate projects that are outside the scope of public-sector budgets but require de-risking/securitization to attract private investment. Without this extra layer of support, private investment will continue to focus predominantly on collateralized loans targeting high-income groups and large businesses, excluding the considerable investment potential from lower-middle income groups and small businesses that can be unlocked through the right packaged deals.

Specific details of the mitigation and adaptation impact potentials are presented below, along with the potential for the BGB to contribute to a green recovery from the COVID-19 pandemic.

Mitigation Potential

The initial BGB pipeline will finance substantial investment in rooftop PV, which will contribute to Barbados' ambitious mitigation goals. Specifically, through the HOPE and SHIFT programmes, BGB finance is expected to increase renewable energy generation by 914 MW – exceeding the minimum 100MW of rooftop PV generation capacity needed (as identified in Barbados' IRRP). This will contribute directly to the GCF mitigation result area (MRA 1), with an expected emissions reduction over 25 years of 16,736,509 tCO₂e. Beyond this initial mitigation potential, the long-term finance mobilised by the GCF is expected to further contribute to GoB's mitigation ambitions through potential investments in vehicle-to-grid charging infrastructure as well as energy efficiency in homes, small businesses and tourism establishments.

Adaptation Potential

In addition to the mitigation potential, the projects supported by the BGB are expected to yield considerable adaptation benefits. Under the initial pipeline of projects, the adaptation benefits will focus primarily on hurricane-resilient housing and water security. Specifically, both the HOPE and SHIFT initiatives include a core focus on hurricane-proof roofs, with HOPE designing resilience into the construction of new homes and SHIFT retrofitting existing structures to be hurricane resilient. The decentralisation of energy generation through distributed rooftop PV will have additional adaptation benefits in relation to hurricanes³³, reducing reliance on vulnerable centralised systems – enabling communities to recover faster following an extreme event.

Water security will be increased under both the HOPE programme — through the water recycling initiatives being built into the housing estates — and the water conservation initiatives, which will promote water conservation practices and water harvesting measures at the household and small business levels. These

³³ Hurricanes can withstand Saffir-Simpson Category 3 hurricanes. Saffir-Simpson Category 3 hurricanes pack wind speeds of 130 mph (on a 3-second gust basis, with a return period of 50 years) and possess extensive damage potential.

water conservation and recycling practices will reduce pressure on already limited freshwater resources in Barbados, increasing overall water security.

The design of the BGB investment criteria is such that it is open to any climate-change-related hazard, provided a clear case can be made for the climate change attribution. While the initial pipeline focuses predominantly on hurricanes and water scarcity, there is potential for a broader range of climate change hazards to be included on a needs basis.

Beneficiaries

The project will support a total of 226,000 direct beneficiaries, of which at least 50% will be women. Beneficiaries include the most vulnerable groups in the population, such as 10,000 low-income households that will benefit from affordable climate-resilient, green homes, as well as 50,000 lower- to middle-income households that will be able to access rooftop solar and water conservation measures (**Error! Reference source not found.**, with an expanded table provided in Section 4.1 of Annex 22).

Table 5. Adaptation beneficiaries

Pipeline Initiative	Who Benefits	Number of beneficiaries	Assumptions
Direct			
HOPE Project's climate-resilient houses	Lower-income first-time homeowners and their immediate families.	50,000 people At least 50% women ³⁴	• 10,000 homes built. • 1-2 people per bedroom • 50% 3-bedroom houses • 50% 2-bedroom houses
SHIFT Project	Lower-income households receiving climate-resilient roofs and de-centralised power production.	176,000 people At least 50% women	• 50,000 roofs made resilient. • Average household size of 3.52 people³⁵ • 35,200 person overlap with Water conservation
Water Conservation	Households with increased access to water during dry periods and improved water-use efficiency.	70,400 At least 50% women	• 20,000 homes targeted for water conservation. • Average household size of 3.52 people • 35,200 person overlap with SHIFT
Indirect			
Water Conservation	Population benefiting from increased water availability resulting from water conservation initiatives	65,200	Introducing water conservation measures will reduce pressures on limited groundwater reserves, thereby increasing the availability of fresh water for domestic use.

Covid-19 Green Recovery

Establishing the BGB in Barbados will play a crucial role in supporting a green recovery from the COVID-19 pandemic. The bank will provide financing and investment opportunities for renewable energy and energy efficiency projects, helping to reduce the island's dependence on fossil fuels and mitigate the effects of climate change. This, in turn, will create new jobs and stimulate economic growth in the green energy sector. Additionally, the bank can leverage private sector capital to maximize its impact, making a green recovery in Barbados more accessible and achievable.

³⁴ Based on gender ration in Barbados (51.5% women)

³⁵ Based on 2010 census numbers, with a total population at the time of 277,821 and a total of 78,936 dwellings

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

The design of the BGB is based on the idea of transforming the climate finance sector in Barbados to facilitate a shift in development pathways towards a focus on climate resilience and low emissions. The intention of establishing a new green bank is not merely to add a new source of finance into the mix, but rather to create an institution that has the mandate and structures needed to innovate, package and securitise investments to transform the market and mobilise finance for climate action at scale. To this end, the role of the BGB as a funder and potentially as an arranger has been designed with the primary aim of securitising revenue streams and making projects bankable to stimulate significantly greater participation by the national capital market. An increase in the number of bankable projects will lead not only to significant climate impacts, but also provide a broader catalytic effect in a market that has historically proven risk-averse and resistant to innovation. The BGB's presence and dynamic role in the market will incentivise and create bankable opportunities for commercial banks to enlarge their green asset portfolios, as well as provide alternative asset classes such as green bonds and affordable green housing. Through its direct participation in the funding of Government-promoted climate change-related projects, alongside deploying its expertise to render projects bankable, the BGB — with its GoB shareholding — can be perceived as playing the part of a small national development bank, established to mobilize resources from third parties for the exclusive benefit of climate change mitigation and adaptation purposes.

Progressive access to affordable and sustainable green finance opportunities made possible through the intervention of the BGB will create a significant opportunity for investment in climate action. Support for the rollout of key national climate change strategies and programmes such as the Roofs to Reefs Programme will shift the development pathway of Barbados. By facilitating access to climate finance for lower- and middle-income households and small businesses, GCF's investment in the BGB will open up a significant new market. This will, in turn, reduce pressure on constrained public finances, and create the fiscal space for the GoB to direct limited concessional and public finance towards public good projects (particularly for adaptation) which are not inherently financially viable.

Potential for innovation, scaling up and replication

The establishment of the BGB will serve to crowd in additional debt and equity finance, thereby extending the available capital far beyond the bank's initial equity investment. Output 1.2 of the programme is specifically directed towards such fund-raising, through direct engagement and negotiation with potential investors and lenders, by creating the right tools and mechanisms for their participation. As a 'mission-driven' institution, BGB will endow Barbados with more effective and agile mechanisms to access finance at scale and for multiple projects. A programmatic approach of the kind foreseen for the BGB will replace the piecemeal, project-by-project approach which is ultimately unsustainable when looking at the ambition of Barbados' NDC.

In addition to the mobilization of new funding, the revolving nature of the BGB will ensure that the investment process is maintained beyond the scope of the GCF-supported programme, creating sustained opportunities to increase the number of bankable projects through the replication of viable financing models in other sectors. The scope for reinvestment is also strengthened through the project's operational and governance arrangements, with shareholder dividends being distributed based on financial performance and all additional profits being re-invested into the capital of the BGB. Moreover, the digital nature of the bank will ensure that operational costs are kept low, maximising the capital available for reinvestment and ensuring affordability while maintaining high standards of financial performance, management and transparency. The total investment potential of the bank over 15 years is anticipated to be ~USD 5.8 billion, enabling a 24-fold scaling of the initial investment.

A further transformative aspect of the BGB's operations is its potential to herald a more holistic approach to ESG. Given GCF's core involvement, BGB has been set up to abide by the GCF's stringent environmental and social standards, which when combined with the 'green' vocation of the bank, will bring additional value by delivering clear and tangible progress in the ESG space and act as a benchmark for the local banking sector.

While the Barbados climate change project market is deemed sufficient to occupy the operations of the BGB in the long term, opportunities also exist for extending the operations of the bank beyond the borders of Barbados to support similar initiatives across the broader Caribbean region. Additionally, the successful rollout of the BGB would serve as a best practice model for other small, middle-income states on the frontline of climate change, facing similar challenges and constraints. Through its central role in championing innovative climate solutions on the global stage²⁶, Barbados is well-positioned to demonstrate the viability of such a model to other developing states and support the dissemination of best practices.

D.3. Sustainable development (max. 500 words, approximately 1 page)

In addition to the primary impacts described in Section D.1 above, the BGB will directly contribute to several Sustainable Development Goals (SDGs) and is expected to yield several environmental, social and economic co-benefits. Details of these are provided below.

Table 6. Contribution of the project to the Sustainable Development Goals (SDGs).

Goal	Relevant Targets	Contribution
SDG 5 – Gender Equality	5.1: End all forms of discrimination against all women and girls everywhere.	The eligibility criteria and frameworks for BGB will be gender inclusive and ensure equal access to finance for women.
SDG 6 – Clean Water and Sanitation	6.1: By 2030, achieve universal and equitable access to safe and affordable drinking water for all. 6.4: By 2030, substantially increase water-use efficiency across all sectors and ensure sustainable withdrawals and supply of freshwater to address water scarcity and substantially reduce the number of people suffering from water scarcity.	The proposed project will promote water conservation and rainwater harvesting initiatives that will improve climate-resilient access to water for low-income households alongside improving water-use efficiency.
SDG 7 – Affordable and Clean Energy	7.1: By 2030, ensure universal access to affordable, reliable and modern energy services 7.2: By 2030, increase substantially the share of renewable energy in the global energy mix 7.3: By 2030, double the global rate of improvement in energy efficiency 7.a: By 2030, enhance international cooperation to facilitate access to clean energy research and technology, including renewable energy, energy efficiency and advanced and cleaner fossil-fuel technology, and promote investment in energy infrastructure and clean energy technology	The proposed project will establish sustainable and affordable financial mechanisms to mobilise finance for the uptake of renewable energy and energy-efficient technologies.
SDG 11 – Sustainable Cities and Communities	11.1: By 2030, ensure access for all to adequate, safe and affordable housing and basic services and upgrade slums 11.5: By 2030, significantly reduce the number of deaths and the number of people affected and substantially decrease the direct economic losses relative to global	The BGB will fund the development of 10,000 affordable green homes, as well as funding the uptake of climate-resilient roofs that will reduce

	gross domestic product caused by disasters, including water-related disasters, with a focus on protecting the poor and people in vulnerable situations	the impact of hurricanes ³⁶ on vulnerable households.
SDG 13 – Climate Action	13.1: Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries 13.3: Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning	The BGB will fund green projects that will improve the climate resilience of vulnerable communities in Barbados. The BGB will also serve as a conduit between local financial institutions and the Government of Barbados, allowing the BGB to identify market opportunities and provide feedback and data to inform policy decisions more effectively. By showing that climate investments are bankable and profitable, disseminating market knowledge, and encouraging best practices, the BGB will close the existing climate finance gap.

Environmental co-benefits

- *Improved air quality:* Supporting the transition from fossil fuel-based to renewable energy production through rooftop PV will reduce air pollution — including particulate matter, nitrogen oxides, and sulphur dioxide, which can cause respiratory problems and contribute to acid rain.
- *Reduced water usage:* Fossil fuel-based energy production requires large amounts of water for cooling and other processes, whereas solar energy sources do not consume water. Therefore, by supporting the energy transition, the project will further contribute to water conservation efforts.
- *Improved groundwater reserves and quality:* Water conservation initiatives will reduce demand on and extraction of limited groundwater resources, increasing the availability and quality of water for terrestrial and freshwater ecosystems. This will have positive impacts on biodiversity. Moreover, increased groundwater reserves will improve the hydraulic pressure of the freshwater lens, helping maintain the equilibrium between fresh, brackish and salt water — thereby reducing the threat of saltwater intrusion linked to sea-level rise.

Social and Economic co-benefits

- *Living standards:* Provision of affordable green housing will help improve the lives of low-income households, many of whom are currently paying more for rent in sub-standard housing than for a mortgage which could be provided under the HOPE programme.
- *Health:* Improved air quality resulting from reductions in fossil fuel use will have a positive impact on human health, which in turn will have a positive knock-on effect on public health systems. Improved water security through water conservation and investment in adapting waste-water systems will further reduce potential health impacts of CC, including through the spread of waterborne diseases, especially after heavy rainfall events.
- *Food security:* Eligible projects for the BGB will influence food security in several ways. First, improved water security provided through the proposed water conservation and recycling initiatives under the first phase of the project pipeline will increase water for agriculture, supporting local food production. This includes direct supply for irrigation, as well as indirect supply through reduced pressures on limited groundwater reserves. Potential projects under the second phase pipeline could include additional support to promote resilient agricultural practices and reduce imports through the promotion of locally produced foods.
- *Job Creation:* The implementation of projects financed by BGB will create employment opportunities for Bajans, while also supporting local contractors and suppliers. Programmes such as HOPE have

³⁶ Hurricanes can withstand Saffir-Simpson Category 3 hurricanes. Saffir-Simpson Category 3 hurricanes pack wind speeds of 130 mph (on a 3-second gust basis, with a return period of 50 years) and possess extensive damage potential.

significant potential to strengthen employment in the construction sector, including by creating opportunities for upskilling young artisans. Moreover, investments directed to small businesses in the tourism industry will contribute to job security and creates employment opportunities that can help lift vulnerable populations out of poverty.

- *Financial security:* Access to affordable finance will reduce debt burdens on low- and middle-income households who otherwise rely on costly short-term finance to adopt resilience measures or low-emissions technologies.
- *Reduced economic downtime:* Building the resilience of the population will reduce economic downtime which typically follows extreme events that disrupt transportation networks, damage power and water infrastructure, increase maintenance efforts at the household level, as well as disrupting the flow of tourists into the country. Reducing this downtime will have a notable benefit to the broader economy.
- *Energy transition:* Decentralised energy production and private investment reduce financial pressure on the government for energy production and the transition to renewables.

Gender-sensitive development

The BGB will promote gender parity in the investment approach, incorporating gender mainstreaming into its operations, governance structure and investment eligibility criteria, with the goal of advancing equality and fighting discrimination. The BGB's Board, staff, trainees and loan applications will have a reservation of 30%–50% for women. Moreover, improved access to affordable energy is an essential prerequisite to achieving economic growth and poverty reduction. Increased access to non-polluting power for lighting, cooking, and other household and productive purposes can have dramatic effects on women's levels of empowerment, education, literacy, nutrition, health, economic opportunities, and involvement in community activities. These improvements in women's lives can, in turn, have significant beneficial consequences for their families and communities.³⁷

D.4. Needs of recipient (max. 500 words, approximately 1 page)

Vulnerability

Barbados, as a SIDS, is highly exposed to the impacts of climate change. Several climate risks constitute significant threats to the country. The North and East coasts of the island are particularly exposed to strong wave action and ocean currents. Tropical storms and hurricanes also have considerable impacts on livelihoods and infrastructure. Flooding, strong winds and storm surges³⁸ are common, with historical data showing that major flooding events occur every 12 years on average. These phenomena can be extremely problematic since most of the critical infrastructure and economic activity are in the coastal zone. For instance, the Deepwater Harbour — which facilitated 80% of the country's imports in 2018 — is a key risk area due to its vulnerability to coastal hazards. It also handles the largest number of tourists per year in the Caribbean (800,000 in 2018).

It is also worth mentioning that rising sea surface temperatures will have adverse effects on coral reefs, causing them to bleach. For Barbados, where foreign exchange earnings are heavily dependent on tourism services, such an occurrence will seriously affect the product offerings of the country. Moreover, the loss of coral reefs will have a detrimental impact on the natural defences against storm surge during a hurricane.

The country is also facing serious water issues due to a combination of droughts, floods and rainfall variability — exacerbating the challenges of already scarce freshwater resources. Sea level rise and saltwater intrusion will also increase salinity, thus affecting the country's water resources and intruding into aquifers. Changes in soil quality are also to be expected with indirect effects on nutrition and well-being due to the impact on agriculture and water resources. The existing water infrastructure is also at risk of destruction from extreme climate events, which creates a lot of uncertainty around water supply. Water scarcity has historically been a major concern in Barbados, with the country ranked 15th in the world. Annual freshwater withdrawals peaked

³⁷ <https://www.undp.org/sites/g/files/zskgke326/files/publications/genderengtoolkit.pdf>

³⁸ an abnormal rise of water generated by a storm

at over 100% in the late 1990s and early 2000s, according to the World Bank. Addressing infrastructure security and making sure it becomes climate-resilient is absolutely fundamental to the country's future.

Economic and social development

Barbados has managed to transform itself from a low-income economy reliant almost solely on sugar production to having one of the highest per capita incomes in the Caribbean. With a GDP per capita PPP of \$13,487 in 2021, Barbados has managed to diversify its economy to become services-driven and rely more on tourism as well as financial services. This reliance on tourism was a major factor behind the considerable negative economic effects of the Covid-19 pandemic, with a 14% contraction of GDP in 2020. However, the Barbadian economy has begun to show signs of recovery, with a GDP growth registered at 11.8% in the first quarter of 2022. The tourism sector has been an important driver of this growth.

The country's economy is marked by a huge amount of public debt. Typically, a nation saddled with debt means that it will have less to invest in its own future. However, efforts by the GoB to manage this debt means that the public debt has been gradually decreasing over the past several years – going from 157% of GDP in 2018 to 131.9% in 2022. It has a debt-to-GDP target of 60% by the 2035/36 fiscal year (Economic Survey of Latin America and the Caribbean, 2022).

The HDI, which measures the country's average achievements in three basic aspects of human development: health, knowledge and standard of living, is relatively high: 0.790 - ranked 70th globally and 9th in the LAC region, but drops 16.8% when adjusted for inequality. The female HDI (0.799) is higher than the male HDI (0.773), creating a GDI of 1.034 – the highest in the LAC region and among the highest in the world.

Multi-dimensional poverty is low, with 2.5% of the population being multidimensionally poor – the equivalent of 7000 people in 2020 (Multidimensional Poverty Index, 2022). There is a narrow job market and a lack of job opportunities with an unemployment rate gradually increasing since 2018, reaching 10.41% in 2021.

Alternative sources of financing

As mentioned above, Barbados has a considerable amount of public debt, which significantly reduces its field of action when it comes to addressing climate mitigation and adaptation issues. Furthermore, given that its income status – which is skewed by the small size of the economy and extensive inequality in the distribution of wealth and assets — Barbados does not meet the eligibility criteria for concessional loans from many lenders, such as Multilateral Development Banks (MDBs). It is also worth mentioning that the local capital market has a fairly short history for climate investment with very few local private financial institutions – the rest of them being foreign-owned. These banks find themselves exposed to local risk and are often reluctant to invest in or lend money to small customers - especially in untraditional, risky or low-profit finance areas such as climate projects.

As a result, available finance is directed to high-income households and focused on collateralised loans, with limited opportunity for most vulnerable groups. Loans outside of the primary offerings from banks are generally extremely high interest and short tenors, creating a barrier to investments in resilience. There is therefore a clear and substantial finance gap that urgently needs to be filled for Barbados to meet its ambitious climate objectives and build a sustainable, climate-resilient future for its people.

D.5. Country ownership (max. 500 words, approximately 1 page)

The establishment of the BGB has been strongly country driven from its inception – holding support from the highest levels of government. The concept of a green bank to transform climate finance in Barbados was first proposed by Prime Minister Mia Mottley — born out of an ambition to help Barbados and other Caribbean countries meet the targets of their NDCs. To initiate the process, the GoB invited a delegation of GCF officials — including the Director of the GCF's Private Sector Facility, Kavita Sinha, and the Fund's Investment Officer, Tom Bishop — to Barbados for a three-day working visit, during which the green bank with the aim of becoming a major institution for financing climate action in the region was proposed and discussed. Since this inception, the GoB has remained closely involved in the development process, taking full ownership of

the programme. To further promote country ownership, the GoB has ensured full engagement with key stakeholders at all stages of development — from the initial workshops with the GCF which were attended by numerous financial and development finance agencies as well as other key players in the renewable energy sector, through to in-depth consultations during the in-country mission by the project development team. This engagement is considered critical, as the success of the bank will be enhanced by developing strong partnerships with the financial sector and potential project intermediaries. Details on the stakeholder engagement process are presented in Annex 7, along with a detailed stakeholder engagement plan to ensure continued engagement throughout the implementation. The project development has also been closely monitored by the NDA — in the Ministry of Finance, Economic Affairs and Investment's Economic Affairs and Investment Division — to ensure that it aligns with the country's climate action priorities and economic strategies. Moreover, because the Government of Barbados will have a vested interest in the bank's success as a junior shareholder, the investment criteria have been designed to support Barbados' NDC priorities ensuring that capital is channelled toward key sectoral priorities and opportunities.

Relevance to and alignment with existing policies

The approach of the BGB, along with its proposed pipeline and project eligibility criteria are all aligned with the policy and planning landscape in Barbados. This alignment has been a core consideration throughout the design process, with the BGB being specifically positioned to fill a niche within the country's resilience and energy transition strategies. Key policies considered in the project are summarised below, with a more comprehensive list provided in Section 3.2 of Annex 2: Feasibility Study.

Table 7. Alignment of the BGB with key national policies, plans and strategies

Sector	Policy or Strategy	Alignment
Climate and sustainable development	Nationally Determined Contribution (NDC) - Update July 2021	A core focus of the BGB pipeline is the potential to contribute to Barbados' NDCs. In particular, the BGB will target contributions towards a 95% share of renewable energy in the electricity mix, and a 20% increase in energy efficiency across all sectors as compared to BAU.
	National Climate Change Policy (NCCP), 2012	The policy aims to establish a national process for adaptation to climate change, reduce the country's greenhouse gas emissions, design appropriate mechanisms to respond to climate change, and move toward renewable energy. The BGB, as a government-driven initiative, has been initiated to support the government in the implementation of the mechanisms proposed under the NCCP.
	Roofs to Reefs Programme (R2RP)	The R2RP is the core resilience strategy for Barbados and provides the vehicle through which public investment for adaptation will be directed. The BGB pipeline is fully aligned with the ambitions of R2RP and has been positioned to fill a specific niche in terms of financing core initiatives that have the potential to yield financial returns. This includes the priority objective of making low- and middle-income homes more resilient to extreme weather events and their impacts, such as possible loss of access to electricity and potable water distribution systems; and to increase freshwater storage capacity and water use efficiency and reduce emissions through the deployment of distributed renewable energy generation.
	Barbados' Sustainable Development Policy	The over-arching goal of this policy is to ensure the optimisation of the quality of life for every person by ensuring that economic growth and development do not occur to the detriment of our ecological capital.
Energy	Integrated Resource and Resiliency Plan for Barbados, 2021	The IRRP provides a roadmap and action plan for the energy transition, which includes the need for utility-scale solar (180MW by 2030), along with rooftop solar (100MW by 2030) and solar thermal energy (60MW by 2030). The plan also identifies a number of pilot projects as a key area for budget allocation, including li-ion batteries, solar PV, V2G technology embedded electric vehicle chargers, and high-power DC vehicle chargers. The BGB will provide finance to contribute to these targets, particularly for the rooftop solar components.

	Barbados Feed-in Tariffs for Renewable Energy, including Accelerated Investment Plan	The BGB will leverage the Feed-in tariff (FIT) programme to establish innovative models, such as those under HOPE and SHIFT to finance the energy transition while simultaneously building climate resilience. The pre-existing RER terms include the \$0.416/kWh rate for solar photovoltaic and the \$0.315/kWh rate for wind.
Water	Barbados Water Conservation Plans (WCPs)	The Barbados Water Conservation Plan is a comprehensive strategy for managing the country's water resources sustainably. The plan recognizes the importance of water as a finite and valuable resource and seeks to ensure its sustainable use and protection for present and future generations. The goal of the Water Conservation Plan is to ensure the long-term availability and quality of water for all the people of Barbados, while protecting the country's natural water resources for future generations. The water conservation projects supported by BGB will align with the priorities identified in the WCPs
	National Water Reuse Policy	This policy examines current water demand and reuse, challenges and policy and legislative instruments in use. It provides a proposed policy to address critical issues within the context of national and international policy instruments and strategic plans.
	Revised Stormwater Management Plan	Stormwater management systems proposed under the Hope initiative are designed in line with the provisions of the national Stormwater Management Plan.
Development Planning	National Strategic Plan of Barbados 2006-2025	The objectives of BGB are aligned with Goal 4 of the National Strategic Plan of Barbados: <i>Building a green economy – strengthening the physical infrastructure and preserving the environment</i> . In particular, the project aligns with three of the core objectives under Goal 4: 1. To promote and facilitate the sustainable use of our renewable resources and the wise management of our non-renewable natural resources. 2. To maintain a safe and reliable water supply. 3. To ensure an efficient and reliable energy sector
	Barbados Physical Development Plan (PDP) 2021	The 2021 Physical Development Plan is based on a vision of sustainable growth and development of Barbados. It addresses the critical impacts of climate change on Barbados through policies and strategies that enable the people to thrive and remain resilient under changing climate conditions. The alignment with the PDP is linked to the R2Rp, which operationalises the objectives of the PDP.
	Economic Recovery and Transformation (BERT) plan	BERT 2022 succeeds BERT 2018 and lays out the GoB's economic and financial programme for the next four years, focussing on achieving inclusive and sustainable growth, while maintaining fiscal and debt sustainability. Pillar 2 of BERT's growth strategy incentivises low- and middle-income housing, with a specific focus on resilient housing stock through the HOPE initiative.

D.6. Efficiency and effectiveness (max` . 500 words, approximately 1 page)

In the context of National fiscal constraints, the design of the BGB as a green merchant bank provides both an expedient as well as financially sustainable mechanism for financing Barbados' climate response activities.

Efficiency

The institutional design and governance structures of the BGB provide a managed vehicle with well-defined mandates that can facilitate risk-taking investments in the form of equity to serve as base capital for the bank. The Government of Barbados has committed equity of up to \$10 million. Discussions are also currently underway with third parties to secure additional equity commitments, including with USAID – from whom the project anticipates receiving US\$5 million. Other equity commitments are being sought from foundations, high-net-worth individuals (HNWIs) and other institutions. This initial investment will be pooled along with the GCFs investment of US\$15 million with the intention of using \$5 million for the BGB's infrastructure and the remaining US\$30 million to finance lending activities.

As a well-governed, equity-backed institution, the BGB's design provides a risk-managed entity that can attract additional capital from private and public sources, while maintaining its ability to respond to Barbados' climate needs. This is enabled by the BGB's ability to play host to financing instruments and mechanism, such as green bonds and structure finance, that is envisioned to unlock an additional \$5.8 billion in incremental financing. This not only enables Barbados to unlock additional funding without burdening the national finances but provides a level of concessionality that enables the realisation of Barbados' climate response initiatives. To this end, the project is targeting a specific gap in the market for climate finance. While many initiatives are already underway, these still do not approach the level of investment required to meet the adaptation and mitigation needs of Barbados. It is therefore deemed essential that private finance be mobilized to increase investments to enable the government to focus the limited funds available through public budgets and international finance to be directed towards public-good projects. For this reason, the BGB will specifically not seek to target public-good projects, or those that are financeable through traditional commercial options, ensuring that the bank does not crowd out investment from other sources but rather fills a specific niche that currently lacks finance.

Financial viability

While the BGB will provide access to adequate financing and at the requisite level of concessionality, the Banks' lending activities are intended to be financially sustainable. All BGB's on-lending activities will be profit-generating with the appropriate measures taken to ensure estimated and actual credit losses are kept to a minimum. The BGB will also operate as a digital bank ensuring that not only are operation expenditures kept to a minimum, but the bank remains agile and responsive to the needs of Barbados. Robust governance and risk management structures will oversee the bank's financial sustainability and ensure stringent regulatory compliance.

Application of Best Practices

The proposed programme has been designed with a specific focus on best practices, both in terms of the design of BGB itself, as well as the pipeline of climate-responsive projects that the bank will finance. A summary of the key considerations is provided below, with a more detailed analysis provided in Annex 2: Feasibility Study.

Design of the BGB

- A market assessment was conducted for the Barbadian financial sector, which identified gaps in the current market that could be filled by BGB to ensure that the new bank complements the existing structures, rather than competing with and potentially crowding out commercial banks.
- An options analysis was conducted to identify the preferable financial mechanism under which the programme should operate. The analysis considered six potential financial mechanisms, namely: i) a green bank; ii) a public-private partnership (PPP); iii) a credit/guarantee facility; iv) a development fund; v) concessional and private equity funds; and vi) a government agency. These mechanisms were assessed against five criteria categories (mandate, function, financing, governance and others) and while each mechanism aligned with some of the criteria, the green bank was the only option that suited all the criteria. Based on this evaluation, the structures and framework of a bank were deemed best suited to facilitate the financial disbursement of funds required for the implementation of blue-green projects in Barbados. A secondary option analysis was then conducted to assess options for the type of bank, including: i) wholesale/investment; ii) commercials and retail; iv) cooperative; and v) credit union. Through this assessment, a wholesale bank was deemed best suited to the objectives of the BGB.
- The governance and operational structures of the bank were compiled following an in-depth review of the relevant regulatory frameworks – as set out by the Central Bank of Barbados – and international best practice to ensure that BGB not only applies to the applicable legal and compliance requirements (as set out in Annex 9), but also ensures that each section/department within BGB exercises its mandate and operates with an aligned vision and ultimately strives towards a common vision and purpose. This

includes consideration of international best practices on the transparency of performance and financial statements.

Project Pipeline

- Detailed investment eligibility criteria have been developed for the BGB to ensure that all supported projects align with the strategic objectives of the BGB and its core shareholders — particularly the GoB and GCF. These eligibility criteria were developed by assessing the mandate and requirements of the BGB s well as best practice standards. The eligibility of initiatives to be funded through the BGB will be determined based on four criteria:
 - *Climate basis*: how does the initiative address a climate change problem, considering both adaptation and mitigation benefits?
 - *Financial feasibility*: is the proposed initiative financially sound, with suitable risk mitigation and potential return on investment to be considered bankable?
 - *Environmental and Social Safeguards*: does the proposed initiative align with the bank's environmental and social safeguard policies and management systems?
 - *Gender Responsiveness*: Is the proposed initiative gender-responsive, equitable and aligned with the bank's gender action plan?
- The governance structures of the BGB includes an investment committee that will be responsible for conducting necessary due diligence on projects to ensure that they are aligned with the BGB's eligibility criteria, and that best practice has been applied in the design of all interventions.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

- ☒ Reduced emissions (mitigation)
- ☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	While there has been an increase in the issuance of green bonds in Barbados in recent years, the capital market for climate investments remains limited. These limitations on the capital market are largely driven by the small number of participants in the market, relying largely on a relatively small banking sector that exhibits considerable market overlap and limited portfolio diversification. Experience to date has demonstrated that the absence of such a market player has left banking liquidity underutilised and market innovation largely stifled (see 'Market Opportunity' under Section B.2). The low participation of the Barbadian banking sector in climate finance can also be ascribed to the historical absence of a broad, coherent package of measures driven by the government to introduce the necessary reforms. As a result of these baseline factors, there is currently a considerable finance shortfall to meet the needs of the country. For example, the government has estimated that Barbados	<u>Low</u>	Progressive access to affordable and sustainable green finance opportunities made possible through the intervention of the BGB will create a significant opportunity for investment in climate action. The BGB will provide opportunities for existing financial institutions and capital markets to mobilise funds, including through conventional bank lending and alternative securitisation of revenue streams deriving from climate change mitigation and adaptation activities, thereby creating stronger and more diversified capital markets. As a partnership which brings together public (Government of Barbados), private and multilateral institutions at the level of equity, the BGB will present a holistic view of the market and the potential for green financing by ensuring that new projects align with Barbados' NDC targets.	The proposed project will establish a sustainable climate finance vehicle — the Blue Green Bank (BGB) — in Barbados to stimulate innovation in the financial sector with the goal of mobilizing the securities and loan markets in order to unlock additional funding for climate-responsive initiatives. The BGB will be a joint public-private sector effort to create an improved financing infrastructure that works closely with other financing players — such as commercial banks, credit unions, pension funds, and insurance companies — to crowd in private capital alongside public capital, rather than crowding out the significant investment opportunities offered by these key financial institutions. By facilitating access to climate finance for lower- and middle-income households and small businesses, GCF's investment in the BGB will open up a significant new market. The establishment of the BGB will serve to crowd in additional debt and equity finance for climate investments, making available extensive capital far beyond the bank's initial equity investment. As a 'mission-driven' institution, BGB will endow Barbados with more effective and agile

	needs at least \$2 billion (40% of GDP) to achieve climate resilience.			mechanisms to access finance at scale and for multiple projects. A programmatic approach of the kind foreseen for the BGB will replace the piecemeal, project-by-project approach which is ultimately unsustainable when looking at the ambition of Barbados NDC.
Replicability	Given the barriers to financing climate-responsive initiatives in Barbados, investments are generally made following a piecemeal, project by project approach which is ultimately unsustainable when looking at the ambition of Barbados NDC. Although notable progress has been made towards Barbados' climate ambitions, without transformative actions in the finance sector, there will continue to be a finance gap preventing the replication of climate initiatives across the country and region.	<u>Low</u>	While the Barbados climate change project market is deemed sufficient to occupy the operations of the BGB in the long term, opportunities also exist for extending the operations of the bank beyond the borders of Barbados to support similar initiatives across the broader Caribbean region. Additionally, the successful rollout of the BGB would serve as a best practice model for other small, middle-income states on the frontline of climate change, facing similar challenges and constraints. Through its central role in championing innovative climate solutions on the global stage, Barbados is well-positioned to demonstrate the viability of such a model to other developing states and support the dissemination of best practices.	Once established, the BGB will engage with investors, including private investment firms, development banks, philanthropic organisations and other sources of finance to continue to capitalise the bank over the long term. Output 1.2 of the programme is specifically directed towards fund-raising, through direct engagement and negotiation with potential investors and lenders, by creating the right tools and mechanisms for their participation. The bank will target at least a 7:1 debt-finance to equity ratio, aiming to mobilise at least USD 5.8 billion over the next 15 years. By focusing on climate resilience and adaptation for middle to low-income households — including a specific focus on women and other highly vulnerable groups — the BGB would aim to attract more concessional or philanthropic providers of capital, expanding the goals of climate action in Barbados without the need for further injections of money by the Government and GCF. This will create long-term scalability in the banking sector, with the potential to scale beyond Barbados into the broader Caribbean region over time (a possible outcome, although not directly intended at the inception of the BGB).
Sustainability	The generally risk-averse nature of the commercial banking sector in Barbados and the resultant focus only on collateralised loans generally excludes the lower- to middle-income groups from accessing adequate finance for climate action. This challenge is exacerbated by	<u>Low</u>	The BGB is designed to become a significant institution for practical and strategic financing of climate change adaptation and mitigation policies and measures, supporting the Governments of Barbados to achieve national resilience and emissions reduction targets. As a	The BGB governance structures are designed to ensure the long-term operation of the bank — beyond the scope of the GCF-supported initiative. The scope for reinvestment is strengthened through the project's operational and governance arrangements, with shareholder dividends being distributed based on financial performance and all

	the limited public resources to support these groups through government programmes. The resultant under-representation of this portion of the population in terms of sizable investments in climate action reduces the overall sustainability of development in the country. While the policy and planning environment in the country actively targets inclusive sustainable action, the absence of innovative financial options stifles sustainable development actions.	partnership which brings together public (Government of Barbados), private and multilateral institutions at the level of equity, the BGB can present a holistic view of the market and the potential for green financing by ensuring that new projects align with Barbados' NDC targets. The revolving nature of the BGB will ensure that the investment process is maintained beyond the scope of the GCF-supported programme, creating sustained opportunities to increase the number of bankable projects through the replication of viable financing models in other sectors.	additional profits being re-invested into the capital of the BGB. Moreover, the digital nature of the bank will ensure that operational costs are kept low, maximising the capital available for reinvestment and ensuring affordability while maintaining high standards of financial performance, management and transparency.
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E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ³⁹	
<u>MRA1 Energy generation and access</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	- Electricity production - Data collected by official representatives who manage household meters (i.e., Barbados Light and Power). - Implementation reports and calculations on the estimated energy produced.	2023 baseline: 721,061 tCO ₂ e	1,252,280 tCO ₂ e reduced emissions after 8 years.	16,736,509 tCO ₂ e reduced emissions after 25 years (lifetime) 6,946,692 tCO ₂ e reduced emissions after 15 years (implementation)	2006 IPCC Guidelines for National Greenhouse Gas Inventories - IRRP energy demand forecast
	<u>Supplementary 1.3: Installed renewable energy capacity</u>	- Site visits to capture data on the number of solar panels per roof and the wattage of panels installed. - Implementation reports. Number of panels installed, the capacity of panels etc.	2023: 44 MW	Additional RE Generation Capacity: 426 MW (8 years)	Additional RE Generation Capacity: 914 MW (15-year implementation)	According to BLPC and sources contained within the Feasibility Study, renewable energy in Barbados is generated only from solar PV facilities (independently and BLPC owned).
	<u>Supplementary 1.4: Renewable energy generated</u>	- Site visits to capture data on the number of solar panels per roof and the wattage of panels installed. - Implementation reports. Number of panels installed, the capacity of panels etc.	2023: 68,658 MWh	Additional RE Generation Capacity: 922,793 MWh (8 year)	Additional RE Generation Capacity: 25,756,583 MWh (20-year lifetime) 10,690,583 MWh (15-year implementation)	According to BLPC and sources contained within the Feasibility Study, renewable energy in Barbados is generated only from solar PV facilities (independently and BLPC owned).

³⁹ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

<u>Core 2: Direct and indirect beneficiaries reached</u>	<u>Total</u>	• Housing surveys • Water Conservation project records HOPE and SHIFT implementation records	0	Direct: 113,360 56,680 Female 56,680 Male Indirect: 32,500 16,250 Female 16,250 Male	Direct: 226,000 113,000 Female 113,000 Male Indirect: 65,200 35,200 Female 35,200 Male	• Combination of water security and infrastructure resilience beneficiaries. • Water security beneficiaries are a subset of the infrastructure resilience beneficiaries.
	<u>ARA1</u>	• Housing surveys • Water Conservation project records	0	Direct: 35,000 17,500 Female 17,500 Male Indirect: 32,500 16,250 Female 16,250 Male	Direct: 70,400 35,200 Female 35,200 Male Indirect: 65,200 35,200 Female 35,200 Male	• 20,000 homes targeted for water conservation, with an average household size of 3.52 people. • Benefits split with ARA 3
	<u>ARA3</u>	• Housing surveys • HOPE and SHIFT implementation records	0	Direct: 113,360 56,680 Female 56,680 Male Indirect: 0	Direct: 226,000 113,000 Female 113,000 Male Indirect: 0	• 10,000 houses built through HOPE, with 4-6 people per house. • 50,000 roofs made resilient through SHIFT, with an average household size of 3.52 people. • Benefits of SHIFT and water conservation split with ARA2
<u>ARA2 Health, well-being, food and water security</u>	<u>Supplementary 2.3: Beneficiaries (female/male) with more climate-resilient water security</u>	• Household Survey of water conservation devices • Site Inspection	0	10,000 households 35,000 people 17,500 women 17,500 men	20,000 households 70,400 people 35,200 women 35,200 men	Assume interest from households in utilizing and maintaining water conservation devices
<u>ARA3 Infrastructure and built environment</u>	<u>Core 3: Value of physical assets made more resilient to the effects of climate change and/or more able to reduce GHG emissions</u>	• SHIFT implementation records • HOPE Implementation Records • Insurance valuation reports	0	USD1.8B	USD5B	The average lower- to middle-income house is valued at US\$100,000

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)					
Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core indicator 7: <u>Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level</u>	While there has been an increase in the issuance of green bonds in Barbados in recent years, the capital market for climate investments remains limited. Although international initiatives directed at the provision of finance for climate change mitigation and adaptation have been present for over twenty years, the national commercial banking market in small economies has, despite possessing adequate liquidity, often remained on the sidelines, with much of the running being made by multilateral institutions such as the IADB or the World Bank. These limitations on the capital market are largely driven by the small number of participants in the market, relying largely on a relatively small banking sector that exhibits considerable market overlap and limited portfolio diversification. While the banks operate legally as local subsidiaries subject to local regulation, the banks are managed and regulated from abroad, which influences their operations and appetite for risk in a significant way. Consequently, the banks operating in Barbados are largely risk-averse. This risk aversion is reinforced by an international bank de-risking process initially spurred by global anti-money laundering efforts. Perceptions of risk and the resulting enhanced due diligence have raised the cost	<u>low</u>	The existing financial institutions and capital markets are supported to mobilise funds, whether through conventional bank lending or alternative securitisation of revenue streams deriving from climate change mitigation and adaptation activities, thereby creating stronger and more diversified capital markets. This will unlock access to finance for low- to middle-income households and small businesses in the interests of wider national economic and social development.	The proposed project will establish a sustainable climate finance vehicle — the Blue Green Bank (BGB) — in Barbados to stimulate innovation in the financial sector with the goal of mobilizing the securities and loan markets in order to unlock additional funding for climate-responsive initiatives. The BGB will create an improved financing infrastructure that works closely with other financing players — such as commercial banks, credit unions, pension funds, and insurance companies — to crowd in private capital alongside public capital, rather than crowding out the significant investment opportunities offered by these key financial institutions. As a partnership which brings together public, private and multilateral institutions at the level of equity, the BGB can present a holistic view of the market and the potential for green financing by ensuring	<u>National level (one country)</u>

	of banking and reduced the attractiveness of small jurisdictions for international banks. As a result, market innovation is largely stifled, leaving banking liquidity underutilised. Aside from a certain institutional prudence described earlier, the low participation of the Barbadian banking sector in activities of this kind can also be ascribed to the historical absence of a broad, coherent package of measures driven by the government to introduce the necessary reforms.			that new projects align with Barbados' NDC targets. This will ensure that all parties value and recognise the financing potential of such investments driven by the local government policy via a regulatory regime for the promotion of climate resilience and transition to low-carbon futures.	
<u>Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards</u>	The packaging of climate investments with other complementary initiatives has been identified as an innovative solution for the financing of what may otherwise appear as un-bankable initiatives when viewed in isolation. However, commercial banks in Barbados have a limited understanding of the opportunities for packaging initiatives. The role of packing initiatives to increase bankability extends to the potential of leveraging economies of scale — for example through bulk procurement. This is particularly important for SIDSs, where emerging technologies generally need to be imported, often at high costs. Bulk procurement of such technologies can significantly decrease costs and increase potential returns; however commercial banks lack the capacity to initiate and operate the schemes necessary to manage bulk procurement.	<u>medium</u>	The finance sector in Barbados has a strong knowledge base for packaging climate investments to strengthen the capital market. Commercial banks, alongside the GoB, are adopting best practices generated through BGB-supported initiatives to inform investment and policy decisions, and to identify opportunities for leveraging finance through innovative packaged deals.	The BGB will serve as a conduit between local financial institutions and the Government of Barbados, allowing the BGB to identify market opportunities and provide feedback and data to inform policy decisions more effectively. By showing that climate investments are bankable and profitable, disseminating market knowledge, and encouraging best practices, the BGB will close the existing climate finance gap — not merely as a tool for deploying capital, but playing a crucial role in changing market behaviour, educating key stakeholders, and informing future policy goals.	<u>National level (one country)</u>

E.5. Project/programme-specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme-specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1: Innovative financial structures operating in Barbados to overcome barriers to climate finance	<i>Type of financial instruments/products on offer from the BGB targeting climate investment.</i>	BGB governance and operations policy documents	No financial instruments/products are available from BGB	BGB offers loan finance	BGB offers loan finance, guarantees and other securities and financial products	The BGB is able to strengthen the capital market and expand its financial offerings over time.
	<i>Amount of additional funding for future pipeline projects unlocked, managed and maintained by BGB</i>	- Official BGB Records - Audit documents	No Finance unlocked by BGB	Year 8: \$48 million Debt-equity ratio of 7:1 CET1 12.5% threshold	Year 15: \$179 million Debt-equity ratio of 7:1 CET1 12.5% threshold	BGB has a process in place to manage and maintain funds BGB undertakes the process to manage and maintain funds Capital isn't spent on other investments
Output 1.1: Blue Green Bank (BGB) established and capitalised	<i>Operational status of the Blue-green Bank</i>	- Regulatory approval documents from the regulator - Company registration documents - BGB online platform	BGB not yet incorporated or licensed by a competent regulator	BGB is incorporated with CAIPO and licensed by the regulator. BGB digital platform developed and operational.	BGB is incorporated with CAIPO and licensed by the regulator. BGB digital platform developed and operational.	The BGB receives the necessary regulatory approval and licenses to operate as a wholesale bank in Barbados.
Outcome 2: Affordable finance crowded in and mobilised for investment in priority government initiatives targeting low-carbon and climate-resilient development in Barbados	<i>Value of finance mobilised for investment in priority government initiatives targeting low-carbon and climate-resilient development in Barbados</i>	- Official BGB Records - Audit documents	0	USD1.47 B investment mobilised by year 8	USD 5.8 B investment mobilised by Year 15	The BGB is able to attract suitable funding to adequately capitalise the bank.
	<i>Ratio of mitigation and projects implemented</i>	- Investment Committee records - Official BGB Records - Audit documents	No projects funded	70:30 climate mitigation vs adaptation projects	50:50 climate mitigation vs adaptation	

	<i>Number of investments made in energy access and power generation (EA&PG) projects, health and well-being and food and water security projects, and infrastructure and built environment projects</i>	- Investment Committee records - Official BGB Records - Audit documents	0	At least 1 EA&PG project At least 1 water security project At least 2 housing projects	At least 1 EA&PG project At least 1 water security project At least 2 housing projects	Money received by lower-income groups and small businesses is spent on appropriate projects. Projects are successfully developed and implemented.
Output 2.1: Blue-green investments identified and executed in Barbados	<i>Number of people living in hurricane-resilient houses, disaggregated by gender, built as part of the HOPE programme</i>	- Property transfer deeds and HOPE records - Construction reports detailing hurricane-proof designs	0	25,000 people 12,500 women 12,500 men In a total of 5,000 homes	50,000 people 25,000 women 25,000 men In a total of 10,000 homes	Houses accommodate 2 people per bedroom.
	<i>Number of houses upgraded with hurricane-resilient roofing through the SHIFT programme</i>	SHIFT Construction reports detailing hurricane-proof roofing design for retrofitted housing	0	25,000 hurricane-resilient upgraded roofs	50,000 hurricane-resilient upgraded roofs	Retrofitting houses with hurricane straps and other structural enhancements will adequately strengthen buildings to withstand hurricane-force winds.
	<i>Number of households with water conservation devices installed funded by the BGB</i>	- Site inspections - Household Survey	0	10,000 households	20,000 households	The public is suitably sensitised to the need for water demand management and willing to adopt water conservation practices.
	<i>Volume of water storage made available through rainwater harvesting tanks</i>	- Site inspections - Installation reports	0	2,000,000 L	4,000,000 L	Each home targeted for water conservation received a 2,000 L rainwater tank
Project/programme co-benefit indicators						
Co-benefit 1. Reduction in air pollution and its associated impacts on public health	<i>Change in the concentration of specific air pollutants such as sulfur dioxide (SO₂), nitrogen oxides (NO_x), and particulate matter (PM) in the atmosphere.</i>	- Air quality sampling - Government records - Hospital records of respiratory illness-related air pollution.	Levels of PM _{2.5} in 2017 were 23 micrograms per cubic meter.	TBD at project inception	TBD at project inception	Increasing clean energy production leads to a decrease in production from burning fossil fuels through an energy transition, as opposed to adding to the total energy production. There will be no other significant polluting factors associated with the installation of clean energy systems.

Co-benefit 2. Gender considerations integrated into climate finance to support that support women-headed households and women-owned businesses to adapt to climate change and/or adopt low-carbon solutions	<i>Percentage of projects funded by the BGB that demonstrate specific gender considerations and create empowerment opportunities for women to invest in climate action.</i>	• Gender action plans (GAPs) for BGB-supported projects • Monitoring reports for the GAPs	0	100%	100%	Effective sensitization and training within the BGB to design products that are gender-responsive. Intermediaries actively implement gender-responsive actions identified in their GAPs. Women are actively engaged by intermediaries and take up finance options made available.
Co-benefit 3. Improved groundwater reserves and quality	<i>Concentration of groundwater pollutants and salinity</i>	• Water quality monitoring at pump stations	TBD at project inception	TBD at project inception	TBD at project inception	Increasing water conservation efforts will reduce extractive pressures on groundwater resources, thereby increasing freshwater reserves, diluting pollutants and reducing saltwater infiltration.
Co-benefit 4. Job creation in the renewable energy and resilient housing sectors.	<i>Number of new jobs created for climate-resilient construction and RE installation and maintenance.</i>	• Employment records for service providers	0	TBD at project inception	TBD at project inception	Increased uptake of RE and resilient construction will create increased demand for services and thereby lead to job creation.
Co-benefit 5. Accelerated Energy transition.	<i>Change in the uptake rate of rooftop solar on private residents and small businesses</i>	• BLPC records of offtake agreements.	TBD at project inception	TBD at project inception	TBD at project inception	Increase accessible finance for lower- to middle-income households will increase the uptake of RE solutions, thereby accelerating the energy transition in Barbados.
Co-benefit 6: Increased Energy Efficiency in Households	<i>Supplementary 1.1: Annual energy savings</i>	• Aligned with CDM methodology AMS-III.AE. • Post-completion inspection. • Household energy consumption data	0	404 tCO ₂ e mitigated	2 273 tCO ₂ e mitigated	Household energy consumption estimated at 41 kWh per month on average. Electricity savings due to natural lighting, ventilation, and access to real-time consumption data estimated at 4.66% compared to the baseline.

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
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Activity 1.1.1: Incorporate the Blue Green Bank	The BGB will be incorporated as a company under the Barbados Companies Act, CAP308, 2002. The incorporation will be done through an application to the Corporate Affairs and Intellectual Property Office (CAIPO). This Activity will be executed by PCA, together with GCF BGB Holdings and Pegasus BGB Holdings GP.	Sub-activity 1.1.1.1. Draft incorporation documents Sub-activity 1.1.1.2. Submit documents to the Corporate Affairs and Intellectual Property Office (CAIPO)	Registration documents for the BGB compliant with the Barbados Companies Act, CAP308, 2002
Activity 1.1.2: Formalise shareholder term sheets and agreements for capitalisation	Shareholders committing equity to BGB will be formally onboarded. The processes will include the finalisation and formalisation of the shareholder term sheets as well as finalising agreements for the capitalisation of the bank. This Activity will be executed by PCA, together with GCF BGB Holdings and Pegasus BGB Holdings GP.	Sub-activity 1.1.2.1. Engage with equity shareholders to finalise the terms of the shareholder agreement and contributions to capitalisation. Sub-activity 1.1.2.2. Shareholder agreement negotiated and executed. Sub-activity 1.1.2.3. BGB is capitalised as per the terms of the shareholder agreement.	Signed agreements with all shareholders.
Activity 1.1.3: Apply for regulatory approval from regulators to operate as a Bank	Once registered as a company, the BGB will be licenced as a merchant bank with the appropriate regulator, providing the necessary regulatory approval to operate as a bank in Barbados. This Activity will be executed by BGB.	Sub-activity 1.1.3.1. Prepare necessary documentation for submission to the regulator. Sub-activity 1.1.3.2. Submit an application for regulatory approval.	Regulatory approval certificate from the Central Bank of Barbados
Activity 1.1.4: Hire a management team and appoint a board	The BGB recruit experienced professionals with a strong track record of managing operational risks and financial risks to form the management team of the BGB. A board of between 5 and 7 members will be appointed, ensuring compositional alignment with the guidelines provided by the CBB. This Activity will be executed by BGB.	Sub-activity 1.1.4.1. Finalise ToRs for core positions and advertise. Sub-activity 1.1.4.2. Engage with shareholders to select suitable Board members Sub-activity 1.1.4.3. Formally appoint board members	• Signed employment contracts for all management positions. • Signed agreements with all board members
Activity 1.1.5: Finalise Governance and Operational Policies	Building on the draft policy recommendations developed during project preparation, the BGB's management team will review and finalise all governance and operational policies. This includes all policies identified under the Legal and Policy review to ensure compliance with regulatory processes. The board will be responsible for the final sign-off and adoption of all policies.	Sub-activity 1.1.5.1. Review draft policy recommendations developed during project preparation. Sub-activity 1.1.5.2. Update and finalise all governance and operational policies, getting final approval from the board	• Corporate Governance Policy • Risk Management Policy • Investment Policy • Loan Policy • Asset/liability management policy • AML/CFT policy • Code of Conduct

	This Activity will be executed by BGB.		• Conflict of Interest Policy • ESG policy. • FACTA policy.
Activity 1.1.6: Build the technical and operational platform for the launch of the bank	As a digital bank, the BGB will require a robust technical platform for its operation. Professional services will be procured to work alongside the BGB management team to develop the technical platform and train all staff on its operation. Once operational, the BGB will be publicly launched. This Activity will be executed by BGB.	Sub-activity 1.1.6.1. Procure the services of a suitable provider to develop the digital platform for the operation of the bank. Sub-activity 1.1.6.2. Launch the platform and train staff on its operation. Sub-activity 1.1.6.3. Publicly launch the bank	Digital operational platform for BGB
Activity 1.1.7: Create tools and mechanisms to increase significantly new private investment through participation and monitoring	The bank will use the information generated through Activity 1.2.1 to structure and package financial products that address investor concerns and unlock the capacities of the local capital market. Through this process, the project will develop a variety of products, tools and mechanisms for non-governmental financing of investments, as well as negotiate with potential investors to co-develop packages that will mobilise private finance for eligible projects. This Activity will be executed by BGB.	Sub-activity 1.2.2.1. Develop suitable tools for attracting investment based on recommendations from 1.2.1. Sub-activity 1.2.2.2. Draft legal structure templates for investor agreements on the financial tools and mechanisms developed.	• Legal structure templates for investor agreements
Activity 2.1.1: Engage with potential investors to negotiate, structure and package financial products which take account of investor concerns and facilitate investments into eligible projects	The BGB Management team will actively engage with potential investors to identify specific needs, concerns and barriers to investment. This information will inform bank operations to enhance the role of BGB as an arranger in the financial sector. The BGB will then structure and package financial products to crowd in investment for eligible projects. The BGB will negotiate terms with investors and enter agreements for investments using the tools/mechanisms developed under Activity 1.2.2. This Activity will be executed by BGB.	Sub-activity 2.1.1.1. Host engagement workshops with potential investors Sub-activity 2.1.1.2. Hold bilateral discussions with individual investors to identify specific needs in the market. Sub-activity 2.1.1.3. Negotiate agreements with investors to secure funding for eligible projects Sub-activity 2.1.1.4. Package deals are based on agreed terms. Sub-activity 2.1.1.5. Sign funding agreements with investors	• Funding agreements signed with investors
Activity 2.1.2: Partner with government programmes and local intermediaries to identify	The BGB investment teams will engage with government programmes and local intermediaries to identify eligible climate projects for investment by the bank. These efforts will initially focus on the	Sub-activity 2.1.2.1. Engage with HOPE, SHIFT and BWA to confirm project pipelines and timelines.	• Published call for proposals. • Refined pipeline of eligible projects

eligible climate projects for investment	three priority pipeline initiatives with HOPE, SHIFT and BWA, but will eventually branch out to identify additional opportunities. A key role of BGB in this process will be as a dealmaker, actively seeking opportunities to package deals that improve the bankability of projects. This Activity will be executed by BGB.	Sub-activity 2.1.2.2. Engage with government agencies to identify additional eligible opportunities for investment. Sub-activity 2.1.2.3. Release a call for applications for eligible projects. Sub-activity 2.1.2.4. Screen all projects against the BGB eligibility criteria and conduct necessary due diligence.	
Activity 2.1.3: Structure optimum solutions by mitigating risks and ensuring financial feasibility	Once opportunities are identified, the BGB investment teams will engage with intermediaries to structure solutions to optimise their bankability. This process will start with HOPE, where the model is already in an advanced stage – looking for opportunities to strengthen the existing model to maximise the benefit offered by BGB over alternative sources. This will be followed by firming up arrangements for SHIFT, and finally taking a broader deal-maker role for water conservation initiatives with BWA as an intermediary partner. In phase 2, the BGB will branch out beyond these initial pipeline initiatives, seeking opportunities to strengthen the capital market and securitise investments for a broader range of climate adaptation and mitigation options. This Activity will be executed by BGB.	Sub-activity 2.1.3.1. Engage with HOPE Inc to package a deal for the financing of hurricane-resilient housing estates, funded through feed-in tariffs for rooftop PV. Sub-activity 2.1.3.2. Engage with the SHIFT programme team to progress the concept and package a deal to finance the retrofitting of houses for hurricane resilience funded through feed-in tariffs for rooftop PV Sub-activity 2.1.3.3. Engage with BWA to identify and package options for the installation of water conservation and harvesting devices and infrastructure on residential homes and small businesses, financed through levies placed on utility bills and offset for end-users by water savings. Sub-activity 2.1.3.4. Identify opportunities for expanding the financial offerings of BGB, including through the issuance of guarantees and green bonds.	• Packaged deals with HOPE, SHIFT and BWA • Mechanisms for packaging future projects • Green bonds issued and/or traded.
Activity 2.1.4: Finance projects through securitisation and other means to upscale climate investments	Once deals are identified and packaged, BGB will mobilise funds to finance eligible and approved projects. Initial agreements will be made with HOPE, which provides a ready-to-finance project model to kickstart bank operations, followed by SHIFT. This Activity will be executed by BGB.	Sub-activity 2.1.4.1. Sign loan agreements with HOPE and SHIFT Sub-activity 2.1.4.2. Release funds in line with the provisions made in each agreement under 2.1.4.1. Sub-activity 2.1.4.3. Sign loan agreements with additional eligible intermediaries. Sub-activity 2.1.4.4. Release funds in line with the provisions made in each agreement under 2.1.4.3.	• Loan agreements signed with HOPE, SHIFT and BWA

Activity 2.1.5: Manage, monitor and report on investments	The BGB management team will take responsibility for actively managing the accounts of all financed projects, including through regular monitoring and reporting on investments. This Activity will be executed by BGB.	Sub-activity 2.1.5.1. Provide active risk management on the loan portfolio. Sub-activity 2.1.5.2. Provide support to borrowers to develop suitable reporting frameworks. Sub-activity 2.1.5.3. Monitor and prepare reporting documents on the performance of investments, ESG and climate performance. Sub-activity 2.1.5.4.	• Reporting frameworks for all borrowers • Quarterly monitoring reports
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E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Monitoring

The M&E structure for the BGB has been developed on two levels, targeting the GCF-supported project Outcome and Output key performance indicators (KPIs; listed above) and the broader bank performance, respectively. The performance monitoring will be conducted by the BGB's dedicated M&E officer with close oversight from Pegasus, with the GCF-supported project KPIs being monitored during the implementation period as a subset of the broader performance monitoring of the BGB operations over the long term. The responsibilities of the M&E officer will include:

- Data collection and performance measurement.
 - The M&E officer will work closely with each project intermediary, to aggregate data on the non-financial fund metrics obtained directly from the intermediaries. This will include data to support the development of site-specific baselines and aggregated investee business plans pertaining to market share, sales, and economic activity in the sector. Additional data will be periodically and systematically aggregated during implementation relating to production, sales volumes and value, human resources, among others.
 - Period site visits will be conducted to assess and verify the results presented by the intermediaries.
 - When appropriate and deemed necessary, external consultants may support the data collection and measurement process.
- Performance assessment and recommending corrective actions: Based on the predefined output and outcome targets and metrics, corrective actions may be defined by engaging with the intermediary.
- Reporting: Performance against the Outcome and Output KPIs are reported to the GCF via the AE annually.

The monitoring and reporting on the Outcome and Output KPIs will be integrated into the broader performance monitoring system for BGB. This M&E system for BGB will be validated and finalised by the BGB Investment Committees during the bank's inception, and will be continually refined as the bank's operations progress. The BGB Board will also work with the M&E Officer to evaluate the causal relationships among the project indicators and results to understand the overall effectiveness of the monitoring system and refine it as needed.

Evaluation

The project performance will be independently evaluated against the Outcome and Output KPIs at the mid-term and final stages of the project implementation period, specifically in years 8 and 15. These evaluations will be conducted by an independent evaluation specialist, under the oversight of Pegasus.

Financial Reporting

The BGB Management, under the oversight of Pegasus, will provide financial information to the BGB's partners each quarter, along with a quarterly report that summarizes material developments of the Bank. Pegasus has established valuation processes and procedures in accordance with US GAAP and fair values for the underlying portfolio investments are done in accordance with Accounting Standards Codification (ACS) 820-10 on a semi-annual basis on June 30th and December 31st.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Selected Risk Factor 1

Category	Probability	Impact
<u>Credit</u>	<u>Low</u>	<u>Medium</u>

Description

There are risks that interest rates rise, and economic activity slows, making it hard to raise social impact investments or to find climate-related assets.

Mitigation Measure(s)

The BGB will mitigate these risks firstly by reducing exposure to the interest rate and economic cycles through long-term contracts not unusual in power purchase agreements around renewables. Moreover, initially, the focus will be on government-supported affordable housing and water-related projects, which will also be, if not entirely so, less sensitive to market risks.

Selected Risk Factor 2

Category	Probability	Impact
<u>Credit</u>	<u>Low</u>	<u>Medium</u>

Description

Retail financial institutions have to assess individual credit risks or even the chance that small and medium enterprises cannot pay their financial obligations.

Mitigation Measure(s)

Credit risks in the initial pipeline are reduced by focusing on income streams from the private power utility (Emera, a highly rated Canadian firm) and water rates and not directly from individuals, and by writing contracts so that these credit risks are not mediated through individual households. Under this model, operational risks behind the assessment of credit risks also will be reduced.

Selected Risk Factor 3

Category	Probability	Impact
<u>Credit</u>	<u>Low</u>	<u>High</u>

Description

A potential risk is that a mismatch between the term periods of assets and liabilities leads to a shortfall of structural liquidity that may result in the bank being unable to honour its funding liabilities.

Mitigation Measure(s)

Liquidity risks will be managed low because the BGB will be funded by patient capital and long-term social impact investors and not by overnight deposits or short-term money markets. This is less speculative than it may be elsewhere. Barbados is already working closely with US-based The Nature Conservancy (TNC) on long-term debt for marine conservation and through this has become exposed to and familiar with long-term social impact investors. Moreover, the assets of the BGB and predictable income from the private electricity and public water utility companies could be securitised into tradeable and marketable securities, increasing the liquidity of the assets on the balance sheet.

Selected Risk Factor 4

Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Medium</u>

Description

Potential risk has been identified linked to the operational structures of the BGB. This includes potentially high operational costs, which reduce the financial viability of the bank, and inexperience of staff in conducting business in a newly formed bank, potentially constraining the banks' ability to identify and package suitable deals.		
Mitigation Measure(s)		
Operational risks will be managed low by establishing the institution as a digital, wholesale institution and not a retail brick-and-mortar firm. The small number of counterparties will also keep operational risks low. And the intention is to recruit experienced professionals with a strong track record of managing operational risks and financial risks.		
Selected Risk Factor 5		
Category	Probability	Impact
<u>Credit</u>	<u>Low</u>	<u>Medium</u>
Description		
A lack of interest from donors and funders to invest in a new programme due to economic and social uncertainty		
Mitigation Measure(s)		
Tracking the economic climate and potential investors' appetite for investment will be crucial for the executive director. This will be reported to the Bank's shareholders. Engagement with donors and financiers to gauge their appetite and engage with financiers to secure funding commitments (or statements of intent) for the establishment and operations of the Bank.		
Selected Risk Factor 6		
Category	Probability	Impact
<u>Credit</u>	<u>Low</u>	<u>Select</u>
Description		
The Bank is unable to establish financial sustainability over the long term due to a slow project pipeline development, or delays from an intermediary in implementing the pipeline projects.		
Mitigation Measure(s)		
The Bank's long-term financial plans are being devised for various sources of future funding, including financing from private-sector investors, and multilateral and relevant government agencies. The Bank will aim to make a sound case for investment that is attractive to and targeted at different financial audiences. Moreover, the Bank will scope a more concrete and committed project pipeline of additional projects that will benefit Barbados as one of its formative steps to establishment once operational. The diversification of the pipeline over time will further mitigate risks of individual projects slowing the rollout of investments, while rigorous due-diligence processes during project formulation will ensure that the BGB engages intermediaries who have adequate systems in place to streamline investments and meet impact targets.		
Selected Risk Factor 7		
Category	Probability	Impact
<u>Reputational</u>	<u>Low</u>	<u>High</u>
Description		
The Bank's credibility and reputation are negatively impacted by errors and omissions in reporting; and/or a lack of results/realization of goals.		
Mitigation Measure(s)		
The Bank's operation processes will be continually enhanced – both at inception and throughout operation), ensuring effective checks and balances are in place to mitigate the risks of errors and omissions. Similarly, these changes create focus and increase the chance of realizing the timely impact. Realistic, yet meaningful objectives will also be established to mitigate the risk of failure.		
Selected Risk Factor 8		
Category	Probability	Impact
<u>Reputational</u>	<u>Medium</u>	<u>Medium</u>

Description		
Existing 'players' construe the BGB as a threat to their position (e.g., local authorities or commercial banks) which could also affect other stakeholder perceptions.		
Mitigation Measure(s)		
The Bank will use the best practice stakeholder engagement process to deeply understand each stakeholder's aspirations, interests and influence coupled with routine and purpose-driven engagement to avoid this potential. The Bank's objectives have also been defined to be complementary to other key stakeholders, thus avoiding overlap (competition).		
Selected Risk Factor 9		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>
Description		
The stakeholder landscape is fragmented with multiple authorities governing the various sectors, resulting in a potential lack of synergy and coordination for effective implementation.		
Mitigation Measure(s)		
The design of the BGB has factored in the need for extensive and continual stakeholder engagement processes that will improve coordination to address these challenges. Moreover, by employing these best practice stakeholder engagement processes — strengthened further by the direct involvement of the GoB in its operation — the BGB will be positioned to not only mitigate this risk but also to contribute to mapping an effective solution for the financial sector as a whole.		
G. GCF POLICIES AND STANDARDS		
G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)		
The Environmental and Social Management System (ESMS) guides BGB in integrating environmental and social risk assessments into its transaction processes. A key component of the ESMS is BGB's Environmental and Social (E&S) Policy, which sets out the organisation's commitment to embedding E&S considerations into its day-to-day business activities and transaction processes. This includes BGB's E&S Guiding Principles, E&S Risk Standards and Requirements and Core E&S Commitments. BGB's E&S Policy is aligned with GCF's E&S requirements. The scope of the ESMS will include all transaction and trading activities of BGB, including loans, guarantees and acquiring bonds. The ESMS identifies four stages at which E&S assessments are conducted throughout the transaction process of loans: (1) E&S Screening, (2) E&S Due Diligence, (3) Transaction Structuring and Approval and, (4) E&S Monitoring: 1. The purpose of E&S Screening is to identify E&S red flags and key E&S risks of the transaction at an early stage of the transaction. There are two types of E&S red flags. The first are business activities that would trigger BGB's Exclusion List and the second are business activities that, based on an initial assessment of key E&S risks associated with the transaction, are categorized as Category A or FI-1 (in accordance with IFC's E&S Risk Categorization Framework). In both cases, BGB will abandon the transaction without further consideration. If the subproject is not categorized as FI-1 or Category A and if the subproject is in alignment with BGB's Eligibility List, BGB will assign an E&S Risk Category (B or C for non-financial intermediaries and FI-2 or FI-3 for financial intermediaries) and proceed to the E&S Due Diligence stage. A key factor in assigning the E&S Risk Category of a transaction is the E&S risks associated with the subprojects that the intermediary will finance. 2. The way E&S Due Diligence (ESDD) is conducted is determined by the E&S Risk Categorization. ESDD for FI-3 or Category C transactions are conducted by BGB internally. ESDD of FI-2 or Category B transactions can be conducted by BGB internally or by an external specialist. An external specialist will be engaged if the E&S risks associated with a transaction trigger any of the IFC Performance Standards		

5-8 or if BGB's internal E&S capacity is insufficient to adequately assess the E&S risks of a transaction. ESDD will result in an E&S Action Plan (ESAP), which lists the identified gaps in the intermediary's E&S risk management practices and/or sub-project-specific E&S risk management as well as proposed measures to close these gaps.

3. During the Transaction Structuring and Approval stage, BGB Investment Committee formally approves the transaction. BGB will then define the E&S conditions of the transaction, which will be highlighted in all relevant legal documents between BGB and the borrower prior to disbursement. An important aspect of the E&S conditions is that BGB's E&S requirements are passed down from the borrower to the subproject.
4. BGB will implement a multi-layered monitoring and reporting process. This includes reporting from subprojects to intermediaries, from intermediaries to BGB and from BGB to investors. These reporting processes are guided by the monitoring and reporting requirements that BGB will impose on the intermediaries and which intermediaries must pass down to the subprojects. These include periodic reporting on E&S KPIs, reporting on ESAP progress and major incident reporting.

As BGB's exposure to E&S risks is smaller with guarantees than with loans, the E&S assessments of guarantees will be slightly lighter than those of loans.

Key requirements for intermediaries and subprojects include requirements on stakeholder engagement and grievance redress mechanisms (GRM). The ESMS includes a Stakeholder Engagement Plan template for developing effective stakeholder engagement mechanisms at the subproject level as well as a GRM appendix that outlines the different steps of the grievance redress process of GRMs at the borrower and sub-project level and the requirements of both GRMs. Further, the ESMS includes safeguarding instruments to be deployed at the sub-project level (when relevant) such as a Land Acquisition and Resettlement Action Framework, a potential Sexual Exploitation, Sexual Abuse, Sexual Harassment (SEAH) report template and an E&S Impact Assessment (ESIA) table of contents.

Full details on the ESMS are provided in Annex 6 of the Funding Proposal.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

A gender assessment was conducted for the BGB, to ensure that women and other vulnerable members of society who are disproportionately impacted by climate change are included and benefit from the programme. The BGB gender assessment has informed the development of the BGB's Gender Framework that will ensure that the principles of gender equality, natural resources management and environmental sustainability are mainstreamed through its operations. This Framework will underpin the BGB's gender policies, which will, in turn, form the basis of one of the BGB's four core investment eligibility criteria. To this end, each project supported by the BGB will have to conduct a suitable project-level Gender Assessment and develop an appropriate Action Plan in line with the principles laid out in the BGB's Gender Framework. The project-level Gender Assessments and Action Plans will ensure that each activity financed by the BGB is designed and implemented in support of/alignment with other national development strategies that promote equal opportunities, poverty reduction and sustainable use of natural resources from a national, regional and sector-specific perspective — including the prevailing national gender strategy of Barbados (once finalised) or other public policies that are relevant to the Bank. The BGB is thus committed to being gender-responsive – through its financed activities – to support gender equality and women's empowerment and to monitor the links between gender equality, climate mitigation and adaptation, and environmental sustainability.

As such, the BGB will commit to the following principles in its investments and engagements, and will ensure that intermediaries, partners, project owners and borrowers apply the subsequent considerations to the extent that is relevant to each of them:

1. Project proposals must consider key issues and requirements of gender-sensitive design and implementation.

2. A sufficient budget is allocated to conduct the gender assessment and compile the action plan.
3. The project owner should establish meaningful gender performance indicators for the project.
4. There is alignment with the Sustainable Development Goals and a commitment to gender equality and women's empowerment (SDG 5) as well as relevant cross-cutting themes.
5. The project/programme is aligned with national development strategies that promote equal opportunities, whether in the intervention region or the sector.
6. Stakeholder consultations and engagements are undertaken and are in line with the BGB's Stakeholder Engagement Plan (Annex 7).
7. The project is aligned with existing country policies and strategies and best practices (in the gender assessment, a policy, legal and regulatory review must be undertaken by the project owner and checked by the intermediary and thereafter reported back to the BGB).

Finally, once the Bank is operational, it will aim to achieve the following targets and incorporate the respective recommendations:

1. Dedicate necessary finances and human resources to implement this Gender Action Plan, with indicator data disaggregated by sex to ensure reportable progress.
2. Develop a BGB Gender Policy and Sexual Harassment Policy (including procedures) and training kit that aids in understanding the nexus between gender and climate finance.
3. Include gender-responsive elements in the eligibility criteria and post-evaluation criteria.
4. Develop mechanisms to assess gender impacts of projects submitted by intermediaries but also to internally monitor, report and verify gender impacts and other co-benefits project levels.
5. Ensure the representation of women in the Board and/or key decision-making and advisory bodies of the BGB.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

Pegasus maintains an Accounting Policy designed to ensure financial transactions are properly authorized, documented and executed. Pegasus' funds are audited by an independent auditor on an annual basis and Pegasus also undergoes a compliance audit conducted by a third-party compliance consultant on an annual basis.

All cash movements and disbursements (e.g., for funding investments and paying fees/invoices) are subject to review by multiple parties and subject to specific approval thresholds and processes depending on the type of disbursement.

The Blue Green Bank, the GCF BGB Holdings, and the Pegasus GCF Holdings GP will all be managed in accordance with industry best practices. The BGB senior management team will draft detailed financial management and procurement policies at the bank's inception, which will be ratified by the BGB Board. The presence of representatives of the Accredited Entity — acting on behalf of Pegasus and the GCF — will ensure that the financial management and procurement policies align with the general principles of both organisations. Once the BGB has been established, the procurement of future goods and services will be managed directly by the BGB management team following the agreed procurement policy. Sound procurement will be characterized by the following standards:

- Fairness, integrity and transparency
- A competitive procurement environment that ensures purchases are made in the best interest of PCA
- Impartial selection process based on price and qualifications
- Quality goods, works and services delivered at the time and place required by responsible and reliable suppliers
- Well-documented selection process

When evaluating a particular need for goods or services, PCA and the BGB will consider, among other things:

- The cost of the provider compared to others in the market and whether multiple bids should be solicited. Generally, most procurement needs should have multiple bids. However, PCA recognizes that for some needs such a bidding process is not appropriate or advisable.
- The reputation of the provider in the marketplace
- PCA's prior experience with the provider
- Where PCA does not have experience with a provider (or may have insufficient experience with such a provider), the feedback from other market participants on the provider's performance
- The required timeframe for procuring the goods or services
- Documentation of decisions and awards - Pegasus and BGB should document the procurement process to memorialize the reason why a particular provider was chosen. All services should be provided pursuant to a binding, executed legal agreement that sets forth the terms for the goods or services to be received.

G.4. Disclosure of funding proposal

Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.

Indicate below whether or not the funding proposal includes confidential information.

☐ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
 - Annex 2.1 Feasibility Study – Barbados Climate Change Country Report
 - Annex 2.2 Feasibility Study – Investment Eligibility Criteria
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☒ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g., Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First-level AML/CFT (KYC) assessment
- ☒ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting
- ☐ Annex X Other references

* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.



GOVERNMENT OF BARBADOS

The Hon. Ryan Straughn, M.P.

MINISTER IN THE MINISTRY OF FINANCE & ECONOMIC AFFAIRS

Ministry of Finance, Economic Affairs and Investment

Government Headquarters,

Bay Street, St. Michael, BB11156, Barbados, W.I.

Our Ref.: I051/1 Vol. I

Tel.: (246)535-5300

Fax: (246)535-5629

Date: April 3, 2023

Mr. Henry Gonzalez
Executive Director (ag.)
The Green Climate Fund
Songdo Business District Incheon
REPUBLIC OF KOREA

Dear Mr. Gonzalez,

Funding Proposal for the Green Climate Fund (GCF)
by Pegasus Capital Advisors Regarding
The Blue Green Bank

We refer to the programme titled *Blue Green Bank* in Barbados as included in the funding proposal submitted by Pegasus Capital Advisors to us on February 23, 2023.

The undersigned is the duly authorized representative of Economic Affairs and Investment Division, Ministry of Finance, Economic Affairs and Investment, the National Designated Authority of Barbados.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The Government of Barbados has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Barbados; and

- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours sincerely,

A handwritten signature in black ink, appearing to read 'Ryan Straughn', with a stylized flourish at the end.

HON. RYAN STRAUGHN, M.P.
Minister

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	The Blue Green Bank (BGB)
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Private
Accredited entity	Pegasus Capital Advisors (PCA)
Environmental and social safeguards (ESS) category	Category I-2
Location – specific location(s) of project or target country or location(s) of programme	Barbados
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	Tuesday, June 6, 2023
Language(s) of disclosure	English
Explanation on language	English is the official language of Barbados.
Link to disclosure	https://d1xeoqaoqyzc9p.cloudfront.net/app/uploads/2023/06/BGB_Annex-6_ESMS-Manual_06.06.2023_Final.pdf
Other link(s)	https://www.pcalp.com/impact/
Remarks	An ESMS consistent with the requirements for a Category I-2 project is contained in the "Environmental & Social Management System Manual".
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A

Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Tuesday, June 6, 2023
Place	The Barbados National Designated Authority (NDA) was provided the link to the ESMS documents, which are also accessible from the Barbados NDA office: Government Headquarters Bay Street St. Michael Barbados, W.I.
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	Monday, July 10, 2023
Date of GCF's Board meeting	Monday, July 10, 2023

Note: This form was prepared by the accredited entity stated above.

Independent Technical Advisory Panel's assessment of FP213

Proposal name:	The Blue Green Bank (BGB)
Accredited entity:	Pegasus Capital Advisors
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. Barbados is a small island developing State (SIDS) that is vulnerable to climate change impacts. These impacts are driven largely by the increased frequency and intensity of tropical storms and hurricanes – along with the associated impacts of sea level rise and storm surge – as well as decreasing rainfall with longer dry spells, lower soil moisture and increased aridity. The national development plans and nationally determined contribution (NDC) of the Government of Barbados include building resilience to climate change and transitioning to one hundred per cent renewable energy by 2030. Mobilization of finance for these efforts will be key to a successful transition.

2. The funding proposal identifies several local market barriers to mobilizing finance notwithstanding Barbados' relatively robust economy. Access to finance for lower income communities, smaller-scale projects and adaptation interventions has proven to be a challenge. To address these barriers, the aim of the funding proposal is to establish a new national climate finance vehicle – the Blue Green Bank (BGB).

3. Green banks are designed to address investment barriers in climate mitigation and adaptation by using limited public funds strategically to leverage far greater private investment. In practice, many of the green banks that have been established are not regulated as banks, nor do they perform the same functions as retail banks, under their respective local laws, but instead are hubs of local climate finance capacity and mission-oriented capital. By developing innovative finance and market development solutions, green banks can address barriers that currently restrict low carbon economic growth. To date, green banks have channelled billions of USD in public and private capital into climate action across the globe.

4. The proposed BGB will be a joint public-private sector effort to be developed by Pegasus Capital Advisors (Pegasus) as the accredited entity (AE) in close coordination with the Government of Barbados. Pegasus will also function as one of four executing entities (EEs), along with three entities still to be established: Blue Green Bank; GCF BGB Holdings LP; and Pegasus BGB Holdings GP, LLC. BGB would be a start-up effort and GCF is being asked to provide both funding support and input into developing the entity's climate policies and structure.

5. BGB would invest in Barbadian national priority adaptation and mitigation projects for primarily micro, small, and medium-sized enterprises (MSMEs) and households. The funding proposal intends that BGB will offer a variety of financial products designed to overcome market barriers, crowd in the local financial sector, and increase access to climate finance for businesses and households. Longer term, the project sponsors hope to expand BGB operation throughout the Caribbean region.

6. The bank will also seek to be a market maker by partnering with existing finance institutions and government programmes to lead on the creation of climate finance opportunities and to broaden and accelerate current programmes. BGB will initially finance rooftop solar, resilient housing and water conservation through national programmes that have already been developed but need financing, including Roofs2Reefs, HOPE and SHIFT (see funding proposal for a description of these programmes). A key feature of BGB (and all green banks) is that it will seek not to replace other financial institutions but instead to function as a partner. To this end, the bank's operations will be wholesale in nature, providing support in the form of funds, guarantees or other mechanisms for eligible projects, directly as well as through intermediaries such as existing lenders, whether banks, credit unions, or insurance companies, to facilitate the funding of climate-responsive programmes contemplated by government climate policy.

7. GCF is being asked to provide USD 15.5 million in finance to BGB in a combination of equity (USD 5.5 million) and convertible loan commitments (USD 10 million). The Government of Barbados will provide an additional USD 10 million in equity on a *pari passu* basis with the GCF equity. As noted above, Pegasus will act as the AE and one of the EEs. Pegasus intends to take an active role in setting up BGB and will maintain a seat on the BGB board of directors.

8. Once established, BGB will engage with investors, including private investment firms, development banks, philanthropic organizations, and other sources of finance to continue to capitalise on the bank over the long term. Several potential investors have already been identified by the AE, with discussions underway around possible contributions. The funding proposal sets a target of raising between USD 30 million and USD 50 million total equity. The BGB sponsors assert that they will maintain a debt-finance to equity ratio of 7:1, aiming to mobilize at least USD 5.8 billion of investment in climate projects over the next 15 years.

1.2 Impact potential

Scale: Low - Medium

9. The funding proposal credibly articulates that Barbados (particularly as a SIDS) is already impacted by climate change, facing increased frequency and intensity of tropical storms and hurricanes, sea level rise and storm surge, decreased rainfall, and worsening soil fertility. These climatic changes threaten lives and livelihoods of vulnerable communities and have a dramatic effect on key economic sectors in the country. Additionally, Barbados has taken on mitigation goals that, if achieved, can generate significant greenhouse gas reductions relative to the size of the Barbadian economy and population base.

10. A well designed and functioning green bank can significantly contribute to financing and mobilizing other sources of funding for interventions that can yield considerable long-term positive impact, both in terms of mitigation and adaptation. Reviewing the indicative BGB pipeline, we can discern that many of the programmes BGB intends to support can derive cross-cutting benefits for the people of Barbados. Moreover, the adaptation impacts are expected to benefit vulnerable groups who otherwise lack access to the funding required to adapt to climate change – particularly in terms of access to hurricane-resilient housing, as well as participation in water conservation initiatives.

11. We also note that the public sector in Barbados is fiscally constrained and on its own is unlikely to fully finance many of the climate programmes described in the BGB pipeline. Additionally, the local financial sector is overweighted to branches of foreign-owned banks, which may be willing to bank larger projects and lower risk projects (highly collateralised standard financial products such as mortgages and car loans) but have been more risk averse when it comes to smaller projects and lower income communities.

12. According to the funding proposal, BGB will initially focus on rooftop solar through the HOPE and SHIFT programmes. The project sponsors claim that BGB is expected to increase

renewable energy generation by 914 megawatts (MW), which in turn will contribute directly to an expected emissions reduction over 25 years of 16,736,509 tonnes of carbon dioxide equivalent (t CO₂ eq). Based on our assessment, we view these mitigation claims to be overstated.

13. Currently in Barbados there are various local initiatives to encourage certain segments of the Bajan population to purchase and install solar panels, including some private sector financing sources. In our view, the realistic size of the finance demand for solar sector investments will be substantially lower and limited to lower income communities who cannot access the commercial products on offer. Additionally, it is worth noting that the installed energy capacity of Barbados in 2021 based on International Renewable Energy Agency information was approximately 300 MW. The funding proposal presumes that BGB will finance 914 MW of renewable energy capacity, which is three times the current total capacity. Total electricity generation to be supported by BGB annually from 2037 onwards is to exceed 1,500 gigawatt hours, which is 40 per cent more than the total electricity generated by Barbados in 2021. Total business as usual CO₂ emissions from the electricity sector are estimated by the AE at 0.8 megatonne of carbon dioxide (Mt CO₂) in 2030, while the project is to reduce 0.98 Mt CO₂ per annum as of 2037. Given these estimates, the project will reduce more than the current emissions of this sector by 2031.¹

14. The funding proposal presumes that the above discrepancies will be addressed by substantial increases in energy demand, as well as an eventual expansion to other Caribbean markets. In our view, the AE was not able to credibly demonstrate the likelihood of energy demand increasing apace in Barbados, nor, importantly, a viable path to expansion of financing services across the Caribbean in the short to medium term.

15. Notwithstanding our above overestimation concerns, we note that some renewable energy finance will be needed in Barbados for it to meet its climate targets. These levels may be less than estimated in the funding proposal, but they are not negligible. Moreover, BGB will not be limited to rooftop solar finance. Successful green banks must be flexible to remain on the cutting edge of local climate finance needs, while ensuring that they do not calcify in a narrow sector, risking crowding out other participants. The funding proposal does demonstrate that Barbados has additional climate finance needs beyond solar energy installation. Notably, there is a significant need around resilience-building initiatives where existing financial products are falling short. There are also other potential mitigation investments in vehicle-to-grid charging infrastructure as well as energy efficiency in homes, small businesses, and tourism establishments.

16. On the adaptation front, the funding proposal makes a clearer case for the role of BGB. Under the initial pipeline of projects, the adaptation benefits will focus primarily on hurricane-resilient housing and water security. Specifically, both the HOPE and SHIFT initiatives include a core focus on hurricane-proof roofs, with HOPE designing resilience into the construction of new homes and SHIFT retrofitting existing structures to be hurricane resilient. Water security is also another area of need, in the face of changing rainfall patterns with longer dry spells. Investments in water recycling initiatives can help to reduce pressure on already limited freshwater resources in Barbados. The adaptation impacts are expected to benefit vulnerable groups who otherwise lack access to the funding required to adapt to climate change.

17. The creation of a functioning green bank in Barbados has the potential to engender positive climate impacts. Green banks, when well-designed and executed, have proven to be

¹ This is an indication that calculations done by the AE are incorrect with the potential mitigation impact being considerably lower than suggested by the AE. It is recommended that the GCF Secretariat reviews emission reduction calculations with more care and assesses their plausibility, to avoid overstatements of mitigation impacts from over optimistic assumptions of the economic cost of reducing carbon emissions.

instrumental in helping jurisdictions achieve their climate goals in a manner that is both financially sustainable and catalytic of other sources of finance. They are not a panacea and not suitable for all jurisdictions, however they can be impactful in markets that have a functioning financial system and dynamic investment opportunities. While we have questions about some of the projections presented in the funding proposal, these concerns touch on the potential scale of impact, and not on whether BGB can have a positive climate change mitigation impact in Barbados. The latter seems clear to us. The bank will provide financing and investment opportunities for renewable energy and energy efficiency projects, and in turn these will help to reduce Barbados' dependence on fossil fuels and mitigate the effects of climate change.

18. The potential benefits of BGB's equity investments could be even more considerable as relates to adaptation interventions. Without support, private investment is likely to continue to focus on collateralized loans that target high-income groups and large businesses, excluding the considerable investment potential from lower-middle income groups and small businesses, which can be unlocked through a combination of BGB finance and BGB ability to act as a facilitator and market educator. It is key that BGB does not become static but should instead have a charter and mission focused on evolving along with market needs. Effectively, green banks should seek to be replaced by other market participants by drawing them into new sectors through a combination of catalytic finance, demonstration effect and investment and capacity-building.

19. Importantly, the funding proposal intends that BGB will act as a market maker or a "dealmaker" within the existing financial landscape. This is an essential role to fill and key to advancing potential climate projects that are outside the scope of public-sector budgets but require de-risking/securitization to attract private investment. This market-making function is a typical remit of green banks. In addition to financial resources provided, impactful green banks understand that their most valuable asset is the willingness and ability to contribute the "sweat equity" that private players are often unwilling to provide, while the public sector often lacks such capacity. Sweat equity in this case involves the time and human capacity contribution to arrange investment opportunities and to structure transactions that would otherwise not be of interest to the market because of the perceived effort needed to reach financial close.

20. We applaud BGB's recognition of the need to play a market-making role. However, a green bank's ability to perform this role well is dependent on a variety of factors, including but not limited to: (i) the professional qualifications and expertise of staff and management; (ii) market perception of the staff and management by developers, other financial institutions and capital markets; (iii) understanding of local market players and dynamics; and (iv) an independent governance system (procedures and individuals) that aligns the bank's activities with the commercial and climate needs of the market. The BGB proposal is very much at the start-up stage and the above issues have not yet been addressed. There is a plan in place to build-out the institution, but considering the importance of human capital to impact, we must significantly discount our impact assessment.

21. Overall, the independent Technical Advisory Panel (iTAP) regards BGB's impact potential as low to medium at its current state of development. This rating of the funding proposal derives primarily from uncertainties around its mitigation impacts described above, and most importantly, the amount of development work that remains to be done in establishing the bank. The entity currently does not have a Board, a leadership team or even a legal personality. However, the plan in place for development, combined with the clear political desire to see this project go forward in Barbados, gives us comfort that it can be fit for purpose eventually. The iTAP believes that if the concerns outlined above can be addressed as per our conditions herein, the rating for impact potential can be much improved.

1.3 Paradigm shift potential

Scale: Medium - High

22. BGB is intended to act as a market maker and market filler in Barbados to address the financial barriers that hamper climate projects. As a market filler it will be able to complement public-sector budgets and provide funding for climate interventions that would normally not be in the purview of private investment. Additionally, as a market maker the bank will work to develop financial products and tools that can partially de-risk transactions on the island to crowd in other sources of finance and, importantly, do the leg work of originating, structuring and shepherding climate impact transactions to closing. Without the financial heft and capacity that BGB can provide, it is likely that private investments in Barbados will continue to focus predominantly on collateralized loans targeting high-income groups and large businesses, excluding the considerable investment potential from lower-middle income groups and small businesses that can be unlocked through the right package of financing.

23. The paradigm shift potential of green banks rests on their ability to address investment barriers by using limited public funds strategically to leverage far greater private investment. A green bank is typically designed to achieve this in a sustainable manner, meaning supporting environmental sustainability but also remaining financially sustainable as an institution. Those institutions that expect periodic infusions of public funds are not truly green banks and have limited paradigm shift potential. The financial returns sought may be more modest than those sought by purely commercial entities, but they should be sufficient to secure the financial health, credit rating and longevity of the green bank itself. BGB is designed to be an institution with indefinite duration and ability to raise additional funding beyond its initial capital raise. The proposed BGB design provides for legal authority to raise capital on the open market once the bank aggregates a performing portfolio of assets. This will require BGB to operate in a financially sustainable way. The investment discipline engendered by having a return requirement will help BGB to become a trusted partner for other sources of co-finance.

24. We note positively that most of the GCF funding requested is in the form of a loan facility which requires payback of equity and interest, albeit highly concessional interest. We also note that the funding proposal anticipates providing shareholders with a dividend return commensurate to their shareholding. Moreover, the proposal foresees a financial exit for GCF in the future, once the BGB can access capital markets for additional finance and refinancing of its portfolio. This pragmatic approach is in line with green bank best practices and indicates that the project sponsors intend to establish a useful entity.

25. BGB also has the potential to drive policy innovation and ambition by demonstrating what is possible in Barbados and providing policymakers with a tool to further accelerate mitigation and adaptation ambitions. An increase in the number of bankable projects will not only lead to significant climate impacts, but also provide a broader catalytic effect by driving finance into 'hard to reach' sectors such as climate adaptation projects.

26. We also note positively that BGB, like other green banks, will be well positioned to develop and disperse market knowledge in the local financial sector as well as drive cross-sector collaboration in project development. Lastly, BGB can also function as a trusted partner to cross-border capital providers. One of the impediments to attracting finance into developing markets from international private sources is country risk perception. These risks can be mitigated by having a local finance partner that understands local political and economic realities.

27. Similarly, BGB can function as a trusted partner to both domestic and international partners as relates to the green or climate credibility of transactions. More investors are looking for assets that have sustainability at their core, while still generating an appropriate risk-adjusted return. Green banks that maintain high impact standards and enforce high quality screening, oversight and reporting can improve the attractiveness of co-investments in their markets. Co-investors will perceive that project climate attributes are credible when BGB is part of the transaction, consequently reducing perceived environmental impact risk.

28. The funding proposal provides that BGB will be established to adopt and adhere to the GCF environmental and social standards. The proposal relies heavily on GCF core involvement to help deliver an institution that is highly credible from a climate impact standpoint. A weakness of the proposal is the expectation that GCF funding will be provided before many of the actual operations and BGB structure are developed. This would leave GCF with few leverage points to ensure that these high standards are truly embedded. Moreover, the AE in this case is also acting as an executing entity, making the alignment of interests even more critical. We suggest that GCF funding be tranching and linked to progress on the development of BGB.

29. The funding proposal also projects that BGB will eventually expand beyond Barbados to support market transformation throughout the Caribbean. We do not assess this possibility as likely or well planned for now. While we discount the probability of a BGB expansion across the region, we do recognize that a successful BGB launch and operation in Barbados can serve as a model for other countries and subnational regions facing similar climate finance barriers.

30. Overall, we rate the potential for BGB to effect paradigm shift as medium to high.

1.4 Sustainable development potential

Scale: High

31. The BGB proposal credibly demonstrated its potential to contribute to the sustainable development of Barbados beyond mitigation and adaptation impacts, including:

- (i) Improved air quality: supporting the transition from fossil fuel-based to renewable energy production through rooftop photovoltaic panels will reduce air pollution;
- (ii) Improved groundwater reserves and quality: water conservation initiatives can reduce demand on and extraction of limited groundwater resources, increasing the availability and quality of water for terrestrial and freshwater ecosystems. Moreover, increased groundwater reserves can improve the hydraulic pressure of the freshwater lens, helping maintain the equilibrium between fresh, brackish, and salt water, thereby reducing the threat of saltwater intrusion linked to sea level rise;
- (iii) Living standards: development of additional housing (built with hurricane resilience in mind) will help improve access to housing for low-income households, many of which are currently paying more for renting sub-standard housing than for a mortgage which could be provided under the HOPE programme;
- (iv) Health: improved air quality resulting from reductions in fossil fuel can have a positive impact on human health, which would also reduce the burden on Barbados' public health system. Similarly, improved water security can further reduce potential health impacts, including through the spread of waterborne diseases, especially after heavy rainfall events;
- (v) Job creation: the implementation of projects financed by BGB will create employment opportunities for Bajans, while also supporting local contractors and suppliers. Programmes such as HOPE have significant potential to strengthen employment in the construction sector, including by creating opportunities for upskilling young artisans. Moreover, investments directed to small businesses in the tourism industry will contribute to job security and create employment opportunities that can help lift vulnerable populations out of poverty; and
- (vi) Reduced economic downtime: building the resilience of the population will reduce the economic downtime which typically follows extreme events that disrupt transportation networks, damage power and water infrastructure, and increase maintenance efforts at the household level, as well as disrupting the flow of tourists into the country. Reducing this downtime will have a notable benefit to the broader economy.

32. Gender: BGB intends to incorporate gender mainstreaming into its operations, governance structure and investment eligibility criteria. BGB's board, staff, trainees, and loan

applications will have target of 50 per cent for women with a minimum reservation of 30 per cent.

33. Overall, iTAP rates the sustainable development potential of BGB as high.

1.5 Needs of the recipient

Scale: High

34. Barbados is highly exposed to the impacts of climate change, including tropical storms and hurricanes that are having increasing negative impacts on livelihoods and infrastructure. Flooding, strong winds and storm surges are common, with most of the country's critical infrastructure and economic activity in the coastal zones.

35. Barbados also faces water issues, due to a combination of droughts and increased rainfall variability. These impacts create stress on the island's scarce freshwater resources. Sea level rise and saltwater intrusion will also increase salinity, thus affecting the country's water resources and intruding into aquifers. The existing water infrastructure is also at risk of destruction from extreme climate events, which creates further uncertainty around water supply. Addressing infrastructure security and making sure it becomes climate-resilient is fundamental to the country's future.

36. Barbados has a diversified economy and a relatively stable economic outlook, including one of the highest per capita incomes in the Caribbean. However, Barbados has a high amount of public debt which constrains public finance efforts around national priority climate projects. Given its income status, Barbados does not meet the eligibility criteria of many concessional lenders, such as multilateral development banks. This leaves the private sector as the only viable option; however, the local financial sector only consists of branches of foreign-owned banks. These banks are highly risk averse and are not financing climate projects for vulnerable borrowers. BGB will specifically target these types of projects and borrowers to function as a dealmaker and crowd in investment from these institutions.

37. Overall, iTAP rates the needs of the recipient as high.

1.6 Country ownership

Scale: High

38. The BGB concept was proposed by the Government of Barbados. We have assessed that there is substantial political support and intention on the part of the Bajan public sector to establish the bank to facilitate their climate goals. We note that the concept was first proposed by Prime Minister Mia Mottley. The Government of Barbados has provided a no-objection letter.

39. We understand from our review that the Government of Barbados has been highly involved from the conceptual phase through to the design of BGB, including the selection of Pegasus as the AE.

40. Importantly, Barbados will provide USD 10 million for the initial capitalization of BGB. This is a high indicator of country ownership. As a result, Barbados will be a significant shareholder in BGB and will have representation on the board of directors on an ongoing basis. Barbadian representation on the Board is useful to provide a strong nexus between national climate policy and the activities of BGB. To succeed and operate effectively and efficiently, BGB should have a current understanding of government climate initiatives and access to government programmes. The difficult balance will be for BGB to have a governance model that can work with government without being part of government. It will be critical that BGB maintains a maximum level of independence from government itself. We would recommend that the BGB board be made up of professional appointments including the appointments made by the Government of Barbados as a shareholder. A useful approach used by other green banks

is to have key government board members serve in an *ex officio* capacity (non-voting) providing a connection to government, while voting board members will be appointed for their specific expertise. We also assess that it will be critical for BGB to be seen as an independent institution in order to attract private sector investment.

41. We also take note that the Government of Barbados has engaged with multiple relevant stakeholders in the course of developing the BGB concept. These engagements have included financial and development entities as well as players in the renewable energy sector. The BGB development team also engaged the private banking community in Barbados. Often, green banks encounter initial resistance from local financial institutions who can be wary of new public institutions that could potentially distort the market through excessive subsidy/concessionality or through blurring lines between political and private realms. Against this backdrop, the BGB concept was generally welcomed, especially once clarifications regarding its wholesale status and business model were provided. According to the AE, the main concern expressed by local financial institutions during their consultations revolved around whether BGB would target large commercial customers – which are currently fully banked. Again, the AE reports that this concern was dispelled once the BGB business model was explained, with the financial institutions engaged during the consultations providing positive feedback about the engagement with HOPE to date and the opportunities generated through it. Lastly, the funding proposal provides a stakeholder engagement plan to ensure continued engagement throughout implementation.

42. We also note that the BGB initiative appears to be aligned with existing Barbadian policies and planning, including its NDC which targets a 95 per cent share of renewable energy in Barbados' electricity mix, and a 20 per cent increase in energy efficiency across all sectors as compared to business as usual projections, and highlights the adaptation goals of the Roofs to Reefs resilience programme.

43. One area of concern is the lack of direct green bank experience on the part of the AE and the lack of direct experience in Barbados on the part of members of the AE's BGB team. Additionally, Pegasus has several other GCF-funded initiatives under development and fundraising concurrently. These elements represent AE capacity risk considering the amount of BGB development work yet to be conducted. However, we take some comfort in the fact that the AE was specifically selected by Barbados after reviewing several other options. The BGB concept did not originate with the AE but instead, Pegasus was specifically recruited by Barbados.

44. The iTAP considers BGB to be aligned with host country priorities, and we assess involvement and ownership on the part of Barbados to be significant. Overall, we rate country ownership of the funding proposal as high.

1.7 Efficiency and effectiveness

Scale: Medium

45. The establishment of a national climate finance vehicle, in the form of a green bank, aligns with GCF's Private Sector Strategy. Specifically, it will build the capacity of national financial institutions for climate finance (through the creation of BGB), and it will mobilize private climate finance at scale by de-risking investments for other commercial institutions. Further, it will primarily aim to increase access to finance for MSMEs and households that struggle to obtain affordable financing due to high collateral requirements from the foreign-owned financial institutions that are currently their only option for financing in Barbados.

46. Other green banks have shown an impressive track record in catalysing new low-carbon markets and crowding in private investment over the past decade and a half. Notably, the "State of Green Banks 2020" report found that twenty-seven existing green banks situated in twelve countries invested nearly USD 25 billion of their own capital since their respective inception

and were supporting projects with a total value of USD 70 billion as of mid-2020. More than USD 45 billion (64 per cent) of this sum came from the private sector.² As specialized financing institutions, green banks function as the focal point for accessing, mobilizing, and scaling up foreign and domestic investment in climate solutions, while simultaneously enabling domestic ownership of project development and NDC implementation.

47. As noted above, green banks can be transformational. However, the efficiency and effectiveness of a green bank model requires that the home jurisdiction be sufficiently developed economically and that the bank be designed with best practice principles in mind. In this case, the Barbadian economy and needs appear to be well suited for a green bank, albeit not necessarily of the long-term target size indicated in the funding proposal. In terms of green bank principles, we note the following key factors described below as they relate to Barbados.

48. Independence: an independent and de-politicised decision-making process is essential. BGB should ensure proper governance, including an appropriate degree of autonomy, transparency, and professional capacity, as this is essential to build confidence in the institution. In the case of BGB, the funding proposal identifies the importance of independence in governance and market perception. However, this is an area where additional clarity will be useful once the actual institution and its governance model is built out. The intention is correct, but independence will only be evident once the entity is legally created (details of its charter), a board appointed, and a process established to select key management roles.

49. Political consensus: stakeholders and donors want to see a strong political consensus around the role of the green bank. The clearer the signal of governmental support from the highest levels, the more likely it is the bank will attract adequate donor support. In the case of BGB there is substantial evidence of political consensus.

50. Financial wherewithal and security: a green bank should have an adequate and sustainable source of operational revenues, including from multiple sources. Relatedly, a green bank should be perceived as having longevity and ultimately seeking financial sustainability. BGB is designed with the goal of financial sustainability in mind and is seeking an initial capitalization that appears to be more than adequate. While BGB will provide access to adequate financing and concessionality when necessary, the funding proposal states that BGB's lending activities are intended to be financially sustainable. All BGB's on-lending activities will be profit-generating, with the appropriate measures taken to ensure estimated and actual credit losses are kept to a minimum. Moreover, BGB is intended to operate as a digital bank to minimize operational expenditures.

51. Clear remit: the mission of the green bank should be well understood and defined around the specific environmental and social challenges that the bank is seeking to solve and how it will measure success. The funding proposal demonstrates effectively how climate impacts will be embedded into the BGB decision-making process. Additionally, the proposed BGB is intended to be a "climate impact first" organization. However, the mechanics of how all this will come together are still a work in progress.

52. Local context: a successful green bank must be fit for purpose considering the local context and reality. Similarly, it should understand the local market (and the local market failures) to have confidence in the project pipeline that the green bank will support. BGB leadership and staff are yet to be identified and the expectation is that, as a virtual bank, it will have a light local touch.

53. Execution capacity calibrated to the green bank's remit: the breadth and scope of the bank's mandate, product offerings and operational modalities should be aligned with the

² Whitney, A., Grbusic, T., Meisel, J., Becerra Cid, A., Sims, D., Bodnar, P. (2020) The State of Green Banks 2020, Rocky Mountain Institute.

available human resources and capabilities. It is better to have a narrow focus that can be properly administered versus a wide mandate that cannot. A green bank should also build flexibility to adapt to changing market conditions and needs, as well as expand the remit over time, based on track record and capacity. The BGB model assumes that it will be an evolving organization and allows for flexibility. The level of alignment between execution capacity and BGB's remit remains unclear, as additional build-out will need to occur; however, the project sponsors have proposed an adequate plan.

54. Human capital: human capital at a green bank is nearly as important as monetary capital. Green banks should recruit top professional talent to lead and staff the institution. This may require seeking talent external to the local market and/or building a transition period for capacity-building of local teams; in both cases competitive remuneration is necessary. As noted in our assessment, the composition of the BGB board and team remains an open question.

55. Complementarity: a green bank should be complementary to the existing financial system. The bank should determine how best to work with existing local financial institutions so as not to compete with, but rather enhance, their capacity to deliver outcomes consistent with the mission of the bank. It is thus important to understand and define the roles of the green bank versus the roles of local financial institutions in delivering the outcomes, so as not to risk crowding out financial partners. BGB in its feasibility studies and design efforts has adequately taken the realities of the local existing financial system into account.

56. Civil society: civil society buy-in is critical for a green bank's credibility and market acceptance. The bank must ensure inclusive stakeholder engagement and buy-in so that it is understood and welcomed by the market. BGB has conducted extensive outreach and has a plan for continued activity.

57. In our view, the BGB design has a well-defined mandate that can facilitate risk-taking investments in support of climate interventions in Barbados. We view very favourably the commitment of the Government of Barbados to provide USD 10 million in equity. We also note that discussions are underway with other third parties to secure additional equity commitments. Additional funding is expected to come from donor countries, philanthropies, and high-net-worth individuals. The commitment by Barbados will be instrumental in attracting other funds. Without such a strong financial signal from Barbados, we suspect that investors and donors would have reason to question the wherewithal and sustainability of the institution.

58. We also applaud the legal ability of BGB to eventually issue securities such as green bonds that can attract additional capital from private and public sources. Raising capital is always a challenge. Many variables will impact the perception of BGB in the future, including whether it has deployment capacity and sufficient demand; however, building in the legal authorities now is an important market signal regarding the BGB trajectory.

59. One concern raised by iTAP is whether BGB runs the risk of crowding out other initiatives (public and private) that are already underway, considering the relatively small size of Barbados. The AE has made the argument that these other initiatives do not approach the level of investment required to meet the adaptation and mitigation needs of Barbados. We agree that there is a gap but, as noted above, we do not agree on the actual size of this financing gap. For this reason, we appreciate the AE's specific commitment that BGB will not seek to target public-good projects, or those that are financeable through traditional commercial options, thereby mitigating the risk of crowding out investment from other sources. BGB will, instead, fill a specific niche that currently lacks finance.

II. Overall remarks from the independent Technical Advisory Panel

60. Green banks are a critical climate mitigation and adaptation financing tool because they open the door for public and private capital for projects that would not have succeeded in the absence of green bank interventions. They do this by providing financing and other assistance to project developers, which can help to reduce the cost and risks (real and perceived) of investments in climate change mitigation and adaptation, making their financial products more accessible to a wider range of people and organizations around the globe. Successful green banks deploy not only their own capital, but also drive significantly higher amounts of private investment, which can help to further accelerate the transition to a more sustainable economy.

61. While BGB has not yet been operationalized, funded by the Government of Barbados or even formally created, the efforts made thus far have moved significantly in the direction of a successful launch of BGB. However, in our view, there are some concerns that have made our assessment decision difficult, as discussed in the body of this document. To summarize a few areas of risk: (i) we note that the finance demand arguments made in the funding proposal appear to be overstated; (ii) the BGB concept is still quite early in development with gaps in personnel, operational tools and underlying legal and institutional frameworks; and (iii) there are questions around the local expertise and ability of the AE to execute. Taken together, and along with the early stage of development, these concerns give us pause on whether the funding proposal is ready for endorsement.

62. Nevertheless, we have ultimately decided to provide an endorsement with conditions (see below). This has been a challenging decision, but several factors tip the scale for endorsement of this particular effort. First, the size of the funding request is relatively modest and matched by commitments from Barbados. Second, it is clear to us that Barbados has a serious political commitment to create BGB and to do so correctly. Third, Barbados is one of the SIDS, which are a GCF priority and particularly susceptible to the impacts of climate change. Lastly, we acknowledge the effectiveness of local green bank efforts and the amount of design that has already taken place. We think it is imperative to provide a path forward so that the work on this important initiative continues.

63. Therefore iTAP endorses this funding proposal subject to the conditions below:

(a) Conditions to be met prior to the effectiveness of the Funded Activity Agreement (FAA):

(i) Delivery by the Accredited Entity to the GCF, in a form and substance satisfactory to the GCF Secretariat, of a disbursement plan under the FAA with the amount and schedule for each disbursement tranche aligned with BGB development milestones as agreed between the Accredited Entity and the GCF Secretariat; and

(ii) Delivery by the Accredited Entity to the GCF, in a form and substance satisfactory to the GCF Secretariat, of the final form of the BGB's Investment Policy, Investment Committee Terms of Reference, Board Terms of Reference, Board Risk Committee Terms of Reference, BGB procurement policy and updated procurement plan, and Board Audit Committee Terms of Reference.

(b) Conditions to be met prior to the first disbursement under the FAA:

(i) Delivery by the Accredited Entity to the GCF, in a form and substance satisfactory to the GCF Secretariat, of evidence of the incorporation and registration of BGB in Barbados and the legally binding subscription for common shares in the share capital of BGB by the Government of Barbados (or an entity representing the Government of Barbados) in the minimum amount of USD10 million; and

(ii) Delivery by the Accredited Entity to the GCF, in a form and substance satisfactory to the GCF Secretariat, of evidence of (A) the appointment of the Board of Directors of BGB, (B) the establishment of the Investment Committee of BGB, which shall include a climate impact specialist with experience in Barbados, whose role is to provide approval authority on the

climate validity of BGB financing; and (C) the appointment of the initial Chief Executive Officer (CEO) and Chief Investment Officer (CIO) of BGB, in the case of the climate impact specialist and the initial CEO and CIO, to be approved by GCF BGB Holdings in its capacity as a shareholder of BGB, as a shareholder reserved matter in accordance with the process set out in the Shareholders' Agreement.

(c) Conditions to be met prior to the second disbursement under the FAA:

(i) Delivery by the Accredited Entity to the GCF, in a form and substance satisfactory to the GCF Secretariat, of a financial breakdown of the start-up costs incurred by BGB or on behalf of BGB prior to the request for second disbursement under the FAA which shall include evidence indicating the status and amount of any Co-financing received and applied to the BGB start-up costs, up to the date of the Request for Second Disbursement;

(ii) Delivery by the Accredited Entity to the GCF, in a form and substance satisfactory to the GCF Secretariat, of a track record of investment activities undertaken in accordance with the BGB's business plan and a pipeline of proposed investment activities that could support the second GCF disbursement;

(iii) Delivery by the Accredited Entity to the GCF, in a form and substance satisfactory to the GCF Secretariat, of evidence that funding (which may be equity, debt or grant funding) in an amount at least equal to the aggregate of (i) the amount of the disbursement being requested from the GCF; and (ii) any amounts previously disbursed by the GCF under the FAA, has been contributed to BGB by the Co-financiers or other funders; and

(iv) Delivery by the Accredited Entity to the GCF, in a form and substance satisfactory to the GCF Secretariat, of evidence that the GCF BGB Holdings is the legal holder of shares in the share capital of BGB, representing not more than 33.33 per cent of the share capital of BGB.

Reply to the Independent Technical Advisory Panel assessment findings

Proposal name: Blue Green Bank

Accredited entity: Pegasus Capital Advisors

Date: 16 June 2023

Impact potential

iTAP's assessment of the impact potential is well noted, particularly regarding the mitigation potential and the highly ambitious targets for energy.

As a way of clarification, you would like to highlight that our assumptions and resulting impact estimates are based on: 1) our use of commonly accepted methodologies regarding the specific capacity factors of different electricity-producing installations (i.e., actual electrical energy output over a given period compared to the theoretical maximum electrical energy output over the same period); 2) Barbados' existing power mix and highly ambitious transition plans; and 3) The land use and grid requirements of different electricity sources compared to available land in Barbados – which makes rooftop solar an appealing technological choice compared to ground-mounted. Most notably, solar has a considerably lower efficiency factor than fossil-fuel-generated power. Therefore, there is a need to install more MW of PV than the existing capacity of fossil-fuel plants – which will need to be phased out for Barbados to meet its pledge to be a 100% renewable energy and carbon-neutral island-state by 2030 (i.e., Zero domestic consumption of fossil fuels economy-wide). This marked difference in capacity factor contributes to the targeted PV capacity exceeding existing fossil-fuel generation capacity – particularly when factoring in other developments such as the transition to EVs or the electrification of other sectors of the economy, all of which will require additional generation capacity from renewable sources.

Another consideration linked to any potential remaining overestimates is that the SHIFT programme is still in the early stages of development, and a detailed feasibility study still needs to be developed prior to investment by the BGB. Given that the primary focus of these initial pipeline initiatives is, in fact, climate resilience rather than energy generation, the amount of PV per targeted house will be optimized to reach the maximum number of houses receiving climate resilience, rather than maximizing energy potential. Such optimization will form part of SHIFT's feasibility study before investment. This will mitigate the risk of energy demand constraining resilience efforts. While it is acknowledged that this might reduce the total mitigation potential of the programme, the impact will still be relatively large given the nature of the country and the size of GCF investment.

Moreover, there is considerable potential with the growth of BGB to expand and replicate the SHIFT and HOPE initiatives beyond Barbados to other Caribbean countries, which will further expand the mitigation potential.

It is anticipated that as the feasibility studies for the programmes supported by BGB are further developed, the scale of impact will become clearer, as noted by iTAP in Paragraph 21. The AE and the Government of Barbados remain firmly committed to establishing the operational systems of BGB in a manner that maintains a strong climate focus, ensuring that systems are in place to not only screen projects for their climate impact potential, but also to provide the necessary expertise to support intermediaries in optimizing the design of initiatives to maximize the impact potential.

Paradigm shift potential

The AE agrees with the views of iTAP on the paradigm shift potential of the BGB, and anticipates that the BGB will continue to play a vital role in mobilizing finance for climate investment in Barbados and the region, particularly with regard to investments in adaptation initiatives.

The concerns on the risks related to GCF investment being mobilised before all the operational and structural elements of the bank are finalized are well noted. This has formed a key discussion point with the GCF secretariat around the terms of the GCF investment, and several points have been added to mitigate this risk. Further risk mitigation is presented in the potential early investment by USAID and GoB, which will be used to initiate the process of establishing the BGB systems once GCF funding is approved and the FAA is finalized. The Government of Barbados also notes that this is a standard issue for all equity investors, and it is conventionally addressed through the equity investor having board representation.

Sustainable development potential

The AE acknowledges and agrees with iTAP's assessment of the high sustainable development potential of BGB.

Needs of the recipient

The AE acknowledges and agrees with iTAP's assessment of the high needs of the recipients in Barbados.

Country ownership

The AE acknowledges and agrees with iTAP's assessment of the existing high-country ownership for BGB. This ownership stems from the Government of Barbados's firm commitment to resilience building and climate action, as evident in the BGB's alignment with the Roof's to Reef resilience strategy and Barbados's NDCs. The proposed new legislation that will underpin the establishment of the bank, which is anticipated to appear before the cabinet in the coming weeks (see below), further demonstrates the strong national ownership of the project.

We acknowledge the iTAP's comment that other green banks often have government board members serve in a non-voting capacity. In this case, however, given how closely we are working with the Government of Barbados and the funding being provided by the Government of Barbados, we believe it is appropriate for the Government to have a voting board seat, but not the ability to control the Board.

Efficiency and effectiveness

The recommendations from iTAP on key factors to consider in relation to the principles of a green bank are well noted. These principles have underpinned the proposed structure of the BGB (as presented in the operations manual) and are expected to become more apparent as the final operational and structural elements of the Bank are defined by its Board. The recommendations from iTAP on this matter are, therefore, appreciated and will be taken into account throughout the process of finalizing the operational structure of the BGB. Regarding the independence of the BGB, it has been concluded that the BGB requires that new legislation be enacted. The first draft of a Brief has already been prepared and circulated for consideration. It is proposed that a review meeting be held on Wednesday June 28, 2023, to finalize the Brief for submission to the Cabinet. When the Brief is approved it will be sent to the Office of the Chief Parliamentary Counsel for the preparation of the draft Bill. Once the legislation is enacted, the entity may be incorporated and the Board and other structures will be put in place.

A central element of ensuring the effectiveness of the BGB is the finalization of the investment criteria that will ensure that climate focus remains at the core of the BGB's operations. While initial investment criteria that establish the climate focus have already been defined, these will be further refined to link the criteria to their potential impacts, creating a space for prioritising high climate-impact projects.

Overall remarks from the independent Technical Advisory Panel:

The AE and the Government of Barbados both applaud the decision of iTAP to endorse the proposal, acknowledging that the risks can be mitigated through the appropriate use of conditions proposed by iTAP, along with those of the GCF Secretariat. It is our firm belief that with the strong commitment from the Government to the BGB and climate action in general that the establishment of the BGB will align with the best practices identified and that the relatively small investment from GCF will unlock considerable climate finance for Barbados and potentially other Caribbean SIDS — for which there is an urgent need.

With respect to the iTAP's proposed conditions, we agree with the intent of iTAP's conditions and the majority are in line with our intended disbursement milestones. We do feel however that the hiring of key management personnel would be better suited as a condition for second disbursement after the initial GCF funding has been invested in BGB.

ANNEX 8

GENDER ASSESSMENT AND ACTION PLAN

Version 1 | GCF Board – June 2023

2023

BLUE GREEN BANK

A GREEN CLIMATE FUND INITIATIVE



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ACRONYMS

AE	Accredited Entity
BGB	Blue Green Bank
BWA	Barbados Water Authority
CARICOM	Caribbean Community
CEDAW	Convention on the Elimination of All forms of Discrimination Against Women
ESMS	Environmental and Social Management System
GCF	Green Climate Fund
GDI	Gender Development Index
GDP	Gross Domestic Product
GFP	Gender Focal Point
GGGI	Global Gender Gap Index
GHG	Green House Gases
GII	Gender Inequality Index
HDI	Human Development Index
HOPE	Home Ownership Providing Energy
HR	Human Resources

ILO	International Labour Organisation
IMF	International Monetary Fund
INDC	Intended Nationally Determined Contribution
NDC	Nationally Determined Contribution
NGO	Non-governmental Organisation
NOW	National Organization of Women
OECD	Organisation for Economic Co-operation and Development
PPP	Purchasing Power Parity
PV	Photovoltaics
R2RP	Roof to Reef Programme
RE	Renewable Energy
UN	United Nations
UNDP	United Nations Development Program
UNFCCC	United Nations Framework Convention on Climate Change
UNWOMEN	United Nations Entity for Gender Equality and the Empowerment of Women
USA	United States America
USAID	United States Agency for International Development
WBL	Women Business Law
WEF	World Economic Forum
WHO	World Health Organisation

1 Introduction

1.1 BACKGROUND AND CONTEXT

The people of Barbados are vulnerable to major climate impacts, which have affected the country's socio-economic and development trajectory thus slowing down economic growth in the past four years (World Bank, 2021). These climate impacts are driven largely by the increased frequency and intensity of tropical storms and hurricanes — along with the associated impacts of sea level rise and storm surge — as well as decreasing rainfall with longer dry spells, lower soil moisture and increased aridity. **Urgent adaptation action is, therefore, needed to reduce the wide-reaching impacts on key economic sectors and secure a sustainable future for all Bajans.** In addition to these core adaptation needs; the country recognises the vital importance of global mitigation efforts to reduce the threat of future climate change exacerbating these impacts. To this end, despite contributing significantly less than 0.01% of global GHG emissions, Barbados' resilience strategy includes highly ambitious mitigation targets, including a transition to 100% renewable energy by 2030. These ambitious targets are intended to demonstrate the country's commitment to climate action, as well as to serve as an example to the rest of the world of the feasibility of the transition when innovative approaches are adopted.

Unfortunately, despite these commitments, progress has been slow — especially in adopting renewable energies and water conservation initiatives by middle- to low-income households. This relates primarily to the structure of ownership of capital in the Barbados economy and the financial ecosystem (see Barriers below for more detail). Given the existing fiscal constraints, Barbados' resilience strategy — encompassed in the Roofs to Reefs Programme (R2RP) — recognises the need to partner with the private sector to mobilise finance for innovative climate investments that are or can be made, bankable. Such an approach will enable the government to direct the limited concessional finance available towards public-good resilience measures that do not have direct financial returns.

There is a discernible gap in the market in Barbados and the Caribbean for a financial institution that can provide opportunities for existing financial institutions and capital markets to mobilise funds, whether through conventional bank lending or alternative securitisation of revenue streams deriving from climate change mitigation and adaptation activities, thereby creating stronger and more diversified capital markets.

In addition to its climate resilience ambitions, Barbados has also made firm commitments with regards to gender, equality and social inclusion - aiming to eliminate all forms of gender discrimination and to ensure that the country attains gender equality in all spheres of development. Over the years Barbados has made great strides in reducing gender inequality, particularly in education attainment. However, 60% of households are female-headed and women are overrepresented among the poor and vulnerable and low-income groups. As a result, these groups are disproportionately affected in accessing economic resources, including land, housing loans, and finance, among other resources. **These gender issues limit the capability of women and low-income groups to own houses and render them vulnerable to climate disasters.** When climate disasters strike poor women find it harder to recover compared to poor men. An innovative and sustainable finance vehicle that targets the low-income groups specifically women is therefore imperative for the development of Barbados and for reducing climate vulnerability on the island.

To address the urgent need for innovative, inclusive finance in Barbados, the proposed project will establish a sustainable climate finance vehicle — the Blue Green Bank (BGB) — to stimulate innovation in the financial sector with the goal of mobilizing the securities and loan markets to unlock additional funding for climate-responsive initiatives. The BGB will be a joint public-private sector effort to create an improved financing infrastructure that works closely with other financing players

— such as commercial banks, credit unions, pension funds, and insurance companies — to crowd in private capital alongside public capital, rather than crowding out the significant investment opportunities offered by these key financial institutions. For such institutions to become involved in climate change mitigation and adaptation activities, the leadership should be provided by a third party putting the deal together, for which it must establish the appropriate capacities in technical and HR terms, as well as appropriate management and institutional arrangements.

To establish the BGB, the Government of Barbados is applying for an initial injection of funds from the Green Climate Fund (GCF). The green climate fund provides finance for climate mitigation and adaptation to transformational and impactful projects and programmes that provide co-benefits to both men and women. The GCF recognises that climate change has a disproportionate impact on women and that gender-responsive approaches are critical to the success of climate action. In its updated Gender Policy, the GCF requires that all projects that it supports implement a gender analysis and Gender Action Plan (GAP). The purpose of this gender analysis and GAP is to identify key gender issues and constraints to be addressed through project implementation as a way of ensuring that women and other vulnerable groups are fully included as beneficiaries and decision-makers and that their capacities are enhanced in building community resilience against climate change risks. For these reasons the proposed Blue-Green-Bank is obliged to ensure that all its operations and those of the intermediary partners benefitting from the funds are gender responsive.

The BGB is designed to become a significant institution for practical and strategic financing of climate change adaptation and mitigation policies and measures, supporting the Governments of Barbados to achieve national resilience and emissions reduction targets. As a partnership which brings together public (Government of Barbados), private and multilateral institutions at the level of equity, the BGB can present a holistic view of the market and the potential for green financing by ensuring that new projects align with Barbados' NDC targets. Because the BGB's goal is national climate adaptation and mitigation, it will seek not to replace other financial institutions but to act as a partner and support system for them in this underdeveloped market segment. To this end, the bank will be wholesale in operation, providing support in the form of funds, guarantees or other mechanisms to existing lenders, whether banks, credit unions, insurance companies etc to facilitate the funding of blue/green programs contemplated by government policy. The BGB will also serve as a conduit between local financial institutions and the Government of Barbados, allowing the BGB to identify market opportunities and provide feedback and data to inform policy decisions more effectively.

Once established, the BGB will engage with investors, including private investment firms, development banks, philanthropic organisations, and other sources of finance to continue to capitalise the bank over the long term. The bank will maintain a debt-finance to equity ratio of 7:1, aiming to mobilise at least USD 620 million over the next 15 years.

Overall, the BGB will promote a paradigm shift towards investment in resilient livelihoods and low-carbon technologies, with emphasis on low-income, vulnerable households and the energy, housing, agriculture and transport sectors. The project will also build the wider community's awareness of climate investment opportunities in support of a more resilient and sustainable Caribbean. Finally, by focusing on climate resilience and adaptation for middle to low-income households — including a specific focus on women and other highly vulnerable groups — the BGB would aim to attract more concessional or philanthropic providers of capital, expanding the goals of climate action in Barbados without the need for further injections of money by the Government and the Green Climate Fund (GCF). This will create long-term sustainability and scalability in the banking sector, with the potential to scale beyond Barbados into the broader Caribbean region over time. Most importantly the BGB will establish a climate finance vehicle that has inclusive systems, is **gender-responsive**, participatory and transparent to address gender issues pertaining to accessing climate funds.

1.2 OBJECTIVES OF THE REPORT

The BGB fully adheres to the principle of country ownership, meaning that for each financed activity that receives funding from the BGB, a Gender Assessment (Annex 8.1) and Action Plan (Annex 8.2) will be in place and alignment with the requirements set out in the GCF updated Gender Policy.

The project-level Gender Assessments and Action Plans will ensure that each activity financed by the BGB is designed and implemented in support of/alignment with other national development strategies that promote equal opportunities, poverty reduction and sustainable use of natural resources from a national, regional and sector-specific perspective.

This gender assessment serves as the foundation for the BGB's gender responsiveness – through its financed activities – to support gender equality and women's empowerment and to monitor the links between gender equality, climate mitigation and adaptation, and environmental sustainability. The assessment also informed the formulation of the project-specific gender action plan (GAP).

The Annex is informed by a review of existing literature and secondary databases, ongoing stakeholder consultations as well as expert input and case study analyses and is considered a living document that will be regularly updated by the BGB's technical team.

This gender assessment aims to:

- Provide an overview of existing **national policies, strategies and action plans** that have integrated gender perspectives, especially in national climate change adaptation strategies and legal frameworks related to built infrastructure, water conservation and renewable energy technologies for low-income Barbadians.
- Identify **gender issues** relevant to the BGB, highlighting the socio-political, economic, and cultural aspects of climate change adaptation and mitigation. It is anticipated that the identification of these issues will highlight potential gaps in the goals of the project that do not align with stakeholder priorities, and that reinforce existing disparities.
- Provide **recommendations that will inform the Gender Action Plan**, and include specific tasks and activities designed to build the capacity of project implementors (i.e. the BGB and its intermediaries) so that they can address climate change and provide gender-responsive investments.
- **Assess the gender dimensions** of the targeted sectors in Barbados, in which the BGB is most likely to channel funding.

This Gender Assessment will be used by the BGB¹, the Accredited Entity (AE, Pegasus Capital Advisors²) as well as project owners to ensure compliance with the GCF's fiduciary principles and standards, environmental and social safeguards standards and Gender Policy.

1.3 GENDER-SPECIFIC CONSIDERATIONS

This Gender Assessment has focussed on identifying gender and social inclusion considerations at a national level, as well as within the BGB's three sectors of interest (that is, the energy, water, and infrastructure and the built environment sectors). As such, there are two layers of gender-specific considerations that have been found through this assessment.

¹ The BGB in this instance will evolve into the executing entity and will oversee that the gender action plans and assessment are submitted, reviewed and aligned with GCF's outcomes.

² The AE is responsible for overseeing and monitoring that the BGB is in compliance with the GCF fiduciary principles and standards, environmental and social safeguards standards and Gender Policy (all of which are standards for GCF accreditation), and other relevant GCF policies and requirements, in accordance with the relevant legal agreements (accreditation master agreements and funded activity agreements).

The first set look at the country's overarching gender context and provide a useful foundation for conceptualising gender-responsive measures at a project-level. The most relevant considerations (for the purpose of this project) include:

- Female headed household and other vulnerable groups are disproportionately impacted by the economic downturns resulting from climate change.
- There is an imbalance of women in senior management and decision-making roles, as well as a wage gap between men and women in the same jobs.
- The country's policy and overarching enabling environment for gender equality and social inclusion is lacking: the National Gender Policy has not yet been ratified; there is an absence of effective gender responsive budgeting in the public sector; and there are insufficient monitoring, evaluation, and sex-disaggregation initiatives to inform gender-responsive decision-making at national, sectoral and industry levels.

The second layer of considerations look more specifically at what challenges / gaps / needs exist from a gender perspective within the three sectors and have been used to guide specific actions in the Gender Action Plan. These include:

- Women are not active stakeholders in the country's energy sector, despite being the primary users of energy at a household level. Specifically, access to affordable electricity is currently a challenge for lower-income and vulnerable households – the majority of which are female headed; and too few women enterprises are involved in the renewable energy sector (i.e., technology development and marketing) due to lack of knowledge and financial resources.
- Extended, and more frequent, droughts have led to major water scarcity issues on the island, which are disproportionately impacting female-headed households and vulnerable communities' socio-economically, physically, and psychologically.
- Due to financial constraints and an inability to secure loans, low-income households (60% of which are female headed) are unable to own houses, nor upgrade or build climate resilient housing.

These findings are elaborated in the sections that follow and culminate in a set of recommendations for the project's Gender Action Plan.

2 Gender in Barbados: legal and institutional framework

2.1 RELEVANT LEGAL FRAMEWORK

The Constitution of Barbados is the Supreme law providing for the rights and freedoms of all its people and outlawing discrimination by race, origin, political opinion, colour and sex. Under the Constitution, women of Barbados enjoy equal rights and all freedoms and social justice. No formal provision protects against discrimination on grounds of sexual orientation.

The Government of Barbados (GoB) ratified the United Nations Framework Convention on Climate Change (UNFCCC) in 1994 and the Kyoto Protocol in 2000. In September 2015, it communicated its Intended Nationally Determined Contribution (INDC) to the UNFCCC Secretariat and six years later its first national Determined Contribution (NDC) in 2021. Barbados signed and ratified the Paris Agreement on 22 April 2016. In the 2021 NDC updates Barbados committed to enhancing its climate response and increasing its climate mitigation contribution in line with the Paris Agreement (UNFCCC report 2021).

Furthermore, in 2021 the Physical Development Plan and the Roofs to Reefs Programme (R2RP) were introduced as the two main national frameworks to achieve Barbados' climate resilience goals.

Barbados main international instrument supportive of gender equality is the Convention on the Elimination of all forms of Discrimination Against Women (CEDAW), which it ratified in 1980. The Government of Barbados also ratified the regional gender equality instrument, the Inter-American Convention on the Prevention, Punishment and Eradication of Violence against Women (known as the Convention of Belem does Para) in 1995 which commits to the elimination of all forms of Violence against Women. In addition, Barbados is a party to the Convention on the Rights of the Child. It is also a signatory to the United Nations Convention on the Rights of Persons with Disabilities. In September 2022, it became the 29th country in the world to ratify the Violence and Harassment Convention, 2019 (No. 190).

The GoB is developing the national policy on gender. The goal of the policy is to create an enabling environment in Barbados for the achievement of gender justice by inculcating values and practices that will eliminate gender stereotypes and discriminatory behavior from all aspects of inter-personal relations and national life. The goal of the policy is to build a culture of gender justice which will facilitate social harmony, sustainable economic development and an enhanced quality of life for all boys, girls, men and women. Some relevant objectives include:

- Provide an institutional mandate and operational framework to initiate, coordinate, implement, monitor and evaluate gender-responsive national development budgets, policies, programmes, plans, projects and activities
- Build awareness and technical competency in gender analysis, planning, budgeting, research, monitoring and evaluation throughout the public service.
- Promote gender mainstreaming and best practices in addressing development issues from a gender perspective in all sectors of government and civil society
- Provide incentives for the use of renewable energy and sustainable development practices.
- Create/expand employment opportunities for women in all sectors of the economy: value added agriculture, alternative energy, new technologies, sustainable micro enterprise.
- Provide adequate financial and technical support from the state and private sectors, in a timely manner, to create decent employment in the formal and informal sectors.

Furthermore, as early as 1992, Barbados made great strides to criminalize gender-based violence (GBV) by enacting the Domestic Violence (Protection Orders) Act. The Act was amended in 2016, making it one of the strongest Protection Orders in the CARICOM region. The amendments simplified GBV case reporting and most importantly, they provided powers of emergency protection orders to the police. The law draws on best practices in the Commonwealth.

Regarding labour rights, Barbados is a party to 36 International Labour Organization (ILO) Conventions concerning the rights and privileges of workers. Among these are the International Labour Organization Employment Policy Convention; ILO Discrimination (Employment and Occupation) Convention, 1958 (No 111); ILO Freedom of Association and Protection of the Rights to Organize Convention, 1948 (No 87) and ILO Right to Organize and Collective Bargaining Convention, 1978 (No 98) and Worst Forms of Child Labour Convention, 1999 (No 182).

In terms of sexual harassment, the Employment Sexual Harassment Act 2017 aims to make provisions for the protection of employees from sexual harassment in the workplace. This includes making provisions for the protection of employees in the

public and private sectors as well as offering an avenue for reporting sexual harassment cases. Read with the Sexual Harassment Act is the Employment (Prevention of Discrimination) Act 2020 that aids in prohibiting employment opportunities based on gender.

Women and men are regarded as equal within the law in the context of the ownership and use of land (OECD, 2019). Married women are protected under the Married Women Act 1978, which enables them to acquire, hold and dispose of by the will of real or personal property as their separate property. Women and men also have equal ownership rights to immovable property under the Property Act of 1980 and can inherit assets equally under the Succession Act of 1979. It should, however, be noted that the lack of sex-disaggregated data in key areas such as ownership of property restricts analysis and evidence-based approaches to gender equality (OECD, 2019).

The Barbados Bureau of Gender Affairs is responsible for the integration of gender in national development policies and programs. The BGA operates under the Ministry of People Empowerment and Elder Affairs. A Draft National Policy on Gender has been formulated and is presently before parliament for consideration.³

2.2 RELEVANT INSTITUTIONAL FRAMEWORK

2.2.1 The Bureau of Gender Affairs

The Bureau of Gender Affairs (BGA) is a government agency under the Ministry of People Empowerment and Elder Affairs (MPEEA) in Barbados, that is dedicated to promoting gender equality and empowering women. Established in 1985, the BGA has played a critical role in advocating for the rights of women and girls in Barbados, and in working to eliminate discrimination and promote gender equality in all areas of society. Falling under the Ministry of People Empowerment and Elder Affairs are the following departments, agencies and related entities: Child Care Board; Welfare Department; National Disabilities Unit; National Assistance Board; Bureau of Social Policy, Risk and Planning; and National HIV AIDS Commission. These entities promote and support social inclusion and address the social welfare issues to empower all people of Barbados.

The BGA serves a variety of stakeholders, including women and girls, families, communities, and organizations. Through its programs and initiatives, the BGA seeks to empower women and girls by providing them with the resources, skills, and support that they need to achieve their full potential. The BGA also works to raise awareness about the importance of gender equality and the challenges facing women and girls in Barbados. This includes providing information, training, and support to organizations and individuals to help them understand and address gender-based barriers and biases.

One of the keyways in which the BGA achieves its goals is through the development and implementation of gender-sensitive policies and programmes. These include initiatives that promote women's economic empowerment, such as providing women with access to finance, training, and markets, as well as programs that support the elimination of violence against women. The BGA also works to ensure that women are fully represented in decision-making processes, including in government and the private sector.

The BGA is made up of several committees, each with a specific focus and responsibility. These committees include the Policy and Planning Committee, the Gender-Based Violence Prevention and Response Committee, the Women's Economic Empowerment Committee, and the Gender Mainstreaming Committee. The Policy and Planning Committee is responsible for developing and implementing gender-sensitive policies and programs that address the needs of women and girls in Barbados.

³ At the time of writing (April, 2023), the draft policy was unavailable for review as part of this assessment.

The Gender-Based Violence Prevention and Response Committee is responsible for addressing violence against women and girls and for developing programs and services to support victims of violence.

The Women's economic Empowerment Committee focuses on promoting women's economic empowerment and addressing the barriers that prevent women from accessing financial resources, markets, and training opportunities. This committee works with government agencies, the private sector, and civil society organizations to ensure that women have access to the resources they need to participate fully in the economy. The **Gender Mainstreaming Committee** is responsible for ensuring that gender considerations are integrated into all areas of government policy and decision-making processes.

The BGA established Gender Focal Points (GFPs) in all the Ministries. The Gender Focal Points play a critical role in advancing gender equality within government ministries. Their responsibilities are to lead, coordinate, and advocate that gender considerations are integrated into all aspects of policy development, program implementation, and decision-making processes for the ministries that they are assigned to.

In addition to the MPEEA, two other government entities have clear inclusion agendas:

- **The Ministry of Labour, Social Security and the Third Sector Promotes** opportunities for the provision of decent work. Through enactment of employment policies and laws it ensures that employment is decent, offers equal opportunities to all citizens, and provides quality social and economic benefits. The Ministry has a gender program to address gender-based violence in the workplace.
- **The National Insurance Department** is mandated among other things, to set up and manage a **national insurance scheme** that benefits those who lose their jobs, pensioners and the unemployed. It also formulates strategies to reduce unemployment.

2.2.2 Non-governmental organisations

Barbados has an extensive network of Non-Governmental Organizations (NGOs), some of which play a significant role in advancing gender equality. These organizations work to empower women and girls by providing them with the resources, skills, and support they need to overcome barriers and biases and achieve their full potential. Some of these NGOs focus on specific issues, such as promoting women's economic empowerment, addressing violence against women, and advocating for even stronger women's political representation. Other organizations provide services, such as counselling, legal support, and job training, to support women and girls. Many of these NGOs also work in partnership with government agencies, such as the Bureau of Gender Affairs, to ensure that gender considerations are integrated into government policies and programs. Through their efforts, these NGOs play a critical role in promoting gender equality in Barbados and in empowering women and girls to achieve their rights and participate fully in society.

Examples include the National Organization of Women (NOW), which coordinates other women's organizations, and the Barbados Christian Council and the Barbados Council of Evangelical Churches, which are the two umbrella bodies in the faith-based community. The Barbados Council for the Disabled and the Barbados National Organization for the Disabled (BARNOD) are umbrella organizations serving the Disabled Community or Persons with Disabilities and both receive assistance from the State.

3 Gender status in Barbados

3.1 GENDER OVERVIEW

3.1.1 Demographics and gender profile

The population estimates for Barbados is 281,200 (The World Bank , 2021) with women making up approximately 52.1% of the population (The World Bank Group, 2021). Barbados obtained a Human Development Index (HDI) value of 0.790 for 2021, placing the country in the High Human Development Category and positioning it at 70 out of 191 countries and territories (UNDP 2021/2022). This high ranking reflects long life expectancy, high levels of school ¹²enrolment and length of education, and high income per capita, all of which improved from 1980. The population growth rate (annual change) is currently estimated as 0.18%% (2021 data point) (The World Bank , 2021), and has been falling since 2008 when it stood at 0.44% (World Bank, 2023).

Table 3-1: Demographic indicators for Barbados (Source: World Bank, Barbados population. World statistical Data)

Indicator	
Percentage of females in the population	52.1%
Sex ratio – Males per 100 females	94.59
% of females who live in urban areas	31.6% (approximately 91000)
Population growth (annual change)	0.18% in 2021

According to the World Economic Forum ‘s “2022 Gender Gap Report,” Barbados ranked 30th globally (among 146 countries) and was the 3rd out of 22 countries in Latin America and the Caribbean countries on the Global Gender Gap Index (GGGI), which benchmarks the evolution of gender-based gaps and tracks progress towards gender parity. The GGGI comprises metrics across four fundamental categories (*Economic Participation and Opportunity, Educational Attainment, Health and Survival, and Political Empowerment*). These are notable achievements. In 2022, Barbados’ Global Gender Gap Index value was 0.765 (the best score is 1 and the least is 0), reflecting a gender gap of 23.5% between men and women (World Economic Forum 2022a). To provide a perspective, Barbados’ ranking was in the same range as the USA and the Netherlands, which ranked 27th and 28th respectively, and far better than other developed countries such as Denmark, Australia and Singapore.

Examining the categories separately sheds more light on the factors that contribute to gender inequality. Barbados ranked 3rd out of 146 countries in Economic Participation and Opportunity (WEF 2022a). Barbados’s weakest performance in the index was on Political Empowerment, scoring 0.27. It should be noted that Barbados still ranked a very respectable 49 out of 146 countries (World Economic Forum 2022a). The performance in political empowerment was indeed above a poor global average of approximately 0.22. Barbados is also one of the countries scoring well in Education Attainment, closely reaching parity with a score of 0.993 and ranking 67 out of 146 countries (World Economic Forum 2022a).

The Gender Development Index (GDI) for Barbados is 1.034 (Table 3-2) placing it in group 2 countries, with a higher female HDI of 0.799 than that of males (0.773) (UNDP 2021/2022). Group 2 countries are those countries with medium to high equality in HDI achievements between women and men (absolute deviation of 2.5- 5 per cent) (UNDP, 20221/2022).

Table 3-2. Gender development indices for Barbados (Source: UNDP 2021/2022)

	Female	Male
Indicator		
Gender Development Index (Value)	1.034. Group 2	
Human Development Index (value)	0.799	0.773
Life Expectancy at Birth (Years)	79.4	75.6
Expected Years of Schooling (Years)	17.7	13.8
Mean years of Schooling (Years)	10.3	9.1
Estimated Gross National Income Per Capita (2017 PPP \$)	10,235	14,555

The Gender Inequality Index (GII) is a measure reflecting inequality in achievement between men and women in three dimensions – reproductive health, empowerment and the labour market. **Barbados has a GII value of 0.268 (1 high inequality, 0 low inequality) for 2021 showing no change from 2020 (UNDP 2021/2022). The low value indicates that there is little inequality in Barbados.**

The indicators that are used for calculating GII are listed below for Barbados:

- 29.4% of parliamentary seats are held by women.
- 95.4% of adult women have reached at least a secondary level of education compared to 86.0% of their male counterparts (ages 25 and older).
- 27.0 women in every 100,000 live births, die from pregnancy-related causes; and the adolescent birth rate is 42.3 births per 1,000 women of ages 15-19.
- Female participation in the labour market is 56.1% compared to 63.7 for men (UNDP, 2021/2022).

These indicators show that there are only a few areas needing further attention, and they should not take away from the achievements made by the country.

The World Bank's Women, Business, and the Law (WBL) 2022 Index further shows the achievements of Barbados in laws that guide/protect gender in the various areas of work life. The indices provide a measure of how conducive the work environment is for women on a scale of 1 to 100%. The overall score for Barbados is 80 (out of a possible 100). Barbados' score is slightly lower than the regional average for Latin America and the Caribbean of 80.4. With that said, Barbados obtained a perfect score of 100 in 4 out of 6 parameters i.e., Workplace, Marriage, Assets and Pensions indicators. Its lowest score was 40 in the Parenthood indicator.

3.1.2 Gender and climate

The Intergovernmental Panel on Climate Change (IPCC) has noted that “[people who are socially, economically, culturally, politically, institutionally, or otherwise marginalized are especially vulnerable to climate change and to some adaptation and mitigation responses [...] This heightened vulnerability is rarely due to a single cause. Rather, it is the product of intersecting social processes that result in inequalities in socioeconomic status and income, as well as in exposure. Such social processes include, for example, discrimination based on gender, class, ethnicity, age, and (dis)ability.” By calling for climate action and requesting finance from GCF, Barbados is not only required to tackle climate change impacts at an ecological level but also recognise the intrinsic link between gender and climate responsiveness. Indeed, the concerns of vulnerable groups and a gender lens must be applied across priority cross-cutting themes within planned adaptation and mitigation initiatives.

Although climate change impacts all Bajans, due to the differing roles of women and men in society, climate change will impact women more differently, requiring a bespoke gendered approach to be applied to address the various challenges and barriers that women may face.

A National Climate Change Committee has been established that encourages gender parity in attendance and assigned roles. The Committee's overarching objective is to uphold "*capacity building, knowledge sharing and communication, gender balance, participation and women's leadership, the strengthening of the integration of gender within the various bodies and provide a gender response to the implementation of the Paris Agreement*". It is also noted that the current administration is crafting additional policies that will contribute towards transformative change including a climate change monitoring report. Despite these initiatives, the collection and reporting of sex-disaggregated data have been stilted and require improvement, however, there are efforts to develop a framework that bolsters the routine collection of sex-disaggregated data.

The draft National Gender Policy includes similar sentiments on the need for improved collection of sex disaggregated data. It also presents several key recommendations focused on climate adaptation and mitigation, which are of relevance to the BGB's areas of interest:

- Increase protection for women and girls in relation to **shelter** situations during occurrences of natural hazards.
- Continue to lobby the international community for **climate change adaptation and resilience-building financing**, particularly for vulnerable communities across Barbados.
- Increase livelihood protection through **micro-insurance solutions** for farmers, fisherfolk and other small businesspeople, many of whom are women whose livelihoods will be most affected by climate change.
- Provide **incentives for the use of renewable energy** and sustainable development practices." (Government of Barbados - Bureau of Gender Affairs, 2021)

3.1.3 Gender in education

Barbados has a well-established education system, with a strong emphasis on providing equal opportunities for education for both men and women. The GoB expenditure on education is distributed among the primary, secondary, and tertiary institutions. The country provides free pre-primary, primary, secondary, and tertiary education, as well as technical and vocational training programs. As a result, the country has a high literacy rate, with 99.6% of the population being able to read and write (Table 3-3. Sex Disaggregated Indicators for Education in Barbados, 2021).

Table 3-3. Sex Disaggregated Indicators for Education in Barbados, 2021 (Source: [USAID. Barbados Gender](#))

Indicator	Male (per cent)	Female (Percent)	Both sexes
Adult literacy rate +15	99.6	99.6	
Total net enrolment rate, primary	100	97.83	
Gross enrolment ratio for tertiary education	40.33	90.57	
School enrolment tertiary (gross) gender parity index (GPI)			1.55
Literacy rate (age 15-24) Youth GPI			1
Completion rate upper secondary	90.4	97.7	
Total net enrolment in upper secondary	92.54	97.32	
Average Years of schooling age 15+		9.69	
Primary completion rate	93.4	98.76	
Labour Force with advanced education	77.4	81.46	
Youth literacy ratio 15-24		99.9	
Enrolment in secondary education		9.5K	

Based on the World Economic Forum Global Gender Gap report (see above) Barbados has a very high score for Education Attainment and has only 0.07% of the educational gender gap to close.

3.1.4 Employment and economic empowerment

The employment sector in Barbados plays a critical role in the country's economic development and the well-being of its citizens. Despite this, women in Barbados face several challenges when it comes to accessing employment opportunities and achieving economic security. The USAID Bridgetown Gender Quick Look (2022) found that only 25.4% of firms have females in top management positions. Furthermore, 84 days are given for paid maternity but there is no parental leave awarded to both men and women (WHO 2022). Women dominate in the service industry, where they occupy middle management jobs, clerical and housekeeping jobs.

Another challenge facing women in the employment sector is the gender pay gap, which refers to the difference in pay between men and women for doing the same work (Jackman & Lorde 2021). In Barbados, for the year 2022 the gender parity indicator for 'Economic Participation and Opportunity' decreased due to loss of incomes and inferring that the woman's earned income dropped by a greater proportion than men's (World Economic Forum, 2022a).

Despite the above gender issues there are initiatives underway to support women in the employment sector in Barbados. For example, the Government of Barbados is placing strong emphasis on facilitating and supporting women entrepreneurship and business development as one way of responding to rising levels of unemployment (Government of Barbados - Bureau of Gender Affairs, 2021). The Government is also working to promote gender equality in the workplace by implementing policies that prevent gender-based violence, and support women in accessing high-paying, high-skill occupations. Additionally, NGOs and civil society organizations are working to provide training and support to women seeking employment, particularly in traditionally male-dominated industries, to help them achieve economic security and success.

3.1.5 Gender, Health & GBV

In comparison to other Caribbean and Latin American countries, Barbados offers fairly good health services to its people and has achieved most of the Sustainable Development Goals (SDG) targets on health. According to WHO Report 2022 the government spent approximately 2.9% of its GDP on health service delivery (WHO 2022). As such, the life expectancy at birth reflects the high quality of health enjoyed by Barbadians.

However, despite achieving most of the SDG targets on health, a few challenges remain in accessing health services as reported by the World Health Organisation 2022 report; these include reducing HIV incidence rate and improving access to universal health. Gender- Based Violence is another ill affecting the Barbadians, with the Royal Barbados Police Force (RBPF) receiving more than one report per day in the period 2016-2018,

3.1.6 Enabling environment considerations

The most recent comprehensive gender assessment for Barbados was implemented in 2016 for the Caribbean Development Bank. The issues raised in the report were discussed with government counterparts working to establish the BGB and provide a useful framing of some key challenges relating to the country's broader enabling environment from a gender and inclusion perspective. That is, while there is no discrimination in terms of seeking healthcare or employment, or participating in politics, there exists intrinsic structural barriers that require intentional affirmative action to address. The gender issues that were identified in this assessment are listed below:

Policy and Strategy: The Government of Barbados committed to the implementation of a National Policy on Gender as a major pathway for achieving long-term sustainable development for all Barbadians: boys, girls, men and women. This commitment is expressed in a number of objectives, targets and strategies outlined in the Medium-Term Growth and Development Strategy 2013 -2020 as follows:

Objectives

- To facilitate the formulation and implementation of a National Policy on Gender in keeping with global developments and the current policies of government to realize equality and equity for women and men
- To advise government agencies and non-governmental organisations on matters of concern to and affecting women and men
- To develop the capacity of the Bureau of Gender Affairs to inform and coordinate policy dialogue on gender issues
- To expand gender awareness, analysis, and training to build capacity within government departments
- To eliminate violence and abuse as a means of control or influence in intimate partner relationships

Strategies

- Create a gender policy on crime and violence that will challenge existing notions of masculinity rather than placing emphasis on socioeconomic or demographic factors for causes of crime and violence
- Create a gender policy on domestic violence to address the power relations between males and females while recognizing that a critical component of the attempt to eradicate gender based and domestic violence lies in the compilation of data to inform appropriate strategies
- Reduce the sexual division of labour which limits a large number of women to primarily service jobs - equal participation of both male and female in all forms of employment and guarantee equal pay for equal work
- Create a virtual space to provide important information on violence against women and a safe place for victims and survivors of violence
- Establish a fund to assist victims of domestic violence
- Undertake gender analysis of the activities of government departments to determine their impact on the pursuit of SDG 5, society and the achievement of national goals
- Advocate for the disaggregation by sex of statistical data to facilitate the comprehensive analysis of information in order to provide accurate and up to date reports on gender and development

Evidence Based Policy Practice. In developing appropriate gender policies, it is imperative to have evidence-based decisions. Unfortunately, in a number of aspects Barbados does not collect sex disaggregated data, be it in wages, health targets, access to land and finances, etc. (CBD 2016) Only when sex disaggregated reporting is institutionalised will the country effectively address all gender issues by putting in place evidence-based policies and strategies. Sex disaggregated data will also facilitate and strengthen monitoring and evaluation of gender outcomes.

Economic activities: Through literature review and engagements with counterparts in Barbados, a number of issues that affect women economically were identified. The issues include: little and no gender budgeting at national level, significant occupational and income gaps between women and men, inflexible working arrangements especially for unskilled and migrant labour, barriers to entrepreneurship among women and limited access to skills and resources for entrepreneurship (CBD,2016). While Barbados has the smallest gender gap among the CARICOM countries, and with high education attainment for women, women still are employed at lower rates than men and earn similar or lower wages than men for the same jobs

(OECS, 2020). A smaller proportion of women own businesses compared to men and these women have limited access to financing (UNWomen 2018).

Social and cultural issues; Women in Barbados, like everywhere else are saddled with care for children and other dependents (CBD, 2016). Social and cultural norms that put responsibilities for care in the women's domain, without affording them the necessary support, impact on their ability to earn and to remain in full-time employment.

4 Sector-level gender considerations

4.1 WOMEN, GENDER EQUALITY AND ENERGY

4.1.1 Renewable energy

Energy is a critical driver of economic growth and renewable energy (RE) in ¹⁷particular is considered a key climate change mitigation technology that fosters a more sustainable development trajectory. Women, as primary users of energy are key to addressing the climate and energy solutions. As caregivers and users of energy to produce food in the home, they are typically most vulnerable to climate change and energy crises, costs, and supply interruptions. And, due to a lack of relevant skills women across the globe also face challenges in becoming active participants in the renewable energy sector other than being end-user customers. In Barbados (CBD, 2018) very few women were and still are, employed in the energy sector and even fewer women have enterprises that deal with RE technologies. Modern energy access is important because women from households with access to reliable affordable clean energy are more competitive and more easily access opportunities to bridge gender gaps through the ability to study, to access information, to use modern appliances and lessen drudgery and even engage in productive work. **Energy access is therefore pivotal to economic empowerment.**

Accordingly, the GCF developed one of its key result areas as **Energy Access and Power Generation**. GCF focuses on energy generation from renewable sources, energy efficiency and reliable energy transmission, distribution, and storage. This aligns with Sustainable Development Goal number 7 (SDG7) which calls for *"affordable, reliable, sustainable and modern energy for all"* by 2030. Projects under this area look to: generate energy from renewable sources such as wind, solar, geothermal, hydro, and sustainable bioenergy; efficiently and reliably transmit, distribute, and store energy; and promote access to clean energy in a way that promotes sustainable development and climate resilience while reducing emissions.

Barbados is highly dependent on the importation of fossil fuels from Trinidad and Tobago thus costing a significant amount of the limited forex available to the country. In turn prices for electricity generation and use are generally higher than for countries connected to a continental land mass. Since the 2017 hurricanes Irma and Maria, and more recent ones that caused excessive damage to the electricity grid system, the government of Barbados has sought to improve the country's energy efficiency and to build a more resilient energy infrastructure. It also sought to expand the use of renewable technologies as well as reduce its energy carbon emissions. One of the areas of development is solar and water heaters for housing. As alluded to before, these programs face several challenges that slow down the process of transitioning to renewable energy (solar, wind, geothermal). The barriers that inhibit fast transition uptake of these technologies include funding, skills and research and development.

Challenges faced with regards barriers to renewable energy uptake.

- Currently, the cost of electricity is too high for the low-income people of Barbados, and this is attributed to importation of fossil energy. As a result, many poor households barely afford electricity. If the bulk of electricity can be generated locally – more funds can be available to the government for its other programmes.

- There are few women enterprises involved in the renewable energy sector (i.e., technology development and marketing) and this is due to lack of knowledge and financial resources. The draft National Gender Policy therefore recommends that targeted effort be invested in creating employment opportunities for women in alternative energy sectors.

Recognizing the gender-energy-poverty-nexus in planning and implementing development programs can contribute to meaningful results that benefit women and address gender energy vulnerabilities such as those identified by GCF (along with capacities for a change) (Table 4-1. GCF Vulnerabilities and Capacities for Change-Energy and Power Generation).

Table 4-1. GCF Vulnerabilities and Capacities for Change-Energy and Power Generation

Vulnerabilities	Capacities for Change
• Share of women who are energy entrepreneurs. • Share of women-headed households without clean energy access • Impacts women are experiencing due to lack of renewable energy access.	• Roles of women in the energy sector • Provide incentives for the use of renewable energy and sustainable development practices. • Create/expand employment opportunities for women in all sectors of the economy: value added agriculture, alternative energy, new technologies, sustainable micro enterprise; • Share of women in the energy labour market and type of occupation • Policies and institutional framework to promote gender equality in energy /power generation. • Share of women households with solar heaters and PV panels on their rooftops

Opportunities for the Blue Green Bank

There are opportunities for BGB to increase women benefiting from climate energy initiatives by ensuring that climate change programmes target women-headed households as beneficiaries, that women businesses benefit from RE technologies, and by providing targeted training to women on technology deployment and maintenance, access to loans etc. Women, as main users of energy in the home, are key stakeholders in deploying and developing energy-efficient practices and therefore should be included in designing and planning the relevant strategies.

4.2 WOMEN, GENDER EQUALITY AND WATER SECURITY

Barbados is ranked as one of the most water-stressed countries in the world and, therefore, has a pressing need to conserve freshwater. Furthermore, frequent drought conditions in the last decade plus ageing water infrastructure have continued exacerbated water unavailability, increasing the cost of the water utility (OCHA, 2022). Isaacs, et. Al. 2017, carried out a comprehensive gender assessment of the issue as part of their PhD Thesis. In study found that, while Barbados has a 97% potable water coverage (Isaacs et al, 2017) it is reported to have experienced unprecedented water interruptions – at times affecting the population. In such situations, the gendered differences are highlighted and those in vulnerable groups suffer disproportionately. This was made visible during the 2018-2020 drought, where anecdotal evidence pointed strongly to the fact that the most significant impacts were felt by female-headed houses with children and other vulnerable groups such as the elderly. Impacts included: increased household costs for basic necessities, challenges with food preparation, poor sanitation, physical and psychological stress and job insecurity. (Government of Barbados - Bureau of Gender Affairs, 2021)

Water security goes hand-in-hand with equitable access to safe potable water, and the draft National Gender Policy therefore advocates for the importance of ensuring equitable access to safe water and sanitary facilities, particularly for densely populated communities in disadvantaged areas, which include many female-headed households.

Water security falls under **GCF's Health, Food and Water Security** results area. The GCF's vision for water security is to secure water resilience and water services under conditions of increased climate change impacts. This vision also aligns with the Sustainable Development Goals SDG6 and SDG13. The SDG goals indicate that *"water security, induced by water scarcity supports fast-tracking climate investment into transformational water projects"*. The proposed integrated water technologies and conservation align well with the integrated housing infrastructure being proposed for the Blue-Green Bank.

The overriding challenge faced in Barbados is water scarcity on the island due to climate change-induced drought, which disproportionately impacts female-headed households and vulnerable communities' access to safe and reliable water supply.

The GCF identifies the vulnerabilities and capacities for change as listed in Table 5 below.

Table 4-2. GCF Vulnerabilities and Capacities for Change - Health Food and Water Security

Vulnerabilities	Capacities for Change
• Share and number of women with increased exposure to water or health-related hazards during water outages • Lack of access to basic services such as safe water and improved sanitation	• Roles of women in targeted communities. • Share of women and men in relevant labour markets and type of occupation. • Share of women enterprises benefiting from the reliable water supply. • Number of women-headed households with houses fitted with water efficient technologies. • Policies and institutional framework to address vulnerabilities. • Equitable access to essential health services, as well as access to safe water and sanitary facilities, must be ensured in disadvantaged areas such as densely populated communities

Opportunities for the Blue Green Bank

The BGB targets integration of housing provision, water conservation and renewable energy technologies ensuring efficiency and cost recovery. Opportunities lie, in the future, in facilitating finance for water conservation.

4.3 INFRASTRUCTURE AND THE BUILT ENVIRONMENT

Climate resilience is critical for the sustainable development of Barbados, especially in terms of infrastructure, including housing. The country is highly vulnerable to the impacts of climate change, such as sea-level rise, increased frequency of natural disasters, and more intense weather events, all of which can cause significant damage to infrastructure, particularly housing. This can result in the displacement of populations, decreased economic productivity, and increased poverty and inequality.

Housing is an important aspect of climate resilience in Barbados. Adequate and safe housing is a fundamental human right and is essential for the health, well-being, and productivity of individuals and communities. **Ensuring that housing is climate resilient is crucial for protecting people from the impacts of climate change, such as the increased risk of flooding and damage from hurricanes and storms.**

Climate-resilient housing in Barbados can take several forms, such as elevating homes to higher ground, constructing homes with reinforced concrete and strong roofs, and designing homes with features that allow for natural ventilation and light. The government, in partnership with NGOs and private sector organizations, can also play a role in promoting and financing the construction of climate-resilient housing, through programs and incentives that encourage the adoption of best practices in building design and construction.

In addition to protecting residents from the impacts of climate change, climate-resilient housing can also provide long-term economic benefits. By reducing the risk of damage to homes, residents can save money on repairs and reconstruction after natural disasters. This can help to maintain the economic productivity of individuals and communities and promote sustainable economic growth in Barbados.

Between 2020 and 2021, Barbados experienced infrastructure damage. Therefore, in responding to these climate hazards and in line with climate commitments the government of Barbados intends to provide funding for hurricane-resilient housing which is integrated into renewable energy and water efficient systems.

Due to its critical role in climate adaptation and mitigation, Infrastructure and built environment are one of the GCF's 8 key result areas. The goals for this results area are advanced across four “transition” areas, namely: i) built environment ii) energy and industry, iii) human security, livelihoods, and wellbeing and iv) land use, forests and ecosystems. GCF aims to build new infrastructure and retrofit old traditional infrastructure to climate-proof it.

This results area aligns with SD 9 “Industry, innovation and infrastructure” and is also linked to SDG 11 “Sustainable cities and communities”, SDG 6 “Clean water and sanitation” and SDG 7 “Affordable and clean energy”. Due to its cross-sectoral nature, it is apparent that the provision of climate-resilient infrastructure has huge potential in addressing climate change and in building climate-resilient communities in the case of Barbados. And since climate-induced disaster events disproportionately negatively impact women and the poor, far more than men and the affluent; mainstreaming gender in the provision of infrastructure is critical. Moreover, women’s empowerment through ownership of housing infrastructure is also flagged as a priority in the draft National Gender Policy. (Government of Barbados - Bureau of Gender Affairs, 2021)

Examples of vulnerabilities that women and men experience due to conventional infrastructure that is not climate resilient are outlined in Table 6.

The main challenges faced in Barbados with regards to infrastructure are:

- Aging infrastructure, which is not climate resilient⁴,
- Inability by low-income households (60% of which are female-headed) to own houses due to financial constraints to secure home loans,
- Addressing the issue of over 90% hotel rooms that are located within the coastline climate risky zone, which has a direct knock-on effect for women given the tourism industry employs a large proportion of low-income workers, many of whom are women
- Lack of financing for climate resilient infrastructure, particularly homes for the low-income and the elderly.

Table 4-3. GCF Vulnerabilities and Capacities for Change-Infrastructure and the Built Environment

Vulnerabilities	Capacities for Change
Share and number of women lacking access to climate-resistant housing and/or infrastructure (efficient water technologies, water heating systems, renewable energy access, etc.)	• Roles of women in the design and maintenance of infrastructure and the built environment • Share of women and men in relevant labour markets and type of occupation. • Share of women-headed households assisted with home loans to acquire green climate housing.

⁴ Damage to infrastructure affects women and the poor mostly

	• Policies and institutional framework to promote equal access to climate-resilient infrastructure.
Share of low-income heads of households who would lose jobs due to location and quality of employer infrastructure (largely hotels); as an aftermath of climate disasters	• Policies and building guidelines that prohibit construction in climate sensitive zones • Policies and codes that enforce construction of climate resilient infrastructure • Adequate insurance policies to cover job losses, and home losses to climate change.

Opportunities for the Blue Green Bank

- The BGG targets and facilitates access to climate-resilient housing at affordable rates through below-market loans that have a quota system to ensure that women and other vulnerable groups benefit equally.
- The BGG and intermediary partners to establish finance packages and lending criteria and processes that are gender responsive and user friendly, to ensure low-income groups and women understand the system and easily apply for loans.
- The FIT program to fully integrate with the BGG climate financing to ensure that earnings from solar energy are sufficient to meet bank home loans repayment.
- The establishment of a robust insurance scheme to cover loans afforded to climate finance beneficiaries.

5 Gender analysis

5.1 OVERARCHING CHALLENGES IDENTIFIED FROM THE COUNTRY AND SECTOR ASSESSMENTS

The Table 7 below serves as a summary of some of the most pressing gender issues in Barbados. Interventions proposed in the table are not necessarily within BGB's scope of influence/impact but serve as useful contextual considerations when thinking about what gender-specific commitments and actions should be taken by BGB. The tables that follow provide sector specific challenges and potential interventions that were considered when developing the project's Gender Action Plan.

Table 5-1. Overarching Issues Affecting Women and Potential Interventions for Barbados

Key Challenge affecting Gender	Possible Cause	Potential Intervention
<i>High vulnerability to climate impacts due to economic downturns</i>	▪ Limited opportunities for income generating projects. ▪ Limited social support ▪ Lack of financial loan resources for the low income (both men and women)	▪ Create opportunities for financing the low-income groups, particularly women. ▪ Support women entrepreneurship
<i>Unacceptable levels of Gender Based Violence</i>	▪ Poverty and vulnerability ▪ Limited awareness of the impacts, both men and women ▪ Limited safe shelters for victims	▪ Community awareness and empowerment women ▪ Programmes targeted at men and boys. ▪ Establishment of safe housing for victims of GBV

<i>Few women represented in top management jobs</i>	▪ Systematic stereotyping in the private sector ▪ Gender neutral policies and plans ▪ Inadequate family support systems at the workplace to push women productivity	▪ Incentivise private firms to employ women and promote women to top position. ▪ Give tax breaks for on-the-job training targeting women. ▪ Government interventions, such as quota systems ▪ Provision of childcare and parental care legislation ▪ Increase affordable day care facilities and hours of operation
<i>Wage gap between men and women in same jobs</i>	▪ Inadequate monitoring of wages and implementation of legislation determining domestic pay or minimum wage	▪ To enact sex disaggregated data reporting requirement on wages for all firms.
<i>Limited access to reproductive health for girls and women</i>	▪ Impacts are HIV risk, early marriages and in some cases large family size	▪ Increase efforts to avail health services for reproductive health

Water security

Table 5-2. Gender issues and potential interventions to attain water security

Key Challenge affecting Gender	Possible Cause	Potential Intervention through the project
<i>Compromised hygiene due to water outages</i>	▪ Inadequate water supply	▪ Ensuring reliable water supply through rainwater harvesting and water-efficient technologies
<i>Lack of fair distribution of service among the population with the poor being disadvantaged (poor maintenance and outages)</i>	▪ Gender-neutral planning by BWA	▪ Maintenance and distribution to be gender sensitive and planned to respond to the needs of specific groups, including the most vulnerable
<i>Lack of new initiatives and programmes that address technological development in water management and conservation</i>	▪ Limited water efficient technologies available ▪ Possibly lack of motivation to conserve water	▪ Facilitating the financing of a water conservation programme

Renewable energy

Table 5-3. Gender issues and potential interventions for renewable energy

Key Challenge affecting Gender	Possible Cause	Potential Intervention by the project
<i>Few women-led enterprises in the RE sector</i>	▪ Limited chances to capitalize on the enterprises. ▪ Gender-blind policies in the climate financing partners	▪ Encouraging intermediaries to target women led Enterprises and creating financing packages to meet women entrepreneurship needs
<i>Few women taking up training in the deployment of RE technologies</i>	▪ Stereotypes on the traditional jobs for women ▪ Gender blind policies in technical colleges or training centres	▪ Ensuring that BGB and intermediary partners address gender biases by putting in place gender policy and guidelines. ▪ Collecting sex disaggregated data of borrowers.

Infrastructure and built environment

Table 5-4. Gender issues and potential interventions in the infrastructure and built environment

Key Challenge affecting Gender	Possible Cause	Potential Intervention by the Project
<i>Few women in the construction and design industry</i>	▪ Women peer pressured into traditional / stereotypical jobs. Lack of effort to market the field for women	▪ Encouraging intermediaries to incentivise women to take up skills training by providing an enabling work environment.
<i>Low-income groups unable to secure home loans</i>	▪ Banks risk averse ▪ Low-income Barbadians lack collateral	▪ Work with intermediaries that structure loans to target the lending to low-income borrowers, particularly women and the vulnerable ▪ The intermediary partners will provide support for applicants to ensure there is financial literacy and understanding of available loan packages

Available housing is not climate resilient	▪ Traditional buildings were not designed to be resilient to current climate impacts	▪ Provision of climate-resilient housing upgrades and new builds.
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5.2 RECOMMENDATIONS TO REDUCE GENDER GAPS

This gender assessment acknowledges the success and efforts that the Government of Barbados has made to reach gender parity in all its economic and development sectors. Some areas will continue to need improvement through the implementation of policies, initiatives, and monitoring. These areas include the gender pay gap, domestic and work security and violence against women (UN Women, 2022). The recommendations given in this report are within the scope of the BGB project.

It is, therefore, recommended that the BGB project ensures that climate finance is accessible by all including the vulnerable and women-headed households who are most in need of homes – which in turn will reduce vulnerability to climate change. To avoid biases and ensure that the targeted beneficiaries apply and obtain the loans, the BGB will train their staff in gender mainstreaming, develop an internal gender policy and guidelines to support gender equality, and as well as develop appropriate financing packages. The comprehensive list of actions that will support this recommendation is listed below.

The Blue Green Bank and its staff must have an appreciation of gender mainstreaming and what it entails. For this to happen the BGB needs to:

- Create a position for a gender specialist (if not in place already- recruit one) to assist in implementing the GAP and ensuring that the BGB and intermediaries have the necessary processes in place.
- Carry out sector-based gender assessments to identify the key barriers to securing finance and use these findings to guide how inclusive financing packages are designed and marketed.
- Develop an **evidence-based gender policy** for staff and intermediary partners.
- Develop as part of the organisation's code of conduct a **Sexual Exploitation Abuse and Harassment Policy** for BGB and its partner programmes.
- Ensures that all its staff attend gender **training /workshops** so that there is an understanding and appreciation of the importance of reducing the gender gap in climate finance.
- Develop and adopt a gender screening tool that all partners supported by the BGB have to satisfy to obtain finances;
- Agree with the partners to put in place **gender-responsive loan/investment products** that target both men and women equally – and develop a funding guideline that is gender-responsive.
- Carry out an awareness-building program to give confidence to potential customers who may need to apply for climate finance.
- Request partners and borrowers to monitor the uptake of financing and collect sex-disaggregated data to measure if progress is being made to reduce the gender gap.
- Put in place customer surveys and complaints mechanisms that allow for learning and improvement, and package and distribute these lessons from the programme to be used as a basis for adjusting and improvements to services.
- Carry out a mid-term evaluation of the programme, using the Gender Action Plan.

6 Gender action plan

Impact: The vulnerable groups in Barbados (the low-income- Women and men) have an opportunity to own their own houses by accessing affordable home loans that are provided through the programme. The resulting climate infrastructure will build climate resilience among the beneficiaries and contribute towards reducing gender gaps. The beneficiaries will also access renewable energy and achieve water security which will improve the health of women and lessen their caregiving burdens. With additional electricity from FIT the Government of Barbados will save costs from importing energy.

Outcome 1: Innovative financial structure operating in Barbados to overcome barriers to climate finance and gender inequalities. The Bank will thus achieve the following gender-responsive outcomes:

- 1.1 The Blue-Green Bank is established and will weave gender responsiveness into its interventions and systems.
- 1.2 Enhanced resilience and improved health for communities, including women and children, through GHG emissions reductions and adaptation measures, relating to both built and natural environments.
- 1.2 Improved access to clean energy and empowerment of local communities, including women, girls, and vulnerable populations, through renewable energy, energy services, enhanced energy efficiency and new technologies.
- 1.3 More equitable decision-making between men and women, reductions in gender-based violence, increased access to and control over resources, and economic empowerment of women and girls through the GGC's Environmental and Social Management System (ESMS), mandatory project safeguards, stakeholder consultation framework (including expert consultation on pressing gender issues, determined on a country-by-country basis), and gender-responsive project design approaches.
- 1.4 Strengthened institutional capacity of project owners in the design and delivery of gender-responsive projects, aligned with GCF requirements.
- 1.5 Improved livelihoods that enhance the resilience of vulnerable communities, including women and girls, to the negative climate impacts currently affecting the critical system.

Output 1: Develop processes and systems that will enhance gender equity at BGB

Activities	Indicator and targets	Timeline	Responsibilities	Budget
i. Recruit a gender specialist for the programme to oversee the implementation of the GAP	Gender specialist recruited by the time the BGB is established	Prior to project implementation	AE- Human Resources	(Salary TBD)
ii. Dedicate necessary human and financial resources to strengthen and evolve the BGB's capacity related to gender equality knowledge management, learning, and implementation through: A. the appointment of a dedicated gender resource to oversee all gender-related activities; and B. the ad hoc appointment of gender specialists with in-country and sector-specific expertise, as needed on a project-by-project basis.	Line items are included in the annual budgets to cover processes and activities that are project-specific and note the appointment of a gender expert, as required.	Year 1 Report: Quarterly and annually	Financial Officer and Human Resources	BGB budget
iii. Include gender-responsive elements in the eligibility criteria and post-evaluation criteria	100% of projects and proposals use the gender criteria prior to submission to the BGB	Year 1	Gender specialist and BGB management team	BGB budget
iv. Develop mechanisms to assess gender impacts of projects submitted by intermediaries but also to internally monitor, report and verify gender impacts and other co-benefits project levels	Mechanisms and guidelines in place Regular reports submitted by intermediaries	Year 1	Gender Specialist	BGB Budget

v. Ensure the representation of women in the Board and/or key decision-making and advisory bodies of the BGB	Percentage of women represented in the Board Management of the BGB approaches 50% (based on senior management status in the country)	Ongoing basis	Chief Executive Officer and Human Resources	BGB Budget
vi. Develop a BGB Gender Policy and training kit that aids in understanding the nexus between gender and climate finance	Human resources equipped to implement. BGB Gender Policy and training manual developed	Within the first 6 months	Gender specialists and Human resources	BGB budget
vii. Develop a Sexual Exploitation, Abuse and Harassment Policy (including complaint procedures)	BGB- SEAH policy and codes of conduct developed	Within the first 6 months	Gender specialist	BGB Budget
viii. Develop a brief, targeted market study to determine the gender landscape in Barbados at the time of implementation including understanding the following: - Identify women's barriers to access finance about the BGB's expected sectors of operations in its first five years. - Identify opportunities to engage in policy dialogue activities to address issues restricting women's ability to adopt climate technologies financed under the programme, such as access to finance, lack of access to collateral etc. - Identify opportunities to market and raise awareness of climate finance opportunities. - Identify capacity needs for the BGB, the intermediaries and the borrowers. Using the evidence gathered develop a gender policy for BGB	Developed market studies for internal and external publications. Identified women's needs and priorities to use for developing finance packages	Year 1	Gender Specialist and consultant	\$10000.00

Output 2: Project-level gender-focused assessments and action plans for financed activities (either directly or indirectly) by BGB are developed				
Activities	Indicators and target	Timeline	Responsibility	Budget
I. The BGB will ensure that prior to finance being issued by an intermediary or directly from the BGB, a robust gender assessment and action plan are developed by the project owner. The gender assessment must contain the following aspects before it's considered complete: - Women's organisations working on gender issues in that sector are identified and engaged. - Gender-related baseline data specific to the sector is established. - Project alignment with / support of the country's SDG, NDC and policy gender-related commitments. - Sector-specific gender vulnerability(ies) are identified.	At least 2 NGOs working on gender identified for partnership and upscaling of BGB interventions Baseline data available for benchmarking progress Revised GAP activities to address vulnerabilities and align with SDGs	In the first year Reporting in the quarterly report to GCF	Project Owner Gender Specialist AE to review the reporting and data produced Intermediary partners	\$60000

Opportunities/solutions for the project to address these vulnerabilities are well-defined and adequately integrated into the project design and delivery processes.				
ii. Establish a gender-responsive environment for staff and borrowers: - Create financial packages that reflect gender considerations. - Disseminate materials, use media, community workshops or other appropriate means. - An operating budget is allotted to implement the action plan and analyses required every year - Stakeholder engagement or consultations were undertaken and informed the design of the project/programme.	Guidelines that address gender are in place. Number of people reached during dissemination activities and indicating number of women and other vulnerable groups- Report. No enquiries from potential borrowers	By the end of the first year of operation Budget on a rolling basis	BGB to appoint GFP	BGB budget for workshops and advertisements
Output 3: Capacity building and knowledge generation				
Output 3a: Capacity building of the intermediary partners' staff and ensure that they have gender-responsive tools and gender-transformative systems/structures to enable borrowers, both men and women to access the climate funds				
I. Produce a gender training manual for staff and train them in gender mainstreaming. ii. During the early days of operation carry out weekly /monthly sessions to discuss progress, and lessons learned from implementing the gender programme	Number of staff trained in line with the gender training module. Number of seminars for senior management to promote women's access to climate funds. Databases created that include the number and type of programme sub-borrowers (disaggregated by gender; loan amount; sector; residential/commercial use)	On a rolling basis, where applicable, as sub-projects come through	Gender Specialist	BGB Budget
Output 3b: Lessons learnt, case studies and product improved based on issues identified. Disseminate information on new products for the benefit of borrowers and programme upscaling				
ix. Document lessons and outcomes through reports or staff briefs compiling and presenting lessons learnt and challenges and areas of success. - Carry out customer service surveys to assess and address issues. - Implement periodic staff feedback on new products and efficacy. - Document frequent complaints/challenges and find ways of addressing these.	Customer surveys are carried out and compiled every month. A register of complaints was maintained. Number of staff meetings that discuss the rollout of products and focusing on gender aspects.	During the first 2 years	Gender Specialist Intermediary partners	BGB Budget
ii. Carry out a midterm project evaluation to assess progress in implementing the Gap and achieving the outcomes	Evaluation reports showing trends and recommendations	End of 2 years	BGB management	\$50000

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